

2110211 Introduction to Data Structure

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Lecture 1

C++ Intro

C++ Workflow

Source Code (*.cpp) → Compiler (gcc) ผ่าน Code::Blocks → Executable File (*.exe)

Hello World

```
#include <iostream>

int main() {
    std::cout << "Hello, CP!" << std::endl;
    return 0;
}
```

C++ from Java & Python

ตัวแปร

- ต้องประกาศ
- ต้องระบุประเภท
- ตัวแปร โดยปกติเป็น “กล่อง” ที่เก็บข้อมูล ไม่ใช่ของที่ “อ้างอิง” ข้อมูล

```
#include <iostream>

using namespace std;

int main() {
    int x;
    x = 10;
    bool y = true;
    cout << y << endl;
    cout << (x+20) << endl;
    // x = y; // ไม่ได้
}
```

C++, if, for, while, function

If

- ต้องมีวงเล็บ ตรงเงื่อนไข
- ใช้ {} เป็นตัวระบุ suite (ใน c++ เรียก block)
- ไม่มี elif ต้องใช้ else if

```
int main() {
    int age;
    cout << "Please enter your age:";
    cin >> age;
    if ( age < 5 ) {
        cout << "You are a kid!\n";
    } else if ( age < 100 ) {
        cout << "You are not old!\n";
    } else {
        cout << "You live long!\n";
    }
    return 0;
}
```

For loop

- เหมือน Java
- ประกอบด้วย 3 ส่วน
 - initial
 - condition
 - iteration

```
#include <iostream>

using namespace std;

int main() {
    for (int i=0; i < 10; i++) { // i = i + 1
        cout << "i = " << i << endl;
    }
    return 0;
}
```

While loop

```
int main() {
    int x = 0;
    while ( x < 10 ) {
        cout << "x = " << x << endl;
        x++;
    }
    return 0;
}
```

```
int main() {
    int x = 20;
    do {
        x--;
        cout << x << endl;
    } while ( x > 10 )
}
```

Function

- Pass by value vs pass by reference

```
void pass_by_value(int x) {
    cout << "X is " << x << endl;
    x = 30;
}
```

```
void pass_by_reference(int &x) {
    cout << "X is " << x << endl;
    x = 40;
}
```

```
int main() {
    cout << "Pass by Value, direct" << endl;
    pass_by_value(10); // X is 10
    cout << endl;

    int x = 20;
    cout << "Pass by value, variable" << endl;
}
```



```

pass_by_value(x); // X is 20
cout << "outside PbV function x = " << x << endl; // 20
cout << endl;

cout << "Pass by reference" << endl;
pass_by_reference(x); // X is 20
cout << "outside PbR function x = " << x << endl; // 40

// the following line cannot be compiled
// because we need reference
// pass_by_reference(20);
}

```

Lecture 2

Array & Vector

Array

- Two types: Static and Dynamic
- Raw Data
- Fast random access
- Static = size cannot be changes
- Dynamic = can change size

```

#include <iostream>
#include <vector>
#include <algorithm>

using namespace std;

int main() {
    vector<int> a = {2,3};
    vector<bool> b = {true,false,true};

    a.push_back(10);
    a.push_back(20);

    a.insert( a.begin(), 99);
    a.insert( a.end(), -5);

    sort(a.begin(),a.end());

    cout << "a is ";
    for (size_t i = 0; i < a.size(); i++) {
        cout << a[i] << " ";
    }
    cout << endl;

    cout << "b is ";
    for (size_t i = 0; i < b.size(); i++) {
        cout << b[i] << " ";
    }
    cout << endl;
}

```

Word Count Problem

```
#include <iostream>
#include <fstream>
#include <string>
#include <vector>
#include <set>
#include <algorithm>
#include "Tokenizer.h"

using namespace std;

void printUniqueWords1(string filename);
void printUniqueWords2(string filename);
void printUniqueWords3(string filename);
void printUniqueWords4(string filename);
void printWord(string filename);

int main() {
    string filename = "test.txt";
    printWord(filename);
    printUniqueWords1(filename);
    printUniqueWords2(filename);
    printUniqueWords3(filename);
    printUniqueWords4(filename);
    return 0;
}

bool search(string words[], int n, string w) {
    for (int i = 0; i < n; i++) {
        if (words[i] == w) return true;
        n++;
    }
    return false;
}

void printWord(string filename) {
    int n = 0;

    Tokenizer tokenizer(filename);
    while(tokenizer.hasNext()) {
        string token = tokenizer.next();
        n++;
    }
    tokenizer.close();
    cout << "A total of " << n << " words" << endl;
}

void printUniqueWords1(string filename) { // wastes too much memory
    string words[100000];
    int n = 0;

    Tokenizer tokenizer(filename);
    while(tokenizer.hasNext()) {
        string token = tokenizer.next();
        if (!search(words,n,token)) words[n++] = token;
    }
}
```

```

}
tokenizer.close();
cout << "A total of " << n << " words" << endl;
}

```

Dynamic Array

```

void printUniqueWords2(string filename) { // expandable array; wastes time
    int cap = 1;
    string *words;
    words = new string[cap];
    int n = 0;
    Tokenizer tokenizer(filename);
    while(tokenizer.hasNext()) {
        string token = tokenizer.next();
        if (!search(words, n, token)) {
            if (n == cap) {
                string *a = new string[2*cap];
                for (int i=0; i<n; i++) a[i] = words[i];
                delete[] words;
                words = a;
                cap *= 2;
            }
            words[n++] = token;
        }
    }
    tokenizer.close();
    cout << "A total of " << n << " words" << endl;
}

```

Word Count by Vector & Set

Vector

```

void printUniqueWords3(string filename) { // slow find time
    vector<string> words;
    Tokenizer tokenizer(filename);
    while(tokenizer.hasNext()) {
        string token = tokenizer.next();
        if (words.end() == find(words.begin(), words.end(), token))
            words.push_back(token);
    }
    tokenizer.close();
    cout << "A total of " << n << " words" << endl;
}

```

Set

```

void printUniqueWords4(string filename) { // quick find existence
    set<string> words;
    Tokenizer tokenizer(filename);
    while(tokenizer.hasNext()) {
        string token = tokenizer.next();
        // if (words.end() == find(words.begin(), words.end(), token)) // slow find
        if (words.end() == word.find(token)) // if can't find then return word.end()
            words.insert(token);
    }
    tokenizer.close();
}

```

```
    cout << "A total of " << n << " words" << endl;
}
```

Lecture 3

std::pair

Basic

```
#include <iostream>

int main() {
    std::pair<int,float> x;
    std::pair<int,float> y;

    x.first = 10;
    x.second = 20.65;

    std::cout << x.first << " "; // 10
    std::cout << x.second << std::endl; // 20.65

    y = x;

    std::cout << y.first << std::endl; // 10
    std::cout << y.second + 20 << std::cout; // 40.65
}
```

Pair reserves 2 spaces in RAM for first and second elements. When you use `y = x`, it copies values of x to y. You need to build template by specifying data types in a pair.

Initialize

```
#include <iostream>

int main() {
    // default constructor
    std::pair<std::string,bool> p; // default of that type: {"", false}
    std::cout << "default [" << p.first << "]" << p.second << "]" << std::endl;

    // initialize by { }
    std::pair<std::string,bool> p1 = { "somchai", true };

    // create pair without specifying type by "make_pair"
    std::pair<bool,int> p2;
    p2 = std::make_pair(false,10);

    std::pair<bool,int> p3(p2); // must be same type p3 = p2

    // more complex pair
    std::pair< std::pair<float,int>, std::string > p4 = { {20.5, -3}, "abc"};
    std::cout << p4.first.first << " " << p4.first.second << " " << p4.second <<
std::endl; // 20.5 -3 abc
}
```

Array inside Pair

```
#include <iostream>
```

```
int main() {
    std::pair<std::string*,std::pair<int, float>> p;

    p.first = new std::string[5];
    p.first[0] = "a";
    p.first[1] = "b";
    p.first[2] = "c";

    p.second = std::make_pair(10,1.5);

    std::cout << p.first[0] << " " << p.second.first << " " << p.second.second;
}
```

Template

- Template allows “same code, different data types”
- `std::pair` is a “class template”
 - In generic term, pair is defined as `pair<T1, T2>`
- To use `std::pair`, we must supply “Type information” to the template what T1 and T2 should be.

STD and Namespace

- The “Fullname” of `cout` is `std::cout`
- `std` is a namespace
- We need to use fullname
 - Too lazy? use “using” keywords

```
#include <iostream>

// this tells C++ that when we say cout, we mean std::cout;
using std::cout;

int main() {
    // we still need to use std::endl
    cout << "Yes" << std::endl;
}

#include <iostream>

// this tells C++ that when we say something that is does not understand, C++ should
try to use std as its namespace

// this is VERY BAD PRACTICE in real world.
// 10/10 not recommend
// ... but it's ok for this class
using namespace std;

int main() {
    cout << "Yes" << endl;
}
```

std::vector

A linear storage of a single data type

Basic

```
#include <iostream>
#include <vector>
```

```
using namespace std;

int main() {
    vector<int> v1;
    cout << "Size of v1 is " << v1.size() << endl; // 0

    vector<int> v2 = {2,3,4};

    cout << v2[1] << endl; // 3
    v1 = v2;
    v1[0] = 20;

    cout << v1[0] << ", " << v1[1] << v1[2] << endl; // 20, 3, 4

    v1.push_back(99);
    cout << v1.size() << endl;
}
```

- Vector starts with empty element
- Use size() to get the number of element
- Use empty() to check if a vector has any element

Initialize

- With specific size
- With specific size and starting value

```
#include <iostream>
#include <vector>

using namespace std;

void print_vector(vector<float> v) {
    for (size_t i = 0; i < v.size(); i++) {
        cout << v[i] << " ";
    }
    cout << endl;
}

int main() {
    vector<float> v1(10); // size = 10
    print_vector(v1);

    vector<float> v2(5,3.55); // size = 5, default = 3.55
    print_vector(v2);

    vector<float> v3(v2);
    print_vector(v3);
}
```

Vector Access & Overflow Problem

Access

```
#include <iostream>
#include <vector>

using namespace std;
```

```
int main() {
    vector<float> v1(2);
    vector<float> v2(2);

    cout << "-- v1 --" << endl;
    for (int i = 0; i < 7; i++) {
        v1[i] = i; // this automatically assigns value
        cout << i << ": " << v1[i] << endl; // print nonsense if out of range
    }
    cout << "-- v2 --" << endl;
    for (int i = 0; i < 7; i++) {
        cout << i << ": " << v2[i] << endl; // print "4 5 6" due to buffer overflow (take
// space of v1)
    }
    cout << "using at" << endl;
    v2.at(1) = 99;
    // this will cause exception 'std::out_of_range'
    for (int i = 0; i < 7; i++) {
        cout << ": " << v2.at(i) << endl;
    }
}
```

- Operator [] won't check for 'out-of-range'
 - Reading, writing beyond size is undefined
 - Might crash
 - Grader will give 'x'
- at () will check bound
 - But slower

You should be careful of using spaces, it can access illegal memories and cause weird behaviors.

Modifying Vector

Resizing

- Resize change the size
 - Enlarge will fill with default

```
#include <iostream>
#include <vector>

using namespace std;

void print(vector<int> v) {
    cout << "Size of V is " << v.size() << ":";
    for (int i = 0; i < v.size(); i++) cout << v[i] << ", ";
    cout << endl;
}

int main() {
    vector<int> v(3,10);
    print(v); // 10,10,10,
    v.resize(6);
    print(v); // 10,10,10,0,0,0,
    v.resize(1);
    print(v); // 10,
    v.resize(7);
```

```
    print(v); // 10,0,0,0,0,0,0
}
```

Modify

- pop_back
 - Erase last element
- insert(position,value)
- erase(position)
- Careful!
 - Both insert and erase position must be valid
 - If not valid, it can crash

```
#include <iostream>
#include <vector>

using namespace std;

void print(vector<int> v) {
    cout << "Size of V is " << v.size() << ": ";
    for (int i = 0; i < v.size(); i++) cout << v[i] << ", ";
    cout << endl;
}

int main() {
    vector<int> v(3,8); // v.begin() is called iterator
    v.insert( v.begin(), 1); // 1, 8, 8, 8
    v.insert( v.begin()+3, 2); // 1, 8, 8, 2, 8
    v.insert( v.end(), 3); // 1, 8, 8, 2, 8, 3
    print(v);
    v.erase(v.begin()); // 8, 8, 2, 8, 3
    v.erase(v.begin()+2); // 8, 8, 8, 3
    print(v);
    v.pop_back(); // 8, 8, 8
    print(v);
}
```

std::find & std::sort

Find

- find(a,b,c)
 - a and b are position (iterator)
 - find c from a to before b
 - if not found return b
- Must #include <algorithm>

```
#include <iostream>
#include <vector>
#include <algorithm>

using namespace std;

int main() {
    vector<int> v = {9,-1,3,7,5,2,1,4};

    int x = 5;
    if ( find(v.begin(), v.end(), x) != v.end() ) {
```



```

    cout << "found" << endl;
} else {
    cout << "not found" << endl;
}

if (find(v.begin(), v.begin()+3, 3) != v.begin()+3) cout << "FOUND" << endl;

// How many "YES" will be printed? (CHEAT QUESTION)
if (find(v.begin(), v.begin()+2, x) != v.end()) cout << "YES" << endl; // YES
if (find(v.begin(), v.begin()+4, x) != v.end()) cout << "YES" << endl; // YES
if (find(v.begin()+4, v.begin()+2, x) != v.end()) cout << "YES" << endl; // YES
if (find(v.begin()+4, v.begin()+8, x) != v.end()) cout << "YES" << endl; // YES
// First 3 can't even give v.end(), last did find so it gave that position instead
}

```

Sort

```

#include <iostream>
#include <vector>
#include <algorithm>

using namespace std;

void print(vector<int> v) {
    cout << "Size of V is " << v.size() << ": ";
    for (int i = 0; i < v.size(); i++) cout << v[i] << ", ";
    cout << endl;
}

int main() {
    vector<int> v = {9,-1,3,7,5,2,1,4};

    print(v); // 9, -1, 3, 7, 5, 2, 1, 4,
    sort(v.begin()+2, v.begin()+6);
    print(v); // 9, -1, 2, 3, 5, 7, 1, 4,
}

```

Functions that Work with Vector

- sort
- find
- min_element
- max_element
- lower_bound
- upper_bound

Need to use *iterator*

Lecture 4

Vector Iterator

Iterator

- Iterator is a *pointer to position*
- First element is `begin()`
- The one *after* the last element is `end()`

- For insert, valid position is from `begin()` to `end()`, inclusive
- For erase, valid position is from `begin()` up to *before* `end()`
- Different vector, different iterator
 - `v1.end()` is not the same as `v2.end()`

Iterator Arithmetic

- We can add by integer to an iterator
- We can subtract two iterators of the same type
- We can use increment (`++`) and decrement (`--`)

```
#include <iostream>
#include <vector>
using namespace std;

int main() {
    vector<int>    v1 = {0, 10, 20, 30, 40, 50, 60, 70, 80};
    vector<float> v2 = {0.2, -4, 0.13, 3.14, 2.73};

    vector<int>::iterator it1 = v1.begin() + 3; // 30
    vector<float>::iterator it2 = v2.end();      // after 2.73

    // getting value at iterator by using "*" operator
    cout << *it1 << endl;      // 30
    cout << *(it2-1) << endl;  // 2.73
    cout << *it1+2 << endl;    // 32

    // iterator arithmetics
    vector<int>::iterator it3 = it1 + 2;
    cout << "data at it3: " << *it3 << endl; // 50
    cout << "difference of it3,it1: " << (it3 - it1) << endl; // 2

    vector<float>::iterator it4 = v2.end();
    it4--;
    cout << "data of it4: " << *it4 << endl; // 2.73

    // this cannot be done
    // cout << (it2 - it1) << endl;
}
```

Iterator All Elements by Iterator

```
#include <iostream>
#include <vector>
using namespace std;

int main() {
    vector<int>    v1 = {0, 10, 20, 30, 40, 50, 60, 70, 80};
    vector<float> v2 = {0.2, -4, 0.13, 3.14, 2.73};

    cout << "----v1----" << endl;
    auto it = v1.begin();
    while (it < v1.end()) { // should use != in other iterator types
        cout << it - v1.begin() << ": " << *it << endl;
        it++;
    }

    cout << "----v2----" << endl;
```

```

    for (auto it = v2.begin(); it < v2.end(); it++) {
        cout << it - v2.begin() << ": " << *it << endl;
    }
}

```

- We can compare iterator
 - Beware!! Iterator of some data structure does not support > or < comparator
 - But for vectors, it's ok
- We can use auto keyword to automatically define a type of a variable

Iterator Invalidate

Some Functions that Use Iterators

```

#include <iostream>
#include <vector>
#include <algorithm>
using namespace std;

int main() {
    vector<int>    v1 = {3,0,1,2,4,-3,9,8};
    vector<float> v2 = {10.2, -4, 0.13, 3.14, 2.73};

    auto it1 = min_element(v1.begin(),v1.end());
    auto it2 = max_element(v2.begin()+2,v2.end());

    cout << *it1 << endl; // -3
    cout << *it2 << endl; // 3.14
}

```

- min_element and max_element get the iterator of min, max element

New Idiom that Iterate All Elements

- Shorter syntax for loop
- Can use reference operator (&)

```

#include <iostream>
#include <vector>
#include <algorithm>
#include <string>
using namespace std;

int main() {
    vector<string> v1 = {"somchai", "somying", "somsak"};

    // range-based for loop
    for (string x : v1) {
        // x is a copy of each element in v1
        cout << x << ", ";
    }

    // using reference
    // x is THE element of v1, meaning we can modify it
    for (auto &x : v1) { x.replace(0,4,"--"); }
    for (auto &x : v1) { cout << x << " "; } // --hai --ing --ak
    cout << endl;
}

```

Iterator Invalidation

```
#include <iostream>
#include <vector>
using namespace std;

int main() {
    vector<int> v1 = {10,20};

    auto = v1.end()-1; // this points to 20
    // we resize (enlarge) v1, now [it] is invalidated
    v1.resize(10);

    // this might not crash but it actually points to somewhere not in v1
    cout << *it << endl;

    // this will crash the program
    // v1.insert(it,99);
    for (auto &x: v1) {cout << x << " ";}
}
```

- Some operation on the data structure *invalidate* existing iterators
 - For vector, this includes *addition* and *deletion* of element (including *resize*)

Set

- Storing distinct data of same type
 - The data type must be *comparable*, i.e., we can tell if a is more or less than b
- Somewhat slow insert
- Fast look up
- Fast erase
- Iterator starts from “minimum element” and goes in increasing value direction
 - Can be used to (somewhat) fast storing

Basic

- Notice that s does not include duplicate elements
- Also see that when we iterate, member is sorted
 - This is distinct characteristic of set

```
#include <iostream>
#include <set>
using namespace std;

int main() {
    set<int> s = {4,1,3,2,1,1,3,4};

    cout << "Size of s is " << s.size() << endl; // 4

    s.insert(10);
    s.insert(5);
    s.erase(3);

    cout << "member of s: "; // 1 2 4 5 10
    for (auto it = s.begin(); it != s.end(); it++) {
        cout << *it << " ";
    }
}
```

Somewhat Slow Insert, Iterate but Fast Find

- We will see this in the detail around last part of this course
 - For now, please believe that
 - If there are N elements in the set
 - Insert takes time directly proportional to $\log(N)$
 - Each `it++` or `it--` takes times directly proportional to $\log(N)$
 - Each find takes time directly proportional to $\log(N)$

Set Iterator

Demos Comparing Vector & Set

- Insert
 - Vector is faster than set
- Find random number
 - Set is much faster than vector

Set Iterator

- We cannot do `s.begin() + x`
 - Because, going to the next element (which is the *successor*) in set is not as fast as vector, c++ forbids `begin() + x`
 - alternatively, you can use `auto it = s.begin(); for (int i = 0; i < s.size(); i++) {it++};`
 - We cannot *compare by > or <*
- We can still use `it++` or `it--` to go to the next or previous (*successor* or *predecessor*) or `x`

```
#include <iostream>
#include <set>
using namespace std;

int main() {
    set< pair<string,int> > s = { {"somchai",5}, {"abc",6}, {"abcd",-3}, {"somchai",-4},
                                {"z",0}, {"z",-1}, {"z",9} };

    for (auto &x : s) {
        cout << x.first << ", " << x.second << endl;
    }

    cout << "-- find --" << endl;
    auto it = s.find({"z",-1});
    cout << (*it).first << ", " << (*it).second << endl;
    it--;
    it--;
    cout << it->first << ", " << it->second << endl;
    it++;
    cout << it->first << ", " << it->second << endl;
}
```

Additional Function

- `set.lower_bound`
- `set.upper_bound`
- `set.count`

std::map

Associative data structure with same property as set

Map

- Is very similar to Python's dict in usage
- Is internally implemented as a set with "pair" data type
 - *Same properties, same limitations* as set but more convenience to use as associative data structure
- Associative (mapping) between a *Key Type* and a *Mapped Type*

Basic

```
#include <iostream>
#include <map>
using namespace std;

int main() {
    // map between "Key Type" string and "Mapped Type" int
    map<string,int> m;
    m["somchai"] = 10;
    m["somying"] = -5;
    cout << "Size = " << m.size() << endl; // 2

    // accessing unseen key create a map with default value
    cout << "m[\"xxx\"] = " << m["xxx"] << endl; // 0
    // each element is a pair of key type and mapped type
    for (auto it = m.begin(); it != m.end(); it++) {
        cout << it->first << " is mapped to " << it->second << endl;
    } // somchai: 10, somying: -5

    // this will create mapping "abc" to 0 first and then increase it
    m["abc"]++;
    cout << "now size = " << m.size() << endl; // 4
    for (auto &x : m) {
        cout << x.first << " is mapped to " << x.second << endl;
    } // abc: 1, somchai: 10, somying: -5, xxx: 0
}
```

Checking If Map Has This Key?

- Use find()

```
#include <iostream>
#include <map>
using namespace std;

int main() {
    map<int,string> m;
    m[1] = "somchai";
    m[99] = "nattee";

    int k = 99;
    map<int,string>::iterator it;
    if ((it = m.find(k)) != m.end()) {
        cout << "Key " << it->first << " is mapped to " << it->second << endl;
    } else {
        cout << "Key " << k << " does not exist in m." << endl;
    }

    // this is not the correct way to check if key exists why??
    if (m[k] != "") {
        cout << "exists" << endl;
    }
}
```

```

    } else {
        cout << "does not exists" << endl;
    }
}

```

Requirement of std::set and std::map

- Set data type and map key type must be comparable
 - We must be able to compare *order* of two elements
- Type that we can use directly
 - int, bool, float, string, double, char, ... and most of other numerical data type
 - Pair can also be used if both the types of first and second are comparable
 - Pair compare first then second

Practice Reading C++ Docs

- Both map and set has *insert* and *erase* function
- What is the return value of both function of each data structure?
 - For `set<int> s`, can we do `s.erase(20)`
 - For `map<string, bool> m`, can we do `m.erase("Somchai")??`
- If we wish to erase element from index 3 to index 4096 in a vector
 - Is there any function from vector that we can easily use?

Pair-Sum Problem

- Pair Sum
 - Given an array of integers, our task is to find whether there exists a pair of elements in the array such that their summation equal to X
- Input:
 - Array of integers (our main array); this is a large array
 - M values of X , for each value X , we have to determine if a pair whose sum equal to X exists.
- Output:
 - For each value of X , print "YES" if we found such pair; print "NO" otherwise

Answer

```

#include <iostream>
#include <array>
#include <algorithm>
#include <cmath>

using namespace std;

bool check(int a[], int n, int X);

int main() {
    int a[15] = {323, 232, 138, 230, 437, 176, 127, 111, 329, 1, 296, 12, 3, 2, 6};
    int n = sizeof(a) / sizeof(a[0]);
    int X = 10000;

    cout << check(a,n,X);
}

bool check(int a[], int n, int X) {
    for (int i=0; i<(pow(2,n)); i++) {
        int t = X;
        for (int j=0; j<n; j++) {
            if (i / ((int) pow(2,j+1)) % 2 == 0) {
                t -= a[j];
            }
        }
    }
}

```

```

    }

    if (0 == t) {
        cout << "YES" << endl;
        return true;
    }
    if (0 > t) {break;}
}
}
cout << "NO" << endl;
return false;
} // My stupid solution where I brute force every possible combinations

```

Lecture 5

Stack

การเพิ่ม / ลบข้อมูลในกองซ้อน

- ข้อมูล เข้าหลัง ออกก่อน (Last_In First-Out)

กองซ้อน : stack

```

bool        empty();
unsigned    size();
T           top();           // ดูบนสุด
void        push(T element); // ใส่ข้อมูล
void        pop();           // เอาออก

```

ข้อควรระวัง

- ใช้ #include <stack>
- top() ตอนไม่มีข้อมูล → พัง
- pop() ตอนไม่มีข้อมูล → พัง
- stack ไม่มี iterator
 - ▶ ดูได้เฉพาะข้อมูลที่เพิ่งใส่เข้าไปล่าสุด (top)
 - ▶ ไม่มี begin(), end() ให้ใช้
 - ▶ ถ้าอยากมาดูข้อมูลทุกตัวใน stack ต้องจำใจ pop ข้อมูลออกมา

ตัวอย่างการใช้งาน Stack

- การตรวจสอบโครงสร้างแบบซ้อนกัน เช่น การใส่วงเล็บ () {} [] ...
- การจัดเก็บตัวแปรและการทำ function calls
- การประมวลผลนิพจน์ทางคณิตศาสตร์ postFix → Discrete Math
- การทำ undo/redo
- การค้นคำตอบแบบ depth-first search
- ...

Parentheses Checking

การตรวจสอบการใส่วงเล็บ

- ถูก : ({ () [{ }] })
- ผิด : เปิดปิดไม่ตรงกัน ({])
- ผิด : มีปิดมากเกินไป ({ () })) }
- ผิด : มีเปิดมากเกินไป ({ () }
- วิธีทำ

- อ่านมาทีละตัว
- ถ้าเป็นวงเล็บเปิด ให้ push ลง stack
- ถ้าเป็นวงเล็บปิด ให้ pop จาก stack มาตรวจสอบว่าเป็นวงเล็บเปิดที่ตรงกันกับวงเล็บปิดหรือไม่
- เมื่อใดที่อยาก pop ถ้า isEmpty แสดงว่า ปิดมีมากไป
- เมื่ออ่านเสร็จหมด stack ยังมีข้อมูล แสดงว่า เปิดมีมากไป

Function Call Stack

Using Stack in Java Virtual Machine

- jvm uses java stack to store
 - state of method calls
 - parameters and local variables
 - temporary memory for computation (operand stack)

```
public class StackFrame {
    public static void main(String[] args) {
        a(3,2);
        b(5);
    }
    static void a(int x, int y) {
        int z = x/y;
        b(z);
    }
    static void b(int x) {
        ++x;
    }
}
```

We make a stack frame of function calls, each element consists of arguments and returning argument of the function. Function main contains args and RA JVM and calls a, a will be added to stack and contains z, y, x, and RA 04:, then it calls b. After b is finished, it will come back to line 9 and popped itself. It will do a and go to line 4, and so on. Arguments in each functions are separated variables.

```
#include <iostream>

using namespace std;

void test(int x) {
    if (x > 0) {
        int y = x-1;
        cout << "I am test(" << x << "). continue... " << endl;
        test(y);
        cout << "now x is " << x << endl;
    } else {
        cout << "I am test(0). stop"
    }
}

int main() {
    test(4);
    return 0;
}
```

Output

```
I am test(4). continue...
I am test(3). continue...
```

```

I am test(2). continue...
I am test(1). continue...
I am test(0). stop
now x is 1
now x is 2
now x is 3
now x is 4

```

Postfix Evaluation

Infix and Postfix Expressions

- infix (เติมกลาง)
 - $a + b * c / d - 2$, $(a + b) * c / (d - 2)$
 - use order of operations
 - use parentheses
- postfix (เติมหลัง)
 - $a b c * d / + 2 -$, $a b + c * d 2 - /$
 - order is from left to right
 - parentheses may be omitted

```

1  2  +  3  *  4  1  -  /
      3  3  *  4  1  -  /
          9  4  1  -  /
              9          3  /
                          3

```

Calculate Postfix Expressions with Stack

- Value of $2 3 + 4 5 - 6 * + ?$
- Use stack to help
- Solution
 - Check each element in postfix from left to right
 - If it is operand, push stack
 - If it is operator, pop elements from stack as many as operator needs, then push the result
 - After complete, answer will be at the top of stack

Convert Infix to Postfix

Use Stack to Convert

- input : infix expression
- output : postfix expression
- Steps
 - Read each element in infix
 - If it is operand, move it to the back of output
 - If it is operator
 - May pop operator to the end of output
 - push operator to stack
 - If all infix is read
 - pop all operators to the end of input

```

string infix2postfix(string &infix) {
    int n = infix.length();
    string postfix = "";
    stack<char> s;
    for (int i=0; i<n; i++) { // Read each element
        char token = infix[i];
        if (token เป็น operand) {

```

```

    postfix += token; // if operand, add to output
} else {
    while ( ??? ) {
        postfix += s.top(); // if operator, pop stack to result,
        s.pop();
    }
    s.push(token); // then push new operator
}
}
while (!s.empty()) {postfix += s.top(); s.pop();}
return postfix;
}

```

Priority of Operators

input 2 * 3 + 4

read	output	stack
2		
*	2	
3	2	*
+	2 3	*
4	2 3 *	+
	2 3 * 4	+
	2 3 * 4 +	

(* comes before and is more important than +)

input 2 + 3 * 4

read	output	stack
2		
+	2	
3	2	+
*	2 3	+
4	2 3	* +
	2 3 4	* +
	2 3 4 *	
	2 3 4 * +	

Code to Compare Priority

```

string infix2postfix(string &infix) {
    int n = infix.length();
    string postfix = "";
    stack<char> s;
    for (int i=0; i<n; i++) {
        char token = infix[i];
        if (priority.find(token) != priority.end()) {
            postfix += token;
        } else {
            int p = priority[token]; // Get priority of the new operator
            while ( !s.empty() && priority[s.top()] >= p ) { // lazy evaluation; left first
                postfix += s.top(); // pop if top operator is not less important than new op
                s.pop();
            }
            s.push(token);
        }
    }
    while (!s.empty()) {postfix += s.top(); s.pop();}
}

```

```
    return postfix;
}
```

Lazy Evaluation

Setting Priority of Operators

- $\wedge$ represents power 7
- $\wedge$ before $+$ $-$ $*$ $/$
- $*$ $/$ before $+$ $-$
- $*$ same as $/$ 5
- $+$ same as $-$ 3

```
map<char,int> priority =
    {{'+',3},{'-',3},{ '*',5},{ '/',5},{ '^',7}};
```

Complications with Parentheses

- In parentheses is like sub-expression
- Open parentheses always push (very high priority)
- Close parentheses is very low priority
- If found, pop until sees open parentheses

input 2 * (3 + 4)

read	output	stack
2		
*	2	
(	2	*
3	2	(*
+	2 3	(*
4	2 3	+ (*
)	2 3 4	+ (* (pop all)
	2 3 4 + *	

```
int p = priority[token];
while ( !s.empty() && priority[s.top()] >= p ) {
    postfix += s.top();
    s.pop();
}
s.push(token);

int p = outpriority[token];
while ( !s.empty() && inpriority[s.top()] >= p ) {
    postfix += s.top();
    s.pop();
}
if (token == ')') s.pop(); else s.push(token);
```

Priority of Operators with Parentheses

	+	-	*	/	$\wedge$	(	)
In expression	3	3	5	5	7	9	1
In stack	3	3	5	5	7	0	

Right to Left Operator

	+	-	*	/	$\wedge$	(	)
In expression	2	2	4	4	8	9	1

In stack	3	3	5	5	7	0
----------	---	---	---	---	---	---

Note Power in expression is more important than in stack

input 2 + 3 - 4 ^ 5 ^ 6

read	output	stack
2		
+	2	
3	2	+
-	2 3	+
4	2 3 +	-
^	2 3 + 4	^ -
5	2 3 + 4	^ -
^	2 3 + 4 5	^ -
6	2 3 + 4 5	^ ^ -
	2 3 + 4 5 6	^ ^ -
	2 3 + 4 5 6 ^ ^ -	

Conclusion

- Implementations of stack to solve problems
- Main operations: push / pop / top
- Make stack with array
- If reserve enough space, all runs is $\Theta(1)$; constant time

Lecture 6

Queue

- Just like a normal queue in real life.
 - First-in-First-Out data structure
 - One way in (back of queue), one way out (front of queue)
 - push = add data to the back of the queue
 - pop = remove data from the head of the queue

Basic

```
#include <iostream>
#include <queue>
#include <vector>

using namespace std;

int main() {
    queue<int> q;
    q.push(1);
    q.push(2);
    q.push(3);
    while (q.empty() == false) {
        cout << q.front() << endl;
        q.pop();
    }
    cout << "-- example 2 --" << endl;
    queue<vector<int>> q2;
    vector<int> v1 = {1,2,3};
    vector<int> v2 = {99,88,-1};
```

```

q2.push( v1 );
q2.push( v2 );
cout << q2.back()[1] << endl;
cout << q2.front().size() << endl;
auto x = q2.front();
q2.pop();
cout << x[0] << endl;
}

size_t    q.size()
bool      q.empty()
void      q.push(T data)
void      q.pop()
T         q.front()
T         q.back()

```

Limitation

- Same limitation as stack
 - No iterator
 - No `begin()`, `end()`
 - Can only access front and back of the queue
 - If we wish to access all members, we have to pop it all
 - Do not call `front()`, `back()`, `pop()` when the queue is empty

Radix Sort

Queue Application: Fast sorting with no comparison

Overview

- Put all data in an array
- For each digit X , from LSD to MSD
 - PUT TO QUEUE step: Sort by digit X by putting all of data from the array into B queues
 - B is the base of the number
 - For example, for a base 10 number, we will have Queue[0] to Queue[9]
 - Put into the queue labelled with that digit
 - GET FROM QUEUE step: Start from queue 0 to queue $B - 1$, remove data from the queue and put back to the array

Example

Input 115 15 42 305 21 8 463

Round 1 (digit 0), from queue

(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
					305				
					15				
	21	42	463		115			8	

⇒ 21 42 463 115 15 305 8

Data is now sorted by the last digits, because we pop from queue 0 to 9

Round 2 (digit 1), from queue

(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
8	15								
305	115	21		42		463			

⇒ 305 8 115 15 21 42 463

Data is now sorted by last two digits, because we goes from queue 0 to 9, which is grouped by digit 1 and the data in each queue is sorted by the last digit.

Round 3 (digit 2), from queue

(0)	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
42									
21									
15									
8	115		305	463					

⇒ 8 15 21 42 115 305 463

Code

```
#define base 10

int getDigit(int v, int k) {
    // return the kth digit of v (MSD is digit 0)
    int i;
    for (int i=0; i<k; i++) v /= base;
    return v % base;
}

// d = number of digits
void radixSort(vector<int> &data, int d) {
    queue<int> q[base];
    for (int k=0; k<d; k++) {
        for (auto &x : data) {
            q[getDigit(x,k)].push(x);
            for (int i=0, j=0; i<base; i++) {
                while(!q[i].empty()) {
                    data[j++] = q[i].front(); q[i].pop();
                }
            }
        }
    }
}
```

Breadth First Search

Queue Application: Gotta generate 'em all

The Problem

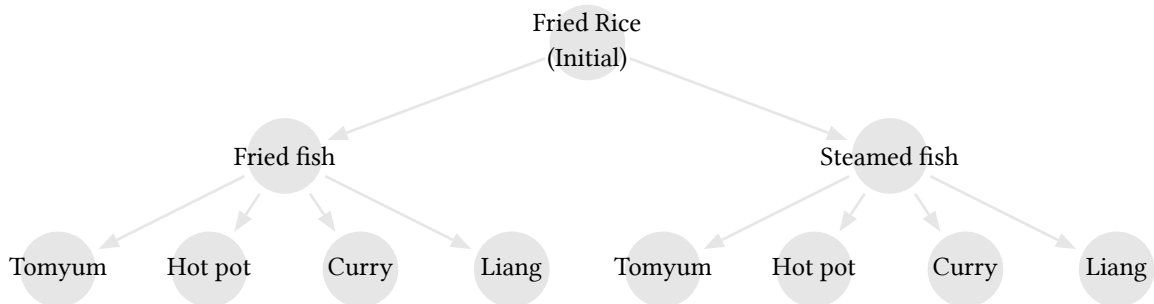
- Given a positive integer X
- Start with a number 1, find a sequence of arithmetics operations, either “ $\times$ 3”, or “ $/$ 2” that makes 1 into X
 - the $/$ 2 is integer division, e.g., $5 / 3 = 1$ (not 1.6666)
 - The sequence must be as short as possible
- Example
 - $10 = 1 \times 3 \times 3 \times 3 \times 3 / 2 / 2 / 2$
 - $31 = 1 \times 3 \times 3 \times 3 \times 3 \times 3 \times 3 / 2 / 2 / 2 / 2 / 2 \times 3 \times 3 / 2$

The Idea

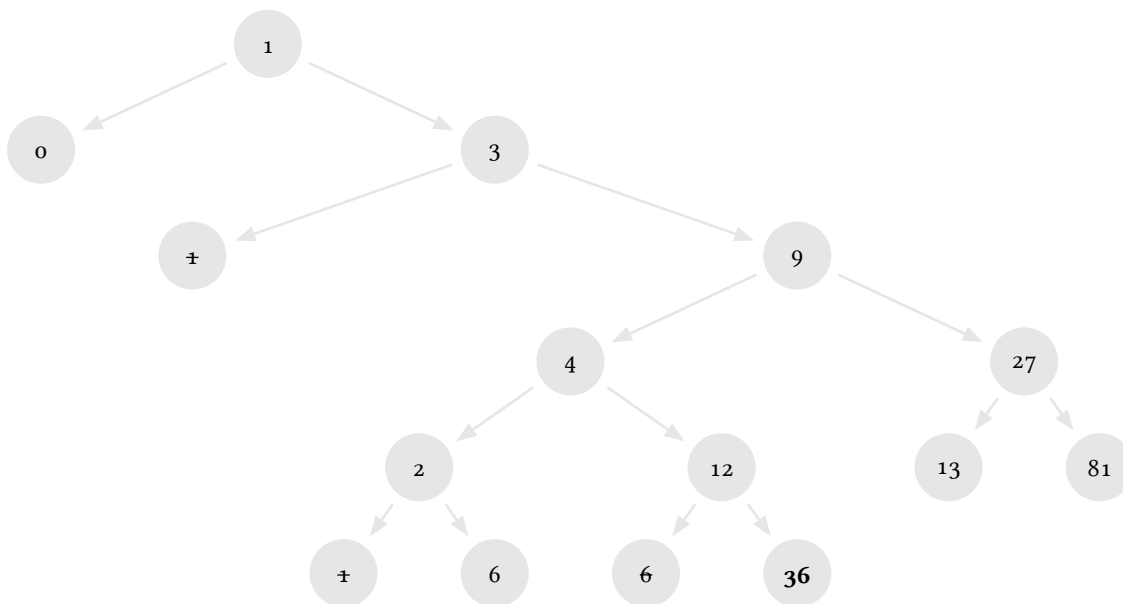
- Generate all possible sequences
 - Start with length 1, 2, 3, ... until we find one that gives X
- This is called an *exhaustive search* algorithm
 - Systematically enumerate all possible somethings

Tree Structure (Search Tree)

- A structure to illustrate *search* algorithm
- Divide into steps
 - Start with *initial solution*
 - For each possible outcome (called a *state*) of each step, *generate all* proper possible *next step*
 - Also, check if the current step is what we needed
- Written as a diagram of *node* and *edge*

Enumerate**M3D2 Problem & Solution****Back to Our Problem**

- Start with 1
- Each step is either * 3 or / 2
- Issue: might get repeated number
 - Solution: if we have found it, do not generate new step

**M3D2 Code****Code**

- `Queue` makes ordering of how we pick a state to enumerate
- From top to bottom and left to right

```

void m3d2(int target) {
    map<int, int> prev;
    queue<int> q;
  
```



```

int v = 1;
q.push(1); prev[1] = -1;
while (!q.empty()) {
    v = q.front(); q.pop();
    if (v == target) break;
    int v2 = v/2;
    int v3 = v*3;
    if (prev[v2] == 0) {q.push(v2); prev[v2] = v;}
    if (prev[v3] == 0) {q.push(v3); prev[v3] = v;}
}
if (v == target) showSolution(v, prev);
}

```

Display Solution

x	0	1	2	3	4	6	9	12	13	27	36	39	40
prev[x]	1	-1	4	1	9	2	3	4	27	9	12	13	81

showSolution

```

void showSolution(int v, map<int,int>& prev) {
    string out = "";
    while (prev[v] != -1) {
        if (prev[v] * 3 == v) {
            out = "x3" + out;
        } else {
            out = "/2" + out;
        }
        v = prev[v];
    }
    out = "1" + out;
    cout << out << endl;
}

```

⇒ 36 = 1 * 3 * 3 / 2 * 3 * 3

Lecture 7

Priority Queue

Queue with privilege

Intro

- Priority Queue is ...
 - A queue with priority
 - Item with high priority is promoted to the front of the queue
 - There is *no back of the queue*
 - Priority is defined by having more value
 - Comparison, by default, is to use operator <, i.e., if item A < B is true, then B *has higher priority*
 - We can have custom comparator
- Has the *same interface* as stack

Example

For intuitive purpose only! This is not really how priority_queue work internally

```

push(10)
< 10      >

```

```
push(30)
< 30 10      >
```

```
push(20)
< 30 20 10   >
```

```
pop()
< 20 10      >
```

```
push(15)
< 20 15 10   >
|
top
```

Basic

```
size_t      q.size()
bool        q.empty()
void        q.push()
void        q.pop()
T           q.top()
* there is no q.back()
```

Limitation

- Same limitation as stack, queue
 - *no iterator*
 - No `begin()`, `end()`
 - Use `#include <queue>`
 - Can only access `top()` of the queue
 - If we wish to access all members, we have to pop it all
 - Do not call `top()`, `pop()` when the queue is empty
- The data type must be *comparable* (similar to set and map)

Class

Quick Summary

- Syntax
 - class declaration must end with `;`
 - Function definition can be outside of the class
 - Access modifier is `public:`, `private:`, `protected:`
 - Constructor is a function with the same name of the class with no return type
- Object is a variable (instantiation) of a class
 - When declared, a constructor is called

Example 1

```
#include <iostream>
#include <string>
using namespace std;

class Student {
public:
    void setFullName(string name, string surname) {
        this->name = name;
        this->surname = surname;
    }
}
```

```

    string getFullName() {
        return "[" + name + " " + surname + "];"
    }
private:
    string name, surname;
};

int main() {
    Student a;
    Student b;
    a.setFullName("nattee", "niparnan");
    cout << a.getFullName() << endl;
    cout << b.getFullName() << endl;
}

```

Declaring class before writing functions

```

class Student {
public:
    void setFullName(string name, string surname);
    string getFullName();
private:
    string name,surname;
};

void Student::setFullName(string name, string surname) {
    this->name = name;
    this->surname = surname;
}

string Student::getFullName() {
    return "[" + name + " " + surname + "];"
}

```

Output

```

[nattee niparnan]
[ ]

```

Example 2: Constructor

```

#include <iostream>
#include <string>
using namespace std;

class Student {
public:
    Student(float score) {gpax = score;}
    void setFullName(string name, string surname) {
        this->name = name;
        this->surname = surname;
    }
    string getFullName() {return "[" + name + " " + surname + "];"}
    bool is1stHonor() {return gpax >= 3.6;}
private:
    string name, surname;
    float gpax;
};

```

```
int main() {
    Student a(2.95);
    a.setFullName("nattee", "niparnan");

    cout << a.getFullName() << endl;
    if (a.is1stHonor()) {cout << "YES" endl;} else {cout << "NO" << endl;}
    // Student b; // <-- cannot compile because there is no default constructor
}
```

Operator Overloading

How C++ has a function for each operator

Overview

- Let say we write `a + b` when `a` and `b` is an object of some classes
 - This can be considered the same as calling a function `plus(a,b)`
 - C++ allows us to write a function for many operator and use it as an operator
 - For example we can write a function `times(a,b)` and let it be used as `a * b`
- This is what we called operator overloading

Instead of writing

```
Matrix multiply(Matrix a1, Matrix a2) {
    multiply(a, multiply(b,c))
}
```

We want to write

```
Matrix a, b, c
a * (b * c)
```

Example

```
#include <queue>
#include <iostream>
#include <string>

using namespace std;

// use operator + symbol that will be overloaded
string operator*(string & lhs, const int & rhs) {
    string result = "";
    for (int i=0; i<rhs; i++) {
        result = result + lhs;
    }
    return result;
}

int main() {
    string a = "abc ";
    cout << (a * 3) << endl; // this gives "abc abc abc "
}
```

- Function must be named operator followed by the operator that we will overload
- Some operator takes two parameters (such as `+`, `-`, `*`, `/`, `%`)
- Some takes one (such as `++`, `--`, `!`, `*`, `&`)

Overloading less-than operator

Using with data structure that require sorting

- We have seen several data structure that requires comparability of the data, such as set, map and priority queue
- If we want to use our class with these data structure, we need to tell them how can we use them
- There are multiple ways to achieve this
 - Let us consider operator overloading

Overloading <

- As stated earlier, set, map, and priority_queue use *operator <* to compare two elements
- It does not work if we overload *operator >*

```
class Student {
public:
    Student(float score, string a, string b) {
        name = a;
        surname = b;
        gpax = score;
    }
    bool is1stHonor() {return gpax >= 3.6;}
    // not good, now our data is public
    string name, surname;
    float gpax;
    // overloading <
    bool operator<(const Student & other) const {
        return gpax < other.gpax;
    }
};

int main() {
    Student a(2.95, "nattee", "niparnan");
    Student b(4.00, "attawith", "sudsang");
    cout << (a < b) << endl; // 1
    priority_queue<Student> pq;
    pq.push(a);
    pq.push(b);
    cout << pq.top().name << endl; // attawith
}
```

Custom Comparator

Why custom?

- By overloading *operator <*, we have defined default ordering of that class
- What if we need another ordering, just for this *priority_queue* only
 - For example, *Student* is ordered by *gpax* by default
 - What if we want our *priority_queue* to order by name instead, while keeping the *Student* default ordering elsewhere
 - Better, can we have multiple *priority_queue* with different ordering?
- Can be done via comparator class

Example

```
#include <iostream>
#include <string>
#include <queue>
```

```

using namespace std;

class Student {...};

// Comparator class use () overloading
class StudentByNameComparator {
public:
    bool operator()(const Student& lhs,
                    const Student& rhs) {
        return lhs.name < rhs.name;
    }
};

class GpaxThenName {
public:
    bool operator()(const Student& lhs,
                    const Student& rhs) {
        if (lhs.gpax == rhs.gpax)
            return lhs.name < rhs.name;
        return lhs.gpax < rhs.gpax;
    }
};

int main() {
    Student a(2.95, "nattee", "niparnan");
    Student b(4.00, "attawith", "sudsang");
    Student c(4.00, "vishnu", "kotrajaras");
    cout << (a < b) << endl; // 1
    StudentByNameComparator comp1;
    GpaxThenName comp2;

    cout << comp1(a,b) << endl; // 0; can use like function

    // Use 3 template parameters
    priority_queue<Student, // class to compare
                  vector<Student>, // vector of that class
                  StudentByNameComparator> pq(comp1); // comparator class

    pq.push(a);
    pq.push(b);
    cout << pq.top().name << endl; // nattee

    priority_queue<Student,
                  vector<Student>,
                  GpaxThenName> pq2(comp2);

    pq2.push(a);
    pq2.push(b);
    pq2.push(c);
    cout << pq2.top().name << endl; // vishnu
}

```

Another Method, lambda-function

```

#include <iostream>
#include <string>
#include <queue>
using namespace std;

```

```
int main() {
    auto compare = [](const string& lhs, const string& rhs) {
        return lhs.size() < rhs.size();
    };

    cout << "Result of compare function = " << compare("xxx","z") << endl;

    priority_queue<string,vector<string>,decltype(compare)> pq(compare);
    pq.push("somchai");
    pq.push("z");
    pq.push("abc");
    while (pq.empty() == false) {
        cout << pq.top() << endl;
        pq.pop();
    }
}
```

Output

```
Result of compare function = 0
somchai
abc
z
```

Priority Queue Template

Templating of priority_queue

- priority_queue requires 3 template parameters
- priority_queue<T, Container = vector<T>, Compare = less<T>>
- The first one is required (which is the type of the data)
- The *second* and the *third* is optional (it has default type)
 - *Second* is the container (for now, just don't think about it)
 - *Third* is the class for comparator (the class that we use to compare)
 - This one is default to less<T>

```
#include <iostream>
#include <string>
#include <queue>
using namespace std;

int main() {
    less<int> x;    // comparator that checks a < b
    greater<int> y; // checks a > b

    int a = 10;
    int b = 3;
    cout << x(a,b) << endl;    // 0
    cout << y(a,b) << endl;    // 1
}
```

Using Comparator for set and map

- To use custom class with set and map, we need to do the same thing, let set and map know how to sort the data
 - Either make default ordering (overload <) in the custom class
 - Or use custom comparator when declare
- For set, the declaration is set<T, Compare = less<T>>
- For map, the declaration is map<Key, T, Compare = less<Key>>

Assignment

- Is any of `vector<int>`, `set<int>`, `map<int, string>`, `queue<bool>`, `stack<vector<int>>` comparable?
 - For any class that is “YES”, how it is ordered?
 - For example, if `vector<int>` is comparable, how `{1, 2, 3}` is compared to `{1, 2, 3, 4}` or `{2, 3, 4}`

Answer

- `vector<int>` compare each element from the left; `{2, 3, 4} > {1, 2, 3, 4} > {1, 2, 3}`
- `set<int>` compare the maximum element () of each set, if equal then look the next max
- `map<int, string>` compare by Key set, then compare max of value set
- `queue<bool>` is not comparable
- `stack<vector<int>>` is not comparable

Summary**Data Structure Summary**

Data Structure	Pro	Cons	Remark
<code>pair<T1, T2></code>	Nothings... It just a pair of two data type		
<code>vector<T></code>	• Fast access [] • Fast append $\mathcal{O}(1)$	• Slow find $\mathcal{O}(N)$ • Slow insert, Slow Erase $\mathcal{O}(N)$	
<code>set<T></code>	• Fast find $\mathcal{O}(\log(N))$ • Item is sorted	• Slower to just append data than vector, stack, queue $\mathcal{O}(\log(N))$ • Iterate is also slow • Takes lots of memory $\mathcal{O}(\log(N))$	Require comparator
<code>map<Key, T></code>			• Associative data type • Also require comparator of Key_Type
<code>stack<T></code>	• Very fast push, pop $\mathcal{O}(1)$	• Very limited functionality but has special uses	• No iterator • Order of data coming out depends on something (stack, queue depends on WHEN it is pushed, pq depends on value)
<code>queue<T></code>			
<code>priority_queue<T></code>	• Fast get max • Fast delete max • Data is sorted • Memory efficient	• Slower to just append data than vector, stack, queue • Very limited functionality	• pq requires comparator

More Data Structure

- C++ has more data structure not really covered right now
 - `list` is a vector with faster input / erase but does not have fast access
 - `unordered_set`, `unordered_map` are set and map that the data is not sorted but is much faster
 - `deque` is a queue that can push, pop at both ends
 - `multiset`, `multimap` are set and map that allows duplicated entries

Lecture 8**Create Your Own CP::pair****Intro**

- We start writing our own data structure
 - In our own *namespace* (CP)
- We start with *pair*
 - Good introduction on writing a class
 - Some C++ feature we need to use

- const
- pass-by-value, pass-by-ref
- Header file

What should we write in a class

- Class name and namespace
- Member *variables* which define what data we want to store in the class
- Member *functions* which define behavior of the class and how member variable is manipulated
 - Include lots of special function such as *constructor*, *destructor*, *overloading*

CP:pair version 0.1 (minimal version)

- Templating
 - Parameter of a code
- No member functions
 - When we declare a pair, T1 and T2 must be declared as a type

```
namespace CP {
    template <typename T1, typename T2>
    class pair{
    public:
        T1 first;
        T2 second;
    };
}

int main() {
    CP::pair<int, string> p1; // In p1, T1 is int, T2 is string
    CP::pair<bool, int> p2;  // In p2, T1 is bool, T2 is int
}
```

Capabilities

- Can hold 2 pieces of data
- Can use template
- Works with operator =
- Has copy constructor
- Cannot check equal
- Cannot check less than

```
#include <iostream>
#include <string>
using namespace std;
namespace CP {
    template <typename T1, typename T2>
    class pair{
    public:
        T1 first;
        T2 second;
    };
}

int main() {
    CP::pair<int, string> p1, p2; // default ctor
    p1.first = 20; p1.second = "somchai";
    CP::pair<int, string> a(p1); // copy ctor
    p2 = p1;
```

```
    cout << p2.first << "," << p2.second << endl;
}
```

```
/*
if (p1 == p2) { // won't compile
    cout << "yes" << endl;
}
```

```
if (p1 < a) { // won't compile
    cout << "yes" << endl;
}
*/
```

CP::pair version 0.2 (comparator overload)

```
#include <iostream>
#include <string>
#include <set>
using namespace std;
namespace CP {
    template <typename T1, typename T2>
    class pair{
    public:
        T1 first;
        T2 second;

        // operator
        bool operator==(const pair<T1,T2> &other) {
            return (first == other.first && second == other.second);
        }

        bool operator<(const pair<T1,T2>& other) const { // operator< must be const
            return ((first < other.first) ||
                    (first == other.first && second < other.second));
        }
    };
}

int main() {
    CP::pair<int,string> p1, p2; // default ctor
    p1.first = 20; p1.second = "somchai";
    CP::pair<int,string> a(p1); // copy ctor
    p2 = p1;
    cout << p2.first << "," << p2.second << endl; // 20, somchai

    if (p1 == p2) {cout << "yes" << endl;} // yes
    if (p1 < a) {cout << "yes" << endl;}
    set<CP::pair<int,int>> s;
    s.insert({1,2});
    // Now we can compare and use less than (and also set, map, priority queue)
    cout << s.begin()->second << endl; // 2
}
```

Const Member

Const Parameter in Function

```
void test(int &x, const int& y) {
    x++;           // Okay; doable
    cout << y << endl; // Okay, we only read y
    for (int i=0; i<y; i++) { // Okay, too
        cout << i << endl;
    }
    y--; // ERROR: we promised NOT to change y
}
```

- Declared by putting a keyword const as a prefix
- Const parameter must not be modified inside the function
- Why? So that we know that the function does not modify the data
 - Especially when we pass-by-reference

Difference of Pass-by-reference && Pass-by-value

Pass-by-value

- The *arguments* can be either constants or variables
- Modifying *parameters* (variable inside the function) does not change the *argument's* value
- The *argument's* value is copied to the parameter
 - **SLOWER!!!** Because we have to copy

Pass-by-reference

- The *arguments* must be variables
- The *parameters* are the argument's variables
 - Modify the *parameter* also modify the *argument's* variable
- **Faster**

Const Member Function

```
class ccc {
public:
    int a,b;

    void inspect() const { // This function promises NOT to change anything
        if (a < b) cout << "yes" << endl; // Okay

        // b += 20; // <-- NOT OKAY
    }

    void mutate() { // This function might change something
        if (a < b) a += 10; // Okay
    }
};

void test2(ccc& changeable, const ccc& unchangeable) {
    changeable.inspect();
    changeable.mutate();
    unchangeable.inspect();
    unchangeable.mutate(); // ERROR: attempt to change unchangeable object
}
```

- Declared by putting a keyword const after the function declaration
- Const member function cannot modify any member data
 - Also cannot call any other function that is not const

- Why? So that we know that the function does not modify its data

Constructor & List Initialization

Custom Constructor

- In STL spec of `std::pair`, it has custom constructor called *initialization constructor*

```
namespace CP {
    template <typename T1, typename T2>
    class pair{
    public:
        T1 first;
        T2 second;

        // Custom constructor, using initialization list
        pair(const T1 &a, const T2 &b) {
            first = a;
            second = b;
        }
        // When we have a constructor, a default constructor is not auto-generated

        // Operator
        bool operator==(const pair<T1,T2> &other) {...}
        bool operator<(const pair<T1,T2>& other) {...}
    };
}

int main() {
    CP::pair<int, bool> p(10,false);
    CP::pair<string, int> q("abc",42), r("",0);

    cout << (q < r) << endl; // 0
    priority_queue<CP::pair<string,int>> pq;
    pq.push(r);
    pq.push(q);
    cout << pq.top().first << endl; // abc

    CP::pair<string, int> x(q);
    CP::pair<string, int> y = x;

    // All below cannot be compiled
    // CP::pair<string, int> w;
    // vector<CP::pair<int, int>> v(10);
}
```

Initialization List

- Instead of writing a code to assign a value to each member, we can use *initialization list*
 - A little bit shorter code
 - Also little bit faster
 - Only way to init *const* member

```
namespace CP {
    template <typename T1, typename T2>
    class pair {
    public:
        T1 first;
        T2 second;
```

```

        // custom constructor, using initialization list
        pair(const T1 &a, const T2 &b) : first(a), second(b) { }
    };
}

```

Default Constructor

- A constructor used when we simply declare an object
- Auto-generated by init all members with its default constructor
- Won't be auto-gen if we have any other constructor

```

namespace CP {
    template <typename T1, typename T2>
    class pair {
        T1 first;
        T2 second;

        // Constructor
        pair() : first(), second() { }
        pair(const T1 &a, const T2 &b) : first(a), second(b) { }
    };
}

```

Include & Header file

Include File

- Usually, our data structure (which is written as a class) will be used by several programs in several files.
 - It is better NOT TO copy our data structure code into each file
- Rather, put our code in a file and include it where it is needed
 - Better if we want to change it
 - Better compilation
- Introducing ".h" files

C++ header file (.h) and #include

- To put a content of one file into another file, we use #include "filename" keyword
- C++ will simply put the content of *filename* into where we #include it
- #include has more benefit
 - Separation of *declaration* (what it is) and *definition* (how it works)
 - Not really explored in this class
- We usually use .h for a file that we will include but that is not a rule, we can use other extension

Problem with include

a.h

```

class a {
    int m1, m2;
};

```

b.h

```

#include "a.h"

```

```

class b {
    a x;
};

```

c.h

main.cpp (expanded)

```

class a {
    int m1, m2;
};

```

```

class b {
    a x;
};

```

```

// There are multiple copies of class a
class a {
    int m1, m2;
};

```

```
#include "a.h"
```

```
class c {
    a y;
};
```

main.cpp

```
#include "b.h"
#include "c.h"
```

```
int main() {
    b b1;
    c c1;
}
```

Fix problem

a.h

```
#ifndef A_H
#define A_H
class a {
    int m1, m2;
};
#endif
```

b.h

```
#include "a.h"
```

```
class b {
    a x;
};
```

c.h

```
#include "a.h"
```

```
class c {
    a y;
};
```

```
class c {
    a y;
};
```

```
int main() {
    b b1;
    c c1;
}
```

main.cpp (expanded)

```
#ifndef A_H
#define A_H
class a {
    int m1, m2;
};
#endif

class b {
    a x;
};

// class a here is taken out of
// compilation because A_H is defined
#ifndef A_H
// #define A_H
// class a {
//     int m1, m2;
// };
#endif

class c {
    a y;
};

int main() {
    b b1;
    c c1;
}
```

Summary

Major

- **Templating**: allow user to write a code that works with different data type
- **Constructor**: a function called when we create a variable
 - Default constructor
- **Operator overloading**: for less-than and equality
- pass-by-ref vs pass-by-value

Minor

- Header file
- const
- List initialization

Exercise (no grader)

1. Modify operator< so that it compare *second* before *first*
2. Modify operator< so that when we call sort(v.begin(), v.end()) where w is a vector of our pair, it is sorted from *Max* to *Min*
3. Write operator!= and operator>=

Final version CP.h

```
#ifndef _CP_PAIR_INCLUDED_
#define _CP_PAIR_INCLUDED_

namespace CP {
    template <typename T1, typename T2>
    class pair {
    public:
        T1 first;
        T2 second;

        pair() : first(), second() {}

        pair(const T1 &a, const T2 &b) : first(a), second(b) { }

        bool operator==(const pair<T1,T2> &other) {
            return (first == other.first && second == other.second);
        }

        // 1.
        bool operator<(const pair<T1,T2>& other) const {
            return ((second < other.second) ||
                    (second == other.second && first < other.first));
        }

        // 2.
        bool operator<(const pair<T1,T2>& other) const {
            return ((first > other.first) ||
                    (first == other.first && second > other.second));
        }

        // 3.
        bool operator!=(const pair<T1,T2>& other) const {
            return !(*this == other);
        }

        bool operator>=(const pair<T1,T2>& other) const {
            return !(*this < other);
        }

    };
}
#endif
```

Lecture 9

CP::vector

Our first “real” data structure

Intro

- Now we will create more complex data structure `CP::vector`
- It can store variable length array
 - Implemented as a dynamic array
- Can be accessed by operator `[]`
 - Additional operator to be overloaded
- We also have to create our own iterator
 - Implemented as a pointer

Key Idea

- Vector stored 3 things (3 member data)
 - `mData`: A dynamic array, large enough space to store current data and might have reserved space
 - `mSize`: Number of data stored
 - `mCap`: Size of the dynamic array (maybe large than `mSize`)
- If the dynamic array is full and more data is being added, we create a new dynamic array and relocate data to the new array
 - This is called `expand`
 - Each `expansion` takes very long time
- Dilemma
 - Large reserve = less often relocation but use more memory
 - Small reserve = more frequent relocation but less memory

Example

```
mSize  mCap      mData
  4      5    [10, 3, 1, 5, ]
```

```
push_back(9)
```

```
  5      5    [10, 3, 1, 5, 9]
```

```
push_back(4)
```

```
  6     10    [10, 3, 1, 5, 9, 4, , , , ]
```

How much reserve should we have?

- Whenever we need to relocate, we `double` the capacity that we currently have
- If we start with a vector with zero size and continuously add data one by one, for example by `push_back`
 - Each expansion will take time `equal` to `the number of data` when we relocate (this is very slow)
- By doubling the size every time we expand, we can show that on average, `each addition of a data` (such as `push_back`, `insert`) `takes constant time!!!`

Pointer

How to implement a dynamic array (and also iterator)

Pointer & Memory

- Each variable is some block in computer memory
 - Programming language just map our `variable name` to that `block of memory`
 - Programming language works with the address of that block
- Pointer variable is a variable that stor `address of memory`

- ▶ Pointer needs type i.e., address of int is not the same as address of bool
- ▶ We can use operator & to ask for the **address** of a **variable**
- ▶ We can use operator * to ask for the **data** of an **address**

Example

```
int x, y;    // this is normal variable x and y
int *a;     // this is int pointer variable a
int *b;     // another int pointer

x = 10;     // say x is at address 3267
a = &x;     // a points to x
b = a;      // b also points to x
*b = 30;    // change the address value to 30
y = *b;     // set value of y to 30 (hard copy)

cout << &x << endl;    // 0x7ffee22885ac
cout << &y << endl;    // 0x7ffee22885a8 +4
cout << &a << endl;    // 0x7ffee22885a0 +8
cout << &b << endl;    // 0x7ffee2288598 +8
cout << sizeof(int) << endl;    // 4 bytes
cout << sizeof(*int) << endl;    // 8 bytes
```

Pointer Arithmetic

- Pointer can be added, subtracted by integer
 - ▶ It **moves the address by the size of the type** of the pointer
 - ▶ For example, when X is an int (which is 4 bytes) X + 10 result in an address 40 bytes away from X
- Two pointers of the same type can be subtracted
 - ▶ The result is the address **difference divided by size of the type** of the pointer

```
int main() {
    bool x, y, z;
    x = 1; y = 2; z = 3;
    bool *a, *b;

    a = &x;
    b = a+2;

    cout << "&x = " << &x << endl;    // &x = 0x6afed4
    cout << "&y = " << &y << endl;    // &y = 0x6afed8
    cout << "&z = " << &z << endl;    // &z = 0x6afedc
    cout << " a = " << a << endl;    // a = 0x6afed4
    cout << " b = " << b << endl;    // b = 0x6afedc
    cout << b-a << endl;            // 2
}
```

Dynamic Array

- **Dynamic Array** variable of type T is a **pointer** to the **starting address** of consecutive block of type T
- Let A be a dynamic array of int
 - ▶ A[x] refer to the x^{th} block starting from A
- Static array works in the same way, that's why accessing A[x] is very fast for an array
 - ▶ It just refers to the address A[x] is $*(A + x * \text{sizeof}(T))$

new and delete operator

- For a datatype T, new T *allocates* a block with a size of T and then call a *constructor* of T, return the address of that memory

- For a datatype T, `new T[n]` allocates n blocks of T and then call a *constructor* of T in each block, return the address of the first block
 - For example, we use `new int [10]` to create a dynamic int array of 10 elements
- For a pointer X, `delete X` calls the *destructor* and then *de-allocates* memory pointed by X
- For a dynamic array X, `delete []` calls the *destructor* of all blocks in the dynamic array and *de-allocates* the memory allocated by X

Example

```
class test {
public:
    // constructor
    test() : data() {cout << "created" << endl;}
    // destructor
    ~test() {cout << data << " destroyed " << endl;}
    int data;
};

int main() {
    test *a, *b;    // pointers of test

    a = new test;    // created new test pointed by a
    a->data = 10;    // (*a).data = 10
    cout << a->data << endl;    // 10
    delete a;        // destroyed

    b = new test[4]; // created 4x
    b[0].data = 10;
    b[1].data = 20;
    b[2].data = 30;
    b[3].data = 40;
    delete [] b;     // deleted 4x
}
```

Memory Leak

- For everything, this is created by new, we must call delete on it
- If you do not, that memory is not deleted until all memory is used up

```
#include <iostream>
using namespace std;

void leaked() {
    int *a;
    a = new int[2000];
}

int main() {
    for (int i=0; i<1000000; i++) {
        int *a;
        cout << i << endl;
        leaked();
    }
}

// terminate called after throwing an instance of 'std::bad_alloc'
```

Smart Pointer (NOT A SUBJECT OF THIS COURSE)

- A better way is to use C++ smart pointer `shared_ptr(T)`
- Smart pointer is a pointer that can delete itself when it go out of scope

- Similar concept to [Java Garbage Collection](#)
- Still possible to have memory leak

vector.h

Version 0.1

- Start with vector that can do push_back, pop_back, and []
- Also with custom constructor

```
namespace CP {
template <typename T>
class vector {
protected:
    T *mData;
    size_t mCap;
    size_t mSize;

    void rangeCheck(int n) {...}
    void expand(size_t capacity) {...}
    void ensureCapacity(size_t capacity) {...}
public:
    vector() {...}
    vector(size_t capacity) {...}
    ~vector() {...}
    // access
    T& at(int index) {...}
    T& operator[](int index) {...}
    // modifier
    void push_back(const T& element) {...}
    void pop_back() {...}
};
}
```

Basic Constructor

```
template <typename T>
class vector {
protected:
    T *mData;
    size_t mCap;
    size_t mSize;

public:
    vector() {
        int cap = 1;
        mData = new T[cap]();
        mCap = cap;
        mSize = 0;
    }

    vector(size_t cap) {
        mData = new T[cap]();
        mCap = cap;
        mSize = cap;
    }
}
```

```
vector<int> v;
```

```
vector<int> w(5);
```

mSize	mCap	mData		mSize	mCap	mData
0	1	[]		5	5	[0,0,0,0,0]

Destructor

- Since we new mData, we have to delete it
 - Or face a memory leak problem

```
~vector() {
    delete [] mData;
}
};
```

Object Life Cycle

- Normal object
 - Object is created (constructor called) when declared
 - Object is destroyed (destructor called) when go out of scope
- Object created by new (both new T or new T[]) Object is created when new Object is destroyed when delete

```
class test {
public:
    // constructor
    test() : data() {cout << "created" << endl;}
    // destructor
    ~test() {cout << data << " destroyed " << endl;}
    int data;
};

int main() {
    cout << "-- Life cycle --" << endl;
    cout << "- normal cycle -" << endl;
    test u;
    u.data = 99;
    for (int i=0; i<5; i++) {
        test t;
        t.data = i*10;
    }
}
```

Output

```
-- Life cycle --
- normal cycle -
created      <-- u
created
0 destroyed
created
10 destroyed
created
20 destroyed
created
30 destroyed
created
40 destroyed
99 destroyed <-- u
```

Accessing Data

- The return type is T& which is a reference
- Same deal as pass-by-reference, this is called **return-by-reference**
 - So we can do `v[i] = 30` or `v[i]++`
- What is returned is actually that variable
- Also notice the difference between `at()` and `operator[]`

```
template <typename T>
class vector {
protected:
    T *mData;
    size_t mCap;
    size_t mSize;

    void rangeCheck(int n) {
        if (n < 0 || (size_t)n >= mSize) {
            throw std::out_of_range("index out of range");
        }
    }

public:
    T& at(int index) {
        rangeCheck(index);
        return mData[index];
    }

    T& operator[](int index) {
        return mData[index]; // return the variable
    }
};
```

Add, remove data

- `push_back` first check of we have reserved space
 - If not, we expand
- Then, the data is put to `mData[mSize]`
- Removing Data is done by just reduce the size

```
template <typename T>
class vector {
protected:
    T *mData;
    size_t mCap;
    size_t mSize;
    void expand(size_t capacity) {
        T *arr = new T[capacity](); // create a new dynamic array
        for (size_t i = 0; i < mSize; i++) {
            arr[i] = mData[i]; // move all data
        }
        delete [] mData;
        mData = arr; // delete old data and point to new one
        mCap = capacity;
    }

    void ensureCapacity(size_t capacity) {
        if (capacity > mCap) {
            size_t s = (capacity > 2 * mCap) ? capacity : 2 * mCap; // double the size
            expand(s);
        }
    }
};
```

```

    }
}
public:
    void push_back(const T& element) {
        ensureCapacity(mSize+1);
        mData[mSize++] = element; // add size and put element
    }
    void pop_back() {
        mSize--;
    }
};

```

Vector Constructors

Problem of vo.1

- Copy constructor and assignment operator is incorrect
 - It is auto generate to copy all variables (but not the data it points to)
- Rule of three in c++
 - Consider destructor, copy constructor, assignment operator
 - If any of them is written in the code, we mostly need all of them
- Since c++11, it's rule of four an a half

```

int main() {
    CP::vector<int> w(5);

    for (int i=0; i<5; i++) w[i] = i*10;
    CP::vector<int> x(w); // this creates shallow copy of w
    CP::vector<int> y = w;
    x[3] = -1; // changes both w and y
    cout << y[3] << endl; // -1
    cout << w[3] << endl; // -1
}

```

vo.2, add small access functions

- empty and size also exists in other data structure
- size_t is non-negative integer type

```

bool empty() const {
    return mSize == 0;
}

size_t size() const {
    return mSize;
}

size_t capacity() const {
    return mCap;
}

```

vo.2 adding copy constructor & assignment operator

```

// copy constructor
vector(const vector<T>& a) {
    mData = new T[a.capacity()]();
    mCap = a.capacity(); // copies only value
    mSize = a.size();
    for (size_t i=0; i<a.size(); i++) {
        mData[i] = a[i];
    }
}

```

```

    }
}

// copy assignment operator
vector<T>& operator=(vector<T> &other) {
    // protect against self-destruct
    if (mData != other.mData) {
        // delete current data
        delete [] mData;
        // copy the new data
        mData = new T[other.capacity()]();
        mCap = other.capacity();
        mSize = other.size();
        for (size_t i=0; i<a.size(); i++) {
            mData[i] = a[i];
        }
    }
}

```

Copy-and-swap

- Utilize written copy-constructor and destructor
- Shorter code

```

// copy assignment operator using copy-and-swap idiom
vector<T>& operator=(vector<T> other) { // notice the pass-by-value!!!
    // other is copy-constructed which will be destruct at the end of this scope
    // we swap the content of this class to the other class and let it be destructed
    using std::swap;
    swap(this->mSize, other.mSize);
    swap(this->mCap, other.mCap);
    swap(this->mData, other.mData);
    return *this;
}

```

Iterator, Insert, Erase, & Analysis

vo.3 Iterator and typedef keyword

```

template <typename T>
class vector {

protected:
    T *mData;
    size_t mCap;
    size_t mSize;
protected:
    typedef T* iterator; // let iterator be type name of T pointer

    // iterator
    iterator begin() {
        return &mData[0];
    }

    iterator end() {
        return begin()+mSize;
    }
}

```

- See that pointer works just like how `std::vector::iterator` works
- In fact, iterator is actually a pointer
- `typedef` keyword allows us to map a type name
 - `CP::vector<int>::iterator` is `int*`
 - `CP::vector<bool>::iterator` is `bool*`

insert

- `push_back` actually call `insert(end(), element)`
- Question: why we need `pos`?

```
iterator insert(iterator it, const T& element) {
    size_t pos = it - begin();
    ensureCapacity(mSize+1);
    for (size_t i = mSize; i > pos; i--) {
        mData[i] = mData[i-1];
    }
    mData[pos] = element;
    mSize++;
    return begin()+pos;
}

void push_back(const T& element) {
    insert(end(), element);
}
```

erase

- See that both `insert` and `erase` also change `mSize`

```
void erase(iterator it) {
    while((it+1)!=end()) {
        *it = *(it+1);
        it++;
    }
    mSize--;
}
```

Exercise

- Read the following function and see how it works in `vector.h`
 - `resize`, `clear`
 - non-stl function
 - `insert_by_pos`
 - `erase_by_pos`
 - `erase_by_value`
 - `contains`
 - `index_of`
- Read in [here](#)

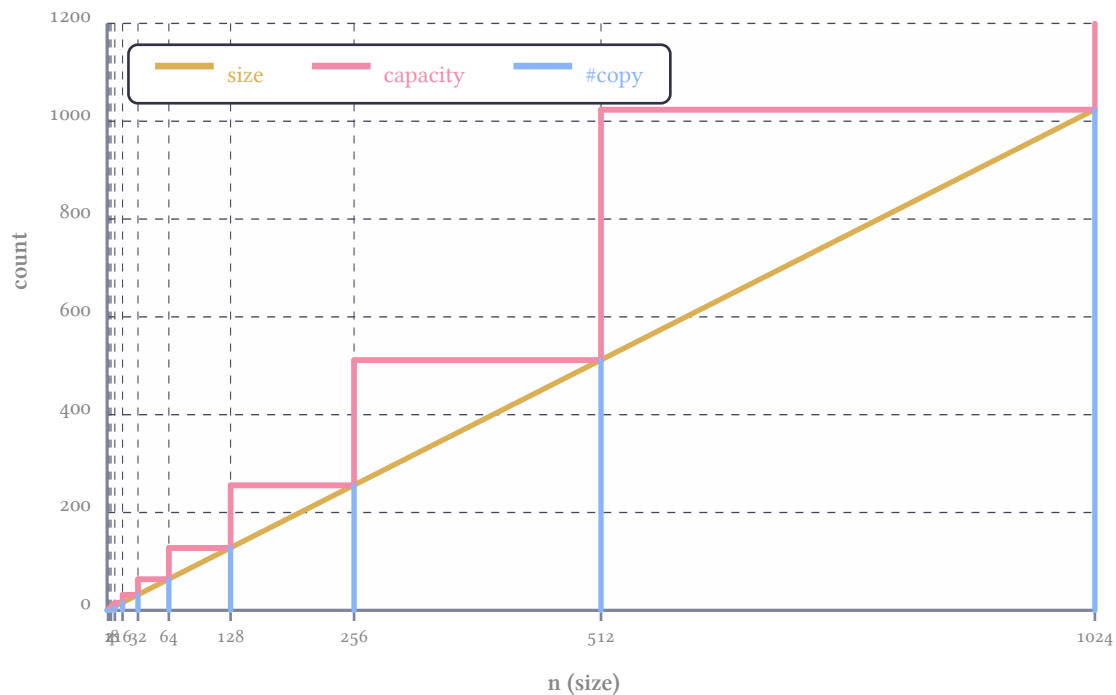
Will do this part later

Analysis of how many data is copied by `push_back`

- When full, `push_back` have to move all data to a new dynamic array
- `ensureCapacity` double the size

Size and Capa & Copy Count

- How much copy we need?



Lecture 10

Stack

Intro

- Now we will create less complex data structure `CP::stack`
- Just like a vector without iterator, insert, erase, resize, at end operator[]
 - Add `top()` which is just a shorthand of looking at the last element
- That's it, really

Key Idea

- The data is stored in the same way as a vector
 - The first element of `mData` is the bottom of stack while the last element is the top of the stack
- We just take `vector.h` and remove unnecessary function

```

| | stack
| |
| A | mSize mCap mData
| X | 4 5 [A, B, X, A]
| B |
| A |
|__|

```

stack.h

```

namespace CP {
    template <typename T>
    class stack {
    protected:
        T *mData;
        size_t mCap;
        size_t mSize;
        void expand(size_t capacity) {...}
        void ensureCapacity(size_t capacity) {...}
    };
}

```

```

public:
    // constructor
    stack(const stack<T>& a) {...}
    stack() {...}
    stack<T>& operator= {...}
    ~stack() {...}
    // capacity operation
    bool empty() const {...}
    size_t size() const {...}
    // access
    const T& top() const {
        return mData[mSize-1];
    }
    // modifier
    void push(const T& operator) {...} // This is push_back
    void pop() {...} // This is pop_back
};
}

```

Speed of each operation

- All read operation always take constant time
 - `size()`, `top()` simply return something that is directly accessible
- All modify operation also take constant time
 - `push()` is constant on average (same as `push_back` of vector)
 - `pop()` is always constant

Stack by Vector

- Instead of writing our own function, there is another way to write a stack
- We simply use `vector` as our sole data member
- Benefit: `code reuse`
- Drawback: `almost none` except that we need one more layer of function call

```

namespace CP {
    template <typename T>
    class stack {
    protected:
        vector<T> v;
    public:
        // default constructor
        stack() : v() { }
        // capacity function
        bool empty() const { return v.empty(); }
        size_t size() const { return v.size(); }
        // access
        const T& top() const { return v.back(); }
        // modifier
        void push(const T& element) { v.push_back(element); }
        void pop() { v.pop_back(); }
    };
}

```

Lecture 11

CP::queue

Will the circle be unbroken?

Intro

- Queue, unlike stack, require more sophisticated technique to achieve fast performance
- We start by writing a simple class that just work (slowly)
- Then we try to improve it

Key Idea

- Just like stack, we will use the same format as vector, using dynamic array to store data
- However we have to somehow manage how we works with `front()` and `back()` of the queue

vo.1 simple implementation of the queue

- To illustrate this idea, we will use a **vector** as our data member
- `push(e)` is simply `v.push_back(e)`, this is fast
- `front()` is `v[0]`, `back()` is `v[v.size()-1]`, this is also fast
- `pop()` is `v.erase(v.begin())`, this is slow (always proportional to `v.size()`)
 - Unlike `std::queue` which has very fast `pop()`

```
namespace CP {
    template <typename T>
    class queue {
    protected:
        std::vector<T> v;
    public:
        // default constructor
        queue() : v() { }
        // capacity function
        bool empty() const { return v.empty(); }
        size_t size() const { return v.size(); }
        // access
        const T& front() const { return v[0]; }
        const T& back() const { return v[v.size()-1]; }
        // modifier
        void push(const T& element) { v.push_back(element); }
        void pop() { v.erase(v.begin()); }
    };
}
```

vo.1 example

```
CP::queue<int> q;      q  v
                        mSize mCap mData
                        0    1    ----> [ ]

q.push(10);            q  v
                        mSize mCap mData
                        1    1    ----> [10]

q.push(20);            q  v
                        mSize mCap mData
                        2    2    ----> [10, 20]

q.push(30);            q  v
                        mSize mCap mData
                        3    4    ----> [10, 20, 30, ]

q.pop();               q  v
                        /      /
```

mSize	mCap	mData	/	/
2	4	----	>	[20, 30, ,]

vo.2 faster queue

- Add more data member mFront, initialized as 0
- push(e) is simply v.push_back(e), this is fast
- front() is v[mFront], back() is v[v.size()-1], this is also fast
- pop() is mFront++, this is fast
 - However, we don't really remove anything when pop

```
#include <vector>
```

```
namespace CP {
    template <typename T>
    class queue {
    protected:
        std::vector<T> v;
        int mFront;
    public:
        // constructor
        queue() : v(), mFront() {}
        // capacity function
        bool empty() const { return v.empty(); }
        size_t size() const { return v.size() - mFront; }
        // access
        const T& front() const { return v[mFront]; }
        const T& back() { return v[v.size()-1]; }
        // modifier
        void push(const T& element) { v.push_back(element); }
        void pop() { mFront++; }
    };
}
```

vo.2 example

CP::queue<int> q;	q	v			
		mFront	mSize	mCap	mData
		0	0	1	----> []
q.push(10);	q	v			
		mFront	mSize	mCap	mData
		0	1	1	----> [10]
q.push(20);	q	v			
		mFront	mSize	mCap	mData
		0	2	2	----> [10, 20]
q.push(30);	q	v			
		mFront	mSize	mCap	mData
		0	3	4	----> [10, 20, 30,]
q.pop();	q	v			
		mFront	mSize	mCap	mData
		1	3	4	----> [10, 20, 30,]

Problem with vo.2

- Fast but **use too many space**
- Queue grows according to **how many time push is called**
 - regardless of how many pop is called
- The data stored in the vector can be **much larger** than the actual data in the queue
- Does not really work in real world

```
for (int i=0; i<1000000; i++) {
    q.push(i);
    q.pop();
}
std::cout << q.size() << std::endl;
```

Circular Queue

Final Idea

mFront	mSize	mCap	mData	*
2	2	4	----> [, , 30, 40]	
q.push(50);				
2	3	4	----> [50, , 30, 40]	
q.push(60);				
2	4	4	----> [50, 60, 30, 40]	
q.pop();				
3	3	4	----> [50, 60, , 40]	
q.push(99);				
3	4	4	----> [50, 60, 99, 40]	
q.push(1);				
0	5	4	----> [40, 50, 60, 99, 1, , ,]	\\\

- We take vo.2 and **reuse** the area at the beginning of mData
 - Expand when necessary
 - Re-arrange when expand

Circular Queue

- We can think of mData to be circular
 - End of the last element of the mData is connected to the first element
- Consider i^{th} element
 - the next element is $(i+1) \% \text{mCap}$
 - The previous element is $(i-1+\text{mCap}) \% \text{mCap}$
 - Next k element is $(i+k) \% \text{mCap}$

Circular Queue Implementation

queue.h

```
namespace CP {
    template <typename T>
    class queue {
    protected:
        T *mData;
        size_t mCap;
        size_t mSize;
        size_t mFront;
        void expand(size_t capacity) {...}
        void ensureCapacity(size_t capacity) {...}
    public:
        // constructor - almost the same but have to take care of mFront
        queue(const queue<T>& a) {...}
        queue() {...}
    };
}
```

```

    queue<T>& operator=(queue<T> other) {...}
    ~queue() {...}
    // capacity function - same as vector
    bool empty() const {...}
    size_t size() const {...}
    // access - circular queue implementation
    const T& front() const {...}
    const T& back() const {...}
    // modifier
    void push(const T& element) {...}
    void pop() {...}
};
}

```

Ctor, Dtor, copy

```

template <typename T>
class queue {
protected:
    T *mData;  size_t mCap;  size_t mSize;  size_t mFront;
public:
    // default constructor
    queue() : mData(new T[1]()), mCap(1),
             mSize(0), mFront(0) { }
    // copy constructor
    queue(const queue<T>& a) : mData(new T[a.mCap]()), mCap(a.mCap),
                             mSize(a.mSize), mFront(a.mFront) {
        for (size_t i = 0; i < a.mCap; i++) {
            mData[i] = a.mData[i];
        }
    }
    // copy assignment operator
    queue<T>& operator=(queue<T> other) {
        using std::swap;
        swap(mSize, other.mSize);
        swap(mCap, other.mCap);
        swap(mData, other.mData);
        swap(mFront, other.mFront);
        return *this;
    }
    ~queue() {
        delete [] mData;
    }
};

```

front(), back(), pop()

```

template <typename T>
class queue {
protected:
    T *mData;
    size_t mCap;
    size_t mSize;
    size_t mFront;
public:
    // access
    const T& front() const {
        return mData[mFront];
    }
};

```

```

    }
    const T& back() const {
        return mData[(mFront + mSize - 1) % mCap];
    }
    // modify
    void pop() {
        mFront = (mFront + 1) % mCap;
        mSize--;
    }
};

```

- back = mFront + mSize - 1
 - Also circular (by % mCap)
- pop = move mFront by 1
 - Also circular
 - Also change size

push, expand

- push add data to (mFront + mSize) % mCap
 - The space just after back()
- Expand re-pack the mData so that mFront is 0
- ensureCapacity is the same

```

template <typename T>
class queue {
protected:
    T *mData;
    size_t mCap;
    size_t mSize;
    size_t mFront;
    void expand(size_t capacity) {
        T *arr = new T[capacity]();
        for (size_t i = 0; i < mSize; i++) {
            arr[i] = mData[(mFront + i) % mCap];
        }
        delete [] mData;
        mData = arr;
        mCap = capacity;
        mFront = 0;
    }
    void ensureCapacity(size_t capacity) {
        if (capacity > mCap) {
            size_t s = (capacity > 2 * mCap) ? capacity : 2 * mCap;
            expand(s);
        }
    }
public:
    void push(const T& element) {
        ensureCapacity(mSize+1);
        mData[(mFront + mSize) % mCap] = element;
        mSize++;
    }
};

```

Analysis

- All access, modification is fast (constant time)

- Space is re-used
 - It is not shrunk when mSize reduce
 - Space is not more than double of maximum mSize during its lifetime

Queue Exercise

- We implement circular queue by maintain mFront and use circular logic ($\% \text{ mCap}$) to calculate the position of back of the queue
 - Can we maintain mBack instead?
 - Can we maintain both mFront and mBack but not mSize?
- How about mCap, if we know mFront, mSize, mBack, can we calculate mCap?

mFront	mSize	mBack	front()	back()	size()
YES	YES	No	v[mFront]	v[(mFront + mSize) % mCap]	mSize()
No	YES	YES	????	v[mBack]	mSize
YES	No	YES	v[mFront]	v[mBack]	????

Answer

- Use $\text{front}() = v[(\text{mCap} + \text{mBack} - \text{mSize}) \% \text{mCap}]$
- Use $\text{size}() = (\text{mCap} + \text{mBack} - \text{mFront}) \% \text{mCap}$, but this will break when $\text{mBack} == \text{mFront}$ because it can't determine whether the queue is empty or full.
- Impossible, mCap can be scaled to any size larger than the current elements.

Now, meet deque

- Can you modify queue to include
 - `push_front()`, add to the front of the queue
 - `pop_back()`, remove from back of the queue
- All operation should still be constant time

Answer

```
void push_front(const T& element) {
    ensureCapacity(mSize+1);
    mFront = (mCap + mFront - 1) % mCap;
    mData[mFront] = element;
    mSize++;
}

void pop_back() {
    mSize--;
}
```

Lecture 12

Complexity Analysis

How fast is our code?

Overview

- Introducing a **measurement** of efficiency of a program
- Why we need measurement
- How we measure
 - How to analyze our code to get a measurement
- What is measurement
 - **Asymptotic Notation**

Key Idea

- We need a useful way to describe efficiency of our code
 - Useful = able to be **easily use to predict** how much resource (time / memory) that our program will need
 - Useful = **not overly complex** in analysis
 - Need to balance between usefulness and complexity
- Ultimately, we introduce a **class of efficiency** that says how our code use resource with respect to size of data
 - Focus on **on growth of resource usage**

Preview

```
int find_max(vector<int> v) {
    int m = v[0];
    for (size_t i = 0; i < v.size(); i++)
        if (v[i] > m)
            m = v[i];
    return m;
}
```

- This code takes time **directly proportional** to **the size of the data**
 - Size N takes time T
 - Size $5N$ should take time $5T$

```
int count_pair_sum(vector<int> v, int k) {
    int count = 0;
    for (size_t i = 0; i < v.size(); i++)
        for (size_t j = 0; j < v.size(); j++)
            if (i != j && v[i] + v[j] == k)
                count++;
    return count/2;
}
```

- This code takes time **directly proportional** to **the square of the size of the data**
 - Size N takes time T
 - Size $5N$ should take time $25T$

Why don't we use real world clock?

- Ultimately, we want to know **how long our program takes** to do each operation
 - Use in design, how much resource we need
 - Help use **choose appropriate data structure**
- **Real world clock measurement is possible** but has **many drawback**
 - System dependency
 - Too complex (we have to build our system)
 - Too specific

Dependency of System

- Same program in different system = different time
 - CPU
 - RAM
 - Compiler
 - Operating parameter
 - Heat? Other running program?

How to measure

- Counting instruction
 - Depend on code only
- Dependency on size of data
 - Focus on large data

```
int count_pair_sum(vector<int> v, int k) {
    int count = 0;
    for (size_t i = 0; i < v.size(); i++)
        for (size_t j = 0; j < v.size(); j++)
            if (i != j && v[i] + v[j] == k)
                count++;
    return count/2;
}
```

// 1
 // 2 + n + n (n = v.size())
 // (2 + n + n) * n
 // 2 * n * n
 // 1 * ??? (<= (d))
 // 1

```
}
```

```
// Total = 1 + 2+n+n + (2+n+n)n + 2n^2 + n^2 + 1
//          = 4 + 4n + 5n^2
```

Time per instruction

- In reality, different CPU instructions use different time
- Same instruction but different CPU also use different number of cycle
- **However, we just ignore it**
 - For now

Focusing on large data

- In many cases, the size of the data we are working with will affect the time our code use
- Large data usually mean longer time
- What matter is when the data is large

Growth rate simplifies analysis

```
int count_pair_sum(vector<int> v, int k) {
    int count = 0;
    for (size_t i = 0; i < v.size(); i++)
        for (size_t j = i+1; j < v.size(); j++)
            if (v[i] + v[j] == k)
                count++;
    return count;
}
```

// i=0: 1 + n-0 + n-1
 // i=1: 1 + n-1 + n-2
 // <-- i=2: 1 + n-2 + n-3
 // . . .
 // i=n-1: 1 + 1 + 0
 // sum = $\Sigma(1+2n-2i-1) = n^2 + n$

```
// Total = 1 + 2+n+n + n^2+n + n^2 + n^2 + 1
//          = 4 + 3n + 3n^2
```

Small Detail

n	5n ² + 4n + 4		3n ² + 3n + 4		n ²	
	raw	growth	raw	growth	raw	growth
10	544		334		100	
20	2084	3.830 88	1264	3.784 43	400	4
40	8164	3.917 47	4924	3.895 57	1600	4
80	32 324	3.959 33	19 444	3.948 82	6400	4
160	128 644	3.979 83	77 284	3.9747	25 600	4
320	513 284	3.989 96	308 164	3.987 42	102 400	4
640	2 050 564	3.994 99	1 230 724	3.993 73	409 600	4
1280	8 197 124	3.9975	4 919 044	3.996 87	1 638 400	4
2560	32 778 244	3.998 75	19 668 484	3.998 44	6 553 600	4
5120	1.31 × 10 ⁸	3.999 37	78 658 564	3.999 22	26 214 400	4
10 240	5.24 × 10 ⁸	3.999 69	3.15 × 10 ⁸	3.999 61	1.05 × 10 ⁸	4
20 480	2.1 × 10 ⁹	3.999 84	1.26 × 10 ⁹	3.9998	4.19 × 10 ⁸	4

When counting instruction, it is usually OK to focus on **most executed line**

Measurement by Growth Rate

What?

- Growth rate = how much resource usage growth with respect to change of input
 - Resource usage = number of instruction used
 - Input = size of data
- Emphasizes long term trend

Why?

- System independent
 - The result can be used to predict behavior on any system
- Focus on change of resource usage with respect to size of input
- Can regard small detail
 - Simple to calculate
 - Applicable in real world

Asymtotic Notation

Classification of growth rate

Overview

- Formally, it is a **set of function** having the **growth rate** related to something
- The definition focus **on growth of the function** while **disregard small detail**
- Also provide some workaround on dependency of value of input

What is?

- A kind notation written as $O(f(n))$
 - O can be one of $\mathcal{O}, \Theta, \Omega, \sigma, \omega$
 - $f(n)$ is some expression
- Example $\mathcal{O}(n)$ or $\Theta(n^2 + 3)$ or $\omega(n^2 \log(n))$
- Usage
 - “This code is $\mathcal{O}(n)$ ” (read as Big-Oh of n)
 - “This function takes time in $\Theta(\log(n))$ ” (read as Big-theta of log n)
 - “Time complexity of this program is $\mathcal{O}(n^2)$ ”

Meaning

- “A is $\Theta(f(x))$ ” means the **growth rate of A is equal** to the **growth rate of $f(x)$**
- “B is $\mathcal{O}(f(x))$ ” means the **growth rate of B is less than or equal** to the **growth rate of $f(x)$**
- For Ω, σ, ω , (**Big-Omega**, **little-oh**, **little-omega**), we won't use it for now but the meaning is similar (which are **more than or equal**, **less than**, **more than**, respectively)
- Conventionally, we usually use **N for the size of the data**

Usage

- Let **$f(n)$** be the number of instruction needed by **code A** when the **size of data is n**
- We will calculate asymptotic notation that **$f(n)$** is a member of
 - Find $g(n)$ such that $f(n)$ is $\mathcal{O}(g(n))$ or $\Theta(g(n))$ (or $\Omega(g(n))$ or ...)
- Let's say we have analyze that **$f(n)$ is $\mathcal{O}(g(n))$**
 - We now understand that the growth rate of instruction required by n grows slower or the same as how $g(n)$ grow

Comparing growth rate of $f(n)$ and $g(n)$

- The relation of growth rate of $f(n)$ and $g(n)$ depends on the value of $f(n)/g(n)$ when $n \rightarrow \infty$

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} = \begin{cases} 0 & \text{if } f(n) \text{ grows slower than } g(n) \\ c & \text{if } f(n) \text{ grows similarly to } g(n) \\ \infty & \text{if } f(n) \text{ grows faster than } g(n) \end{cases}$$

Example

- $f(n) = 4 + 3n + 4n^2$

- $g(n) = n^2$

$$\lim_{n \rightarrow \infty} \frac{4n^2 + 3n + 4}{n^2} = \lim_{n \rightarrow \infty} \left(4 + \frac{3}{n} + \frac{4}{n^2} \right) = 4$$

- Hence $f(n)$ grows similarly to $g(n)$
 - Therefore, $f(n) = \Theta(n^2)$

Another Example

- $f(n) = 0.00005n^2$
- $g(n) = 100000n$

$$\lim_{n \rightarrow \infty} \frac{0.00005n^2}{100000n} = \lim_{n \rightarrow \infty} 10^{-10}n = \infty$$

- Hence $f(n)$ grows faster than $g(n)$ (also means $g(n)$ grows slower than $f(n)$)
 - Therefore $g(n) = \mathcal{O}(n^2)$
 - and $f(n) = \Omega(100000n)$

Dependency on the value of input

- Consider `vector::insert(iterator it, T value)`
- The time it takes depends on both the size of the vector and the value of it
 - Larger size $\rightarrow$ more time
 - Closer to `end()` $\rightarrow$ less time

Big-O describes upper bound

- `vector::insert` is $\mathcal{O}(n)$
 - Its growth rate does not exceed n
 - There are cases that it may grow less than n (insert at `end()`)
- This is very useful in real world
 - Knowing maximum load
 - Not overly complex in analysis

Big-O Example

- Find is $\mathcal{O}(n)$
 - At worst, it can't find value and a & b points to `begin()` & `end()`
 - This case, find growth rate is n
 - At best, it always find value at the first position (a)
 - This case, find growth as 1

```
bool find(iterator a, iterator b, T value) {
    while (a < b) {
        if (*a == value)
            return true;
        a++;
    }
    return false;
}
```

L'Hôpital's Rule

$$\lim_{n \rightarrow \infty} \frac{f(x)}{g(x)} = \lim_{n \rightarrow \infty} \frac{f'(x)}{g'(x)}$$

If $f(x), g(x)$ must be diffable, $g(x)$ is non-zero, $\lim_{n \rightarrow \infty} f'/g'$ exists

- Can help
- $F(n) = \log n$

- $g(n) = n^{0.5}$

$$\begin{aligned}
 \lim_{n \rightarrow \infty} \frac{\log(n)}{\sqrt{n}} &= \lim_{n \rightarrow \infty} \frac{\ln(n)}{\ln(10)\sqrt{n}} \\
 &= \frac{1}{\ln(10)} \lim_{n \rightarrow \infty} \frac{\ln(n)}{\sqrt{n}} \\
 &= \frac{1}{\ln(10)} \lim_{n \rightarrow \infty} \frac{\frac{1}{n}}{\frac{1}{2\sqrt{n}}} \\
 &= \frac{1}{\ln(10)} \lim_{n \rightarrow \infty} \frac{2}{\sqrt{n}} \\
 &= 0
 \end{aligned}$$

Exercise

- $f(n) = \lg^c n$
- $g(n) = n^k$
- We know that $c > 0, k > 0$
- Does $f(n)$ grow slower than $g(n)$?

Will do this part later

Big Theta

Big-Theta is tight bound

- `std::count` always go through entire array
- Regardless of the value in the array, it always perform `if (*first == value)`

```

size_t count(iterator first, iterator last, const T& value) {
    size_t ret = 0;           // 1
    for (; first != last; ++first) { // n + n
        if (*first == value)    // 1 * n
            ret++;              // 1 * ??? ( <= (d) )
    }
    return ret;               // 1
}

```

```

// Total = 1 + n+n + 1*n + 1*n + 1
//         = 2 + 4n

```

More Example

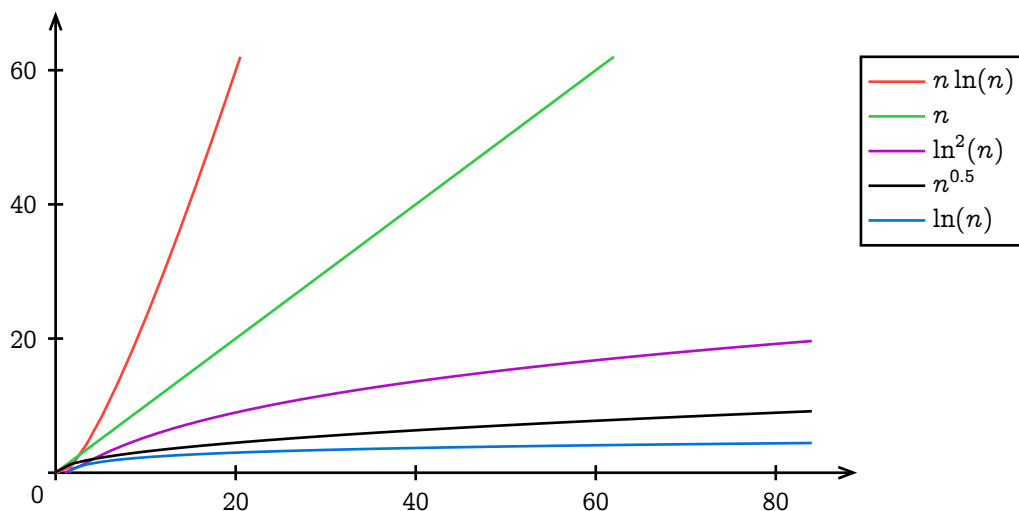
$\Theta(1)$	$\Theta(n)$	$\Theta(n^2)$
$f(n) = 5$	$f(n) = n$	$f(n) = n^2$
$f(n) = 0$	$f(n) = n + 3$	$f(n) = \binom{n}{2} = \frac{n(n-1)}{2}$
$f(n) = c$	$f(n) = \frac{n}{1200} + 86$	$f(n) = 400n^2 + an + b$
	$f(n) = 40000000n$	

Observation: multiplicative and addition constants in $f(n)$ can usually be ignored, since it will be disregarded by $\lim$, Degree cannot.

Well known growth rate class

$\Theta(1)$	Constant
$\Theta(\log(n))$	Logarithm
$\Theta(\log^c(n)), c \geq 1$	Polylogarithm

$\Theta(n^a), 0 < a < 1$	Sublinear
$\Theta(n)$	Linear
$\Theta(n \log(n))$	Linearithmic
$\Theta(n^2)$	Quadratic
$\Theta(n^c), c \geq 1$	Polynomial
$\Theta(c^n), c > 1$	Exponential
$\Theta(n!)$	Factorial



$\ln^2(n)$ grows slower than $n^{0.5}$. At around $n = 56000$, $\ln^2(n)$ is less than $n^{0.5}$.
 x^4 grows slower than 2^n . At around $n = 16$, x^4 is less than 2^n .

Beware

- It is **wrong** to say that `vector::insert` is $\Theta(n)$
 - Because there is a case that it grows slower than n
- It is **wrong** to say that `vector::push_back` is $\Theta(n)$
 - Because there is a case that it grows slower than n
- It is **ok** to say that `std::count` is $\mathcal{O}(n)$
 - Because while it always grows as n , **it does not grow faster than n**
 - $\mathcal{O}(n)$ is upper bound
 - But it is better to say that `std::count` is $\Theta(n)$

How to analyze using asymptotic notation

1. Write a code
2. Calculate the function $f(n)$ that counts the number of instruction of the code when the data is size n
 - Usually, just focus on **most executed line**
3. Find $g(n)$ and a notation X such that $f(n)$ is $X(g(n))$
 - It's either **Big-O** or **Big-Theta**. If there is a case that it can grow slower than $G(n)$, use Big-O

Another Example

- Let's analyze `vector::push_back`

```
iterator insert(iterator it, const T& element) {
    size_t pos = it - begin();
    ensureCapacity(mSize + 1);
    for (size_t i = mSize; i > pos; i--) {
        mData[i] = mData[i-1]; // <----- Most executed line: 1 (b)
    }
}
```

```

    mData[pos] = element;
    mSize++;
    return begin()+pos;
}
void push_back(const T& element) {
    insert(end(), element);
}

void expand(size_t capacity) {
    T *arr = new T[capacity]();
    for (size_t i = 0; i < mSize; i++) {
        arr[i] = mData[i]; // <----- Most executed line: 0, 0, 0, 0, 2, 4, 8, 16 (a)
    }
    delete [] mData;
    mData = arr;
    mCap = capacity;
}

void ensureCapacity(size_t capacity) {
    if (capacity > mCap) {
        size_t s = (capacity > 2 * mCap) ? capacity : 2 * mCap;
        expand(s);
    }
}

// (a) can be 0 to n
// (b) can also be 0 to n
// Best case 0 + 0
// Worst case n + n
// vector::push_back is O(n)

```

Another Definition for $\mathcal{O}$ and Θ

- Using set builder notation
- $\mathcal{O}(g(n)) = \{f(n) \mid \text{there exists } c > 0 \text{ and } n_0 \geq 0 \text{ such that } f(n) \leq cg(n) \text{ for } n \geq n_0\}$
- $\Theta(g(n)) = \{f(n) \mid \text{there exists } c_1, c_2 > 0 \text{ and } n_0 \geq 0 \text{ such that } c_1g(n) \leq f(n) \leq c_2g(n) \text{ for } n \geq n_0\}$
- The result is the same as definition using lim

Summary

- We use Asymtotic Notation to describe efficiency of a program
 - Measure instruction count instead of time
 - Focus on growth rate of instruction count
- Find most frequently executed line in the code and count it
- Maps to Big-Theta if we have tight bound
- Use Big-O if we have upper bound

Example

Showing $\log_2(n!)$ is $\Theta(n \log_2(n))$

- Will use set definition of Big-Theta
 - $\{f(n) \mid \text{there exists } c_1, c_2 > 0 \text{ and } n_0 \geq 0 \text{ such that } c_1g(n) \leq f(n) \leq c_2g(n) \text{ for } n \geq n_0\}$
- Need to find c_1, c_2 and n_0

Finding c_1 and c_2

$$n! = n \times (n-1) \times (n-2) \times (n-3) \times \dots \times 1$$

$$n! \leq n \times n \times n \times n \times \dots \times n$$

$$\log n! \leq \log n^n \quad \boxed{\log n^n = n \log n \text{ when } n \geq 1}$$

$$\text{Found } c_2 = 1 \text{ for upper bound} \quad \boxed{\log n! \leq n \log n \text{ when } n \geq 1}$$

$$n! = n \times (n-1) \times \dots \times \left\lfloor \frac{n}{2} \right\rfloor \times \left(\left\lfloor \frac{n}{2} \right\rfloor - 1 \right) \times \dots \times 1$$

$$n! \geq \left\lfloor \frac{n}{2} \right\rfloor \times \left\lfloor \frac{n}{2} \right\rfloor \times \dots \times \left\lfloor \frac{n}{2} \right\rfloor \times 1 \times \dots \times 1$$

$$n! \geq \left(\frac{n}{2} \right)^{n/2} \quad \boxed{\log \left(\frac{n}{2} \right)^{n/2} = \frac{n}{2} \log n - \frac{n}{2}}$$

$$\log n! \geq \log \left(\frac{n}{2} \right)^{n/2} \quad \boxed{0.5n \log n - 0.5n \geq 0.4n \log n \text{ when } n \geq 32}$$

$$\text{Found } c_1 = 0.4 \text{ for lower bound} \quad \boxed{\log n! \geq 0.4n \log n \text{ when } n \geq 32}$$

$$\boxed{0.4n \log n \leq \log n! \leq n \log n \text{ when } n \geq 32, c_1 = 0.4, c_2 = 1, n_0 = 32}$$

$$\Rightarrow \boxed{\log n! \text{ is } \Theta(n \log n) \text{ and } n \log n \text{ is } \Theta(\log n!)}$$

Lecture 13**Priority Queue Simple Implementation**

Featuring Binary Heap

Overview

- Simple implementation of `priority_queue`
- Quick intro to **Graph** and **Tree**
- **Binary Heap**
- `priority_queue` with **Binary Heap**

`priority_queue`

- Queue by value
- Max-in-First-Out

```
int main() {
    priority_queue<int> pq;
    pq.push(4);
    pq.push(20);
    pq.push(3);

    while (pq.empty() == false) {
        cout << pq.top() << endl;
        pq.pop();
    }
}
```

Vo.1, `priority_queue` by vector

- Use **vector** to store data
- push = simply `push_back`
- top, pop = find the max value and return / erase
- `max_element` returns iterator to max element

```
namespace CP {
    template <typename T>
```



```

class priority_queue {
protected:
    std::vector<T> v;
public:
    bool empty(); { return v.empty(); }
    bool size(); { return v.size(); }

    void push(const T &e) { v.push_back(e); }

    T top() { return *std::max_element(v.begin(), v.end()); }

    void pop() { v.erase(std::max_element(v.begin(), end())); }
};

```

max_element

```

iterator max_element(iterator first, iterator last) {
    if (first == last) return last;
    iterator largest = first;
    ++first;
    for (; first != last; ++first)
        if (*largest < *first)
            largest = first;
    return largest;
}

```

Vo.1 complexities

push	$\mathcal{O}(1)$ *amortized
top()	$\Theta(n)$
pop()	$\Theta(n)$

Vo.2 faster pop, top (and push??)

- vo.1 has many drawbacks
 - Consecutive call of top is slow (it shouldn't)
 - Both pop and top works almost the same
- vo.2 focus on **slower push** while keeps **pop, top fast**
- Make the array **sorted**
 - Max will be at the back
 - Fast pop, top

```

namespace CP {
    template <typename T>
    class priority_queue {
    protected:
        std::vector<T> v;
    public:
        bool empty(); { return v.empty(); }
        bool size(); { return v.size(); }

        T& top() { return v[v.size()-1]; }

        void pop() { v.erase(v.end()-1); }

        void push(const T &e) {
            // do something

```

```

    }
};
}

```

vo.2 push

- Maintain that the vector is sorted at every push

```

void push(const T& e) {
    v.push_back(e); // <-- O(1)
    sort(v.begin(), v.end()); // <-- O(n log(n))
}

void push(const T& e) {
    auto it = v.begin();
    while (it < v.end() && *it <= e) // <-- O(n)
        it++;
    v.insert(it, e); // <-- O(n)
} // Total of this program is O(n)

void push(const T& e) { // upper_bound is O(log(n))
    v.insert(std::upper_bound(v.begin(), v.end(), e), e); // <-- O(n)
}

```

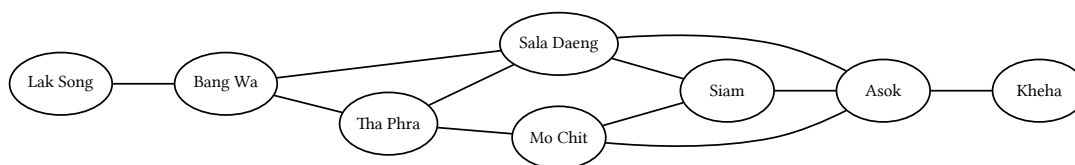
Which one is better?

- vo.1 fast push
- vo.2 fast pop, top
- Depends on which operation we use most often
- The real version works like vo.2, we maintain some **rules** of the data that is stored in the vector such that
 - We know where max is (for fast pop)
 - Much faster push, a little bit slower pop
 - Use structure called **Binary Heap**

Graph & Tree

Graph

- Discrete Math Graph
- A math model that describe entities and connectivity between them



Graph Model

- Graph consists of two things
 - **Nodes (vertex, vertices)** are **things** we want to connect
 - **Edges** are pairs, each pair is a **connectivity** between two nodes
- Graph $G = (V, E)$ where V is a set of nodes and E is a set of edges

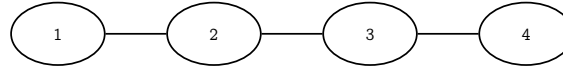
$V = \{\text{"Mo Chit", "Siam", "Asok", "Sala Daeng", "Tha Phra", "Lak Song", "Bang Wa", "Kheha"}\}$

$E =$

$\{(\text{"Mo Chit", "Asok"}), (\text{"Mo Chit", "Siam"}), (\text{"Tha Phra", "Mo Chit"}), (\text{"Sala Daeng", "Tha Phra"}), \dots\}$

Tree

- A special kind of graph
 - Has N nodes and $N - 1$ edges
 - Every nodes must be connected (we can start from any node and can walk through edge to reach any node)

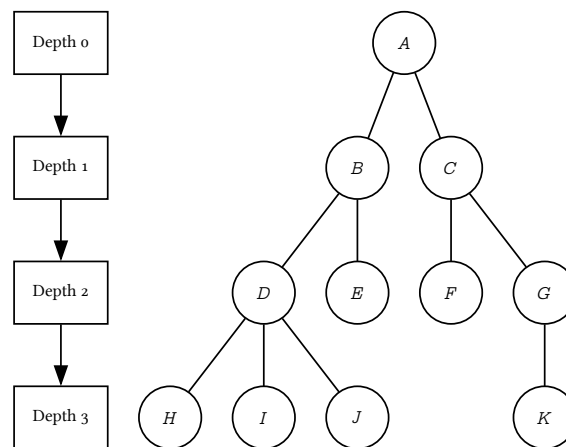


$$V = \{1, 2, 3, 4\}$$

$$E = \{(1, 2), (2, 3), (3, 4)\}$$

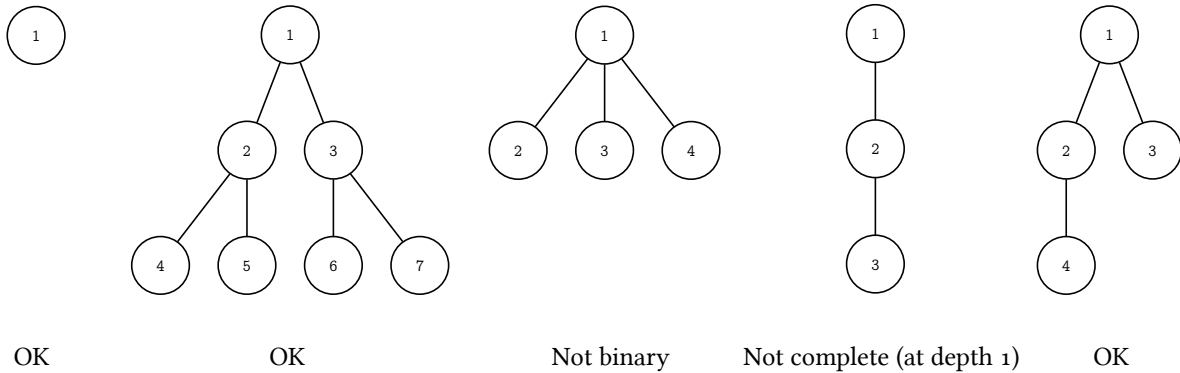
Rooted Tree

- Tree where one node is defined as **root**
- For an edge in a rooted tree, a node that is closer to the root is called **parent** while the other node is called **child**
- **Ancestor** of node A = parent of parent ... of A
- **Descendant** of node A = child of child ... of A
- Root is usually drawn at the top and is considered as the **starting point**
 - Root is at level 0 (depth 0)
 - Children of root are drawn at the same level at **level 1** (depth 1)
 - Children of children of root are at **level 2** (depth 2)



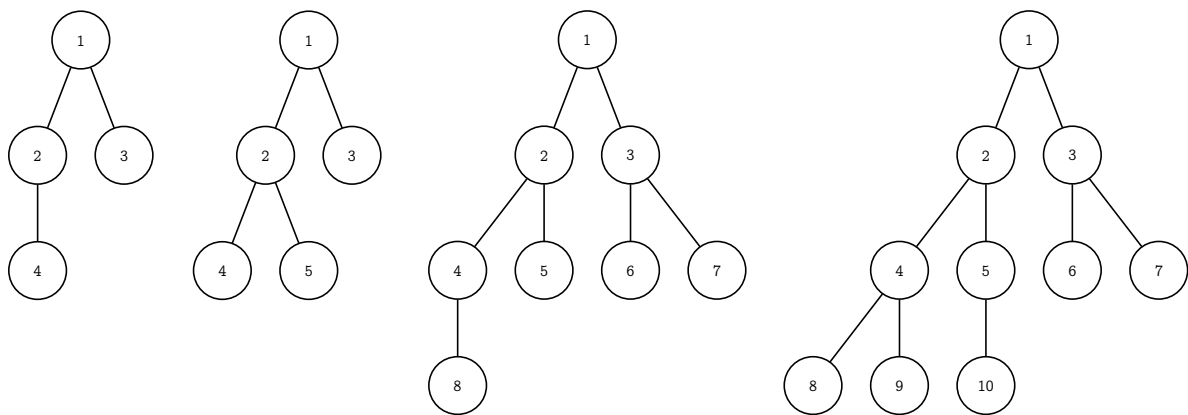
Complete Binary Tree

- Binary Tree = a tree that every node has at most 2 children
- Complete tree = the tree must be filled with every possible node at every level (except the deepest level which must be filled as far to the left as possible)
 - Blank tree is considered a complete binary tree

**Exercise**

- Draw a Complete Binary Tree that has 4, 5, 8, 10 nodes

Hint: The answer is unique (There are exactly 1 way to draw a complete binary tree of size k)

**Special Property of a Complete Binary Tree**

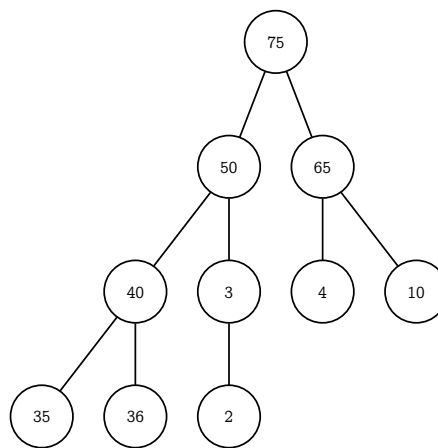
- There is exactly one way to go from any node to any node
- Maximum depth is $\log_2 n$ where n is the number of nodes
 - Because we require completeness, and we have 2 possible children

Binary Heap

Use Complete Binary Tree to make priority_queue

**Binary Heap**

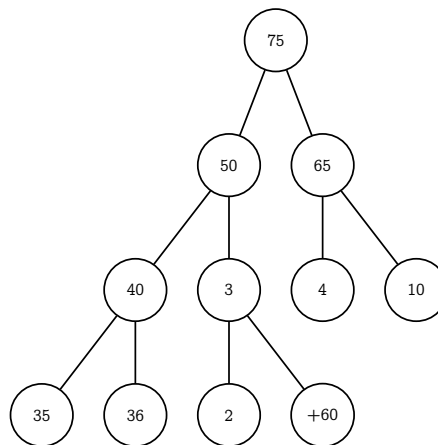
- We use **Complete Binary Tree** to store data
 - A **value** is stored at the **node**
- When data is modified (via push or pop), we must maintain these rules
 1. Tree must always be **Complete Binary Tree**
 2. For any node, its **value must be greater or equal that of its children**



Adding data to Binary Heap

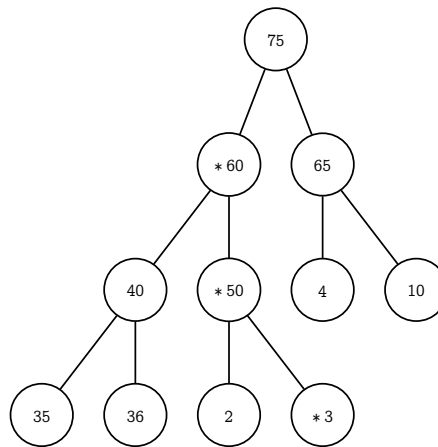
- Maintain Binary Heapness
 - structure of data
 - value of data
- Structure rules says where the new nodes should be
 - Next to **right-most child of deepest level**
 - But if we put new data there, the **value rules might be broken**
 - Fix it

push(60)



Fix from adding a new node

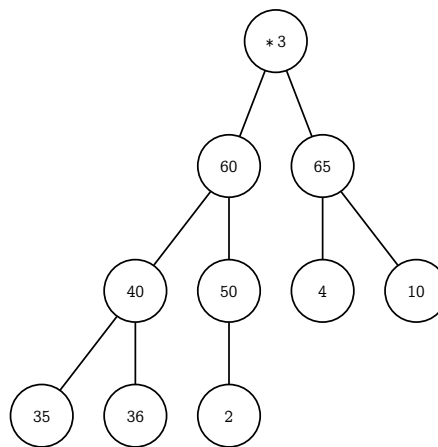
- Fix:
 - Check where we just add a node, if value rules is broken, swap with parent
 - After swap, re-check with new parent
 - Kepp doing until correct or at root



See that: after each swap, the swapped nodes does not violate with **its new children**

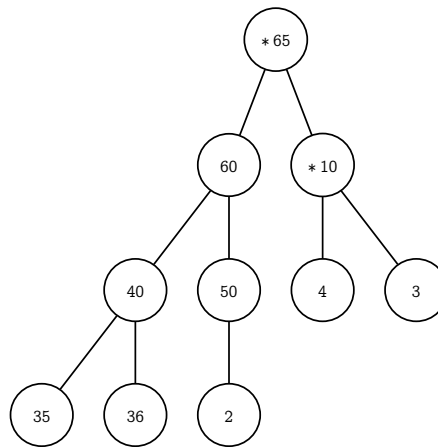
Delete maximum data

- Similar to push, we will try to maintain structure first
- delete will remove root, find something to replace
 - Use the lowest, right-most node
- Value rules might be broken
 - Fix it



Fix from deleting root node

- Fix:
 - Start at replaced root, if value rules is broken, swap with maximum child
 - After swap, re-check with new children
 - Beware! There is a case where we might have only one child
 - Keep doing until correct or has no children



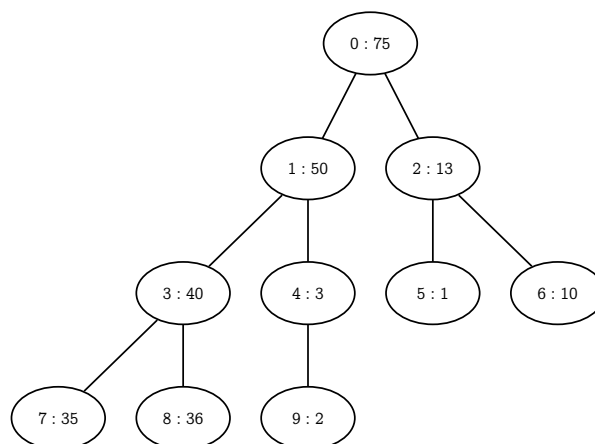
Analysis

- How fast is push, pop
- push
 - Add to vector is $\mathcal{O}(1)$ amortized
 - Fixing value rules is $\mathcal{O}(h)$ where h is the maximum number of depth of the tree (we call this value tree height)
 - Notice that tree height is $\mathcal{O}(\log(n))$
- pop
 - Fixing value rules is $\mathcal{O}(h)$ where h is the maximum number of tree height

Push + Fix Up

How to store a tree?

- Use **dynamic array**
 - Each node can be labelled from 0 to $n-1$
- **Root** is at 0
- **Left child** of node i is at $(i*2)+1$
- **Right child** of node i is at $(i*2)+2$
- **Parent** of node i is at $(i-1)/2$



Dynamic array: [75, 50, 13, 40, 3, 1, 10, 35, 36, 2]

Layout

```

template <typename T, typename Comp = std::less<T>>
class priority_queue {
protected:
    T *mData;
    size_t mCap;
    size_t mSize;
    Comp mLess;
    void expand(size_t capacity) {} // Fix value rules
    void fixUp(size_t idx) {}
    void fixDown(size_t idx) {}
public:
    // constructor
    priority_queue(priority_queue<T,Comp>& a);
    priority_queue(const Comp& c = Comp());
    priority_queue<T, Comp>& operator=(priority_queue<T,Comp> other);
    ~priority_queue();
    // capacity function
    bool empty() const;
    size_t size() const;
    // access
    const T& top();
    // modifier
    void push(const T& element);
    void pop();
};

```

Constructor

- Using list initialize
- See that mData is dynamic array in the same way as vector
- mLess is something that is just either copied or default initialize
 - Will talk about it later

```

priority_queue(const Comp& c = Comp()):
    mData( new T[1]() ),
    mCap( 1 ),
    mSize( 0 ),
    mLess( c )
{ }

```

```

priority_queue(priority_queue<T, Comp>& a) :
    mData(new T[a.mCap]()),
    mCap(a.mCap),
    mSize(a.mSize),
    mLess(a.mLess)
{
    for (size_t i = 0; i < a.mCap; i++) {
        mData[i] = a.mData[i];
    }
}

```

Destructor and Copy Assignment Operator

- Using standard copy-and-swap idiom

```

~priority_queue() {
    delete [] mData;
}

```



```
}
```

```
priority_queue<T,Comp>& operator=(priority_queue<T,Comp> other) {
    using std::swap;
    swap(mSize,other.mSize);
    swap(mCap,other.mCap);
    swap(mData,other.mData);
    swap(mLess,other.mLess);
    return *this;
}
```

Push

- See that the **right-most child of the deepest level** is at `mData[mSize-1]` and the new node should be at `mData[mSize]`
- We do the same thing as vector's `push_back`
- Then fix the value rule

```
void expand(size_t capacity) { // Same as CP::vector
    T *arr = new T[capacity]();
    for (size_t i = 0; i < mSize; i++) {
        arr[i] = mData[i];
    }
    delete [] mData;
    mData = arr;
    mCap = capacity;
}
```

```
void push(const T& element) {
    if (mSize + 1 > mCap)
        expand(mCap * 2);
    mData[mSize] = element;
    mSize++;
    fixUp(mSize-1);
}
```

Fix Up

- Instead of actual swap, we perform insert and find appropriate position at the same time

```
void fixUp(size_t idx) {
    T tmp = mData[idx];
    while (idx > 0) {
        size_t p = (idx - 1) / 2;
        if ( tmp < mData[p] ) break;
        mData[idx] = mData[p]
        idx = p;
    }
    mData[idx] = tmp;
}
```

```
mData p = /, idx = *, tmp = 60
  0  1  2  3  4  5  6  7  8  9 10
[ 75, 50, 13, 40, /3,  1, 10, 35, 36,  2,*60]

[ 75,/50, 13, 40, *3,  1, 10, 35, 36,  2,  3]

[/75,*50, 13, 40, 50,  1, 10, 35, 36,  2,  3]
```

```
tmp = 60 < 75 = mData[p] => break; mData[idx] = tmp;
[ 75, 60, 13, 40, 50, 1, 10, 35, 36, 2, 3]
```

mLess

- priority_queue allows a custom comparator
- Custom comparator X
 - We must be able to $X(a, b)$ where X will compare a and b and return true only when a is less than b
 - X is a variable that implement operator()
- Initialize at constructor as variable mLess to be of type Comp in template
- Any comparison of our data (type T) must be done by mLess

```
void fixUp(size_t idx) {
    T tmp = mData[idx];
    while (idx > 0) {
        size_t p = (idx - 1) / 2;
        if ( mLess(tmp, mData[p]) ) break;
        mData[idx] = mData[p];
        idx = p;
    }
    mData[idx] = tmp;
}
```

Pop + FixDown

Pop

```
void pop() {
    mData[0] = mData[mSize-1];
    mSize--;
    fixDown(0);
}
```

```
void fixDown(size_t idx) {
    T tmp = mData[idx];
    size_t c;
    while ((c = 2 * idx + 1) < mSize) {
        if ( c + 1 < mSize && mLess(mData[c], mData[c+1]) ) c++;
        if ( mLess(mData[c], tmp) ) break;
        mData[idx] = mData[c];
        idx = c;
    }
    mData[idx] = tmp;
}
```

- while loop checks if we have at least one child
- c is the index of highest value child
 - Must consider the case where we have only one child
- Exercise: read the rest yourself

Fast Construction

Construct PQ from n data

```
priority_queue(std::vector<T> &v, const Comp& c = Comp()) :
    mData( new T[v.size()]() ), mCap( v.size() ), mSize( 0 ), mLess( c ) {
    for (size_t i = 0; i < mSize; i++) push(v[i]); // n times and each log(n)
}
```

$$\begin{aligned} \text{Total} &\leq \lfloor \log_2 1 \rfloor + \lfloor \log_2 2 \rfloor + \dots + \lfloor \log_2 n \rfloor \\ &\leq \lfloor \log_2 (1 \times 2 \times 3 \times \dots \times n) \rfloor = \lfloor \log_2 n! \rfloor \end{aligned}$$

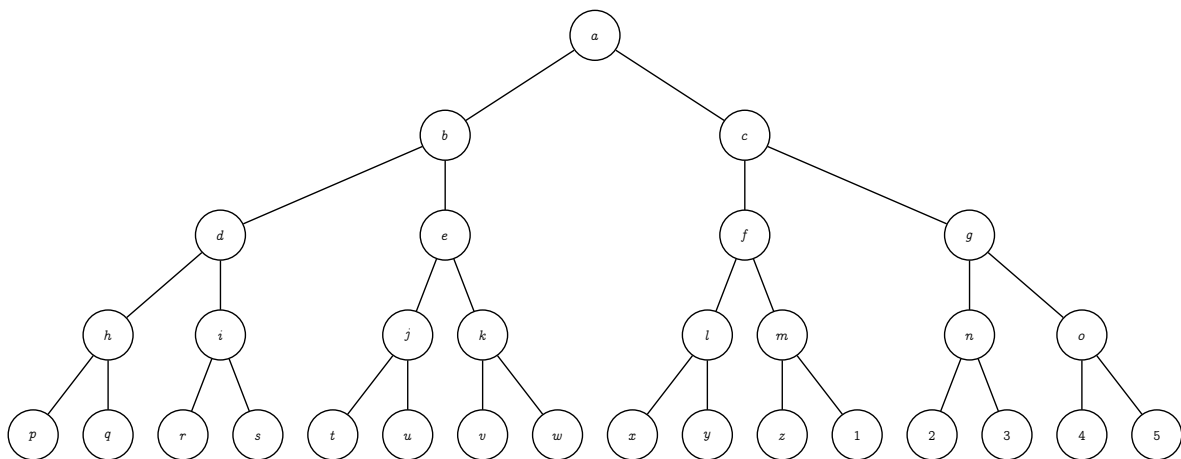
$$\lfloor \log_2 n! \rfloor \text{ is } \mathcal{O}(n \log n)$$

Better Method

```
priority_queue(std::vector<T> &v, const Comp& c = Comp()) :
    mData( new T[v.size()]() ), mCap( v.size() ), mSize( 0 ), mLess( c )
{
    for (size_t i = 0; i < mSize; i++) mData[i] = v[i];
    for (int i = mSize/2-1; i > 0; i++) fixDown(i);
}
```

- Consider each node to be a binary heap
- Fix down from back to front

How fast?



- Binary Tree Property:
 - There are at most 2^k nodes at depth k
 - For a tree of height h , at depth k , fix down needed at most $h - k$ iterations

Total nodes = 31

Tree height = $\log_2(31) = 4$

Depth: k , Nodes: 2^k , Max fix: $h - k$
 Total = $\sum_{k=0}^h k 2^{h-k}$

Depth	Nodes	Max fix per node
0	1	4
1	2	3
2	4	2
3	8	1
4	16	0

$$\sum_{k=0}^h k 2^{h-k} = 2^h \sum_{k=0}^h k 2^{-k} < 2^h \underbrace{\sum_{k=0}^{\infty} k 2^{-k}}_{\text{This is 2}} = 2^{h+1} = 2^{\log_2 n + 1} = \mathcal{O}(n)$$

Exercise

If we use different number of children per node, say 4-ary heap, with same logic still holds.

1. How to keep data in dynamic array?

2. Write the new functions `fixUp` and `fixDown`
3. Is it faster?? Consider any n -ary heap

Will do this part later

Lecture 14

Linked List

Faster insert and erase but slower access

Overview

- Vector takes $\mathcal{O}(n)$ in insert / erase at position specified by an iterator
 - But it is very fast to access any member, $\mathcal{O}(1)$
- List gives better performance on insert / erase as $\mathcal{O}(1)$, providing that we have iterator to the position of insertion / erase
- Achieved by storing data into a **node** and each **node** use a **pointer** to identify the **next element**
 - Access to any elements is $\mathcal{O}(n)$, if we don't have an iterator to that element
- Use more memory than vector

List vs Vector

List	Vector
• Allocate each data separately ▸ Each data points to where is the next data ▸ Very fast insert / erase (just change some pointer) ▸ Very slow access because we don't know where k-th element is	• Allocate data as a consecutive block ▸ Very fast to access any element ▸ Very slow insert / erase requires every element after point of insertion

vo.1 node

- Simple object that stores a **data** and a **link to another node**
- **NULL** is a special value for any pointer that points to nowhere
 - Draw as a ground

```
template <typename T>
class node {
public:
    T data;
    node *next;
    node() :
        data( T() ), next( NULL ) { }

    node(const T& data, node* next) {
        data( T(data) ), next( next ) { }
    }
};
```

Pointer to Node

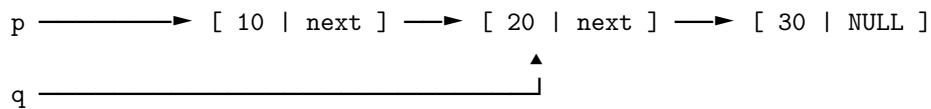
- Working with linked list needs a pointer to a node

```
int main() {
    CP::node<int> *p = NULL;
    p = new CP::node<int>(10, NULL);
    CP::node<int> *q;
```

```

q = new CP::node<int>(20, NULL);
p->next = q;
q->next = new CP::node<int>(30, NULL);
}

```

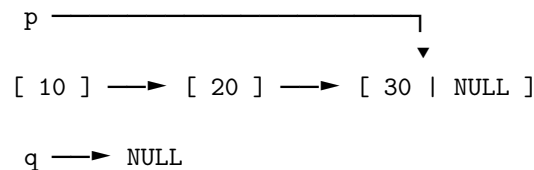


What is the result of?

```

p = p->next->next; // p points to [30, NULL]
q = p;           // q points to [30, NULL]
q = q->next;      // q points to NULL

```



vo.1 Simple (Singly) Linked List

```

template <typename T>
class list {
protected:
    class node {
    public:
        T data;
        node *next;
        node() :
            data( T() ), next( NULL ) { }

        node(const T& data, node* next) {
            data( T(data) ), next( next ) { }
        }
    };
protected:
    node *mFirst;
    size_t mSize;
public:
    list() : mFirst( NULL ), mSize( 0 ) { }
    ~list() { clear(); }
    // ... more function
};

```

- Node is a inner class

Insertion

```

Linked List
mSize mFirst      node
[ 4,  ] → [20, next] → [ 3, next] → [16, next] → [77, NULL]

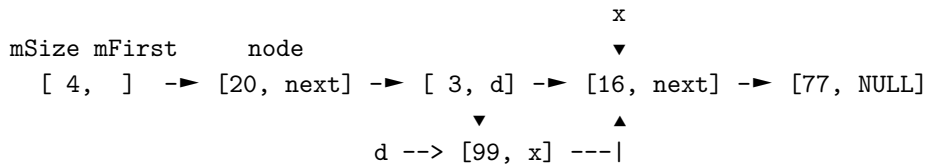
d → [99, NULL]

```

- To insert a value 99 before value 16, let x be a pointer that point to the node of 16
 1. Create a new node d containing a data to be inserted
 2. Change pointer of a node before x (which points to x) to point to d instead

3. Make pointer of d points to x

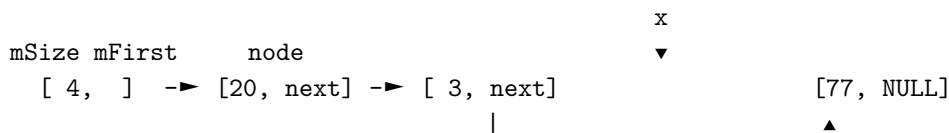
```
CP::node<int> *x = mFirst->next->next;
CP::node<int> *d = new CP::node<int>(99, NULL);
mFirst->next->next = d;
d->next = x;
```



Erase

- To delete a value of 16, in the node pointed by x
 1. Change pointer of a node before x (which points to x) to point to the node that x points to instead
 2. Don't forget to delete x

```
CP::node<int> *x = mFirst->next->next;
mFirst->next->next = x->next;
delete x;
```



Doubly Linked List

Problem with SLL

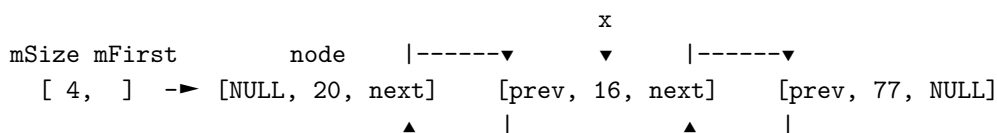
- erase / insert at iterator X is hard
 - If we have an iterator point to X, we cannot easily go to the node before X
 - Cannot go backward
 - Need to start from mFirst and move on
- Adding data to the end takes long time
 - We have to get iterator that points to the last element (which is $\mathcal{O}(n)$)

Finding node before X

```
node *p = mFirst;
while (p != NULL && p->next != x) p = p->next;
```

vo.2 Doubly Linked List

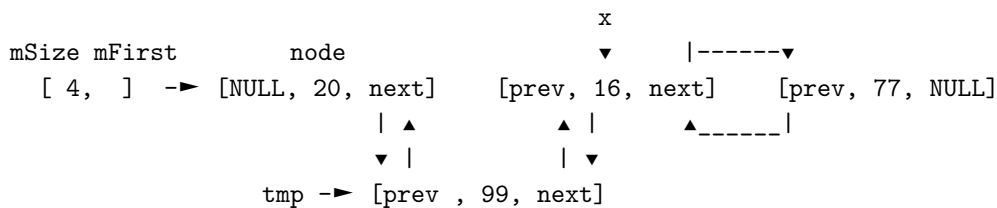
- Each node has 2 pointers
 - next and prev
- Can now move forward and backward
- Now, if X is a pointer to a node, we can easily go to node before X and then erase or insert



Insert in Doubly Linked List

```
CP::node<int> *tmp =
    new CP::node<int> (99, NULL, NULL);
tmp->next = x;
tmp->prev = x->prev;
```

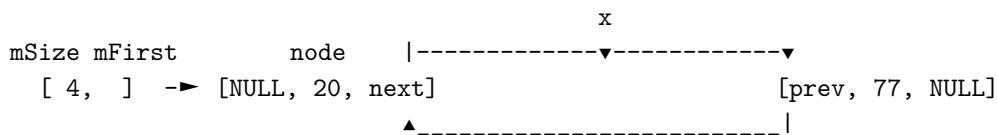
```
x->prev->next = tmp;
x->prev = tmp;
```



Erase in Doubly List

```
x->prev->next = x->next;
x->next->prev = x->prev;
delete x;
```

Note: When we refer to a pointer, x is the variable itself, but the value of x is the destination address it is pointing to. Say x holds a node containing [x->prev, 16, x->next]. In this set up, x->prev->next represents the next pointer inside the node before x. If we write x->prev->next = x->next;, it means that the previous node's next pointer now stores the destination of the node after x. In other words, it successfully skips x from the sequence.



Circular Linked List

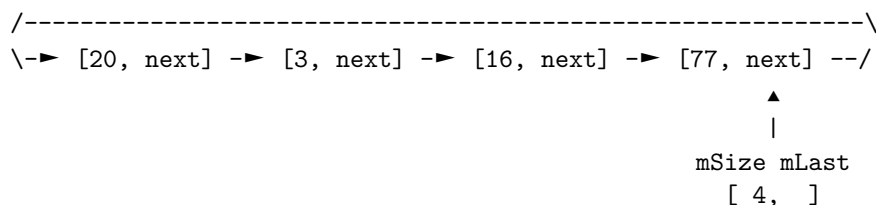
Problem Solved

- erase / insert at iterator X is now **easy**
- But, adding data at the end (push_back) is still hard
 - Need to get X to point to the last element
 - push_back is popular in real world
 - Right now we have only push_back (fast addition to the first)
- Also some minor issue about the **code cleanliness**
 - insert / erase the first / last node

vo.3 Circular Linked List

- Use mLast instead of mFirst
- Fast access to both first and last element

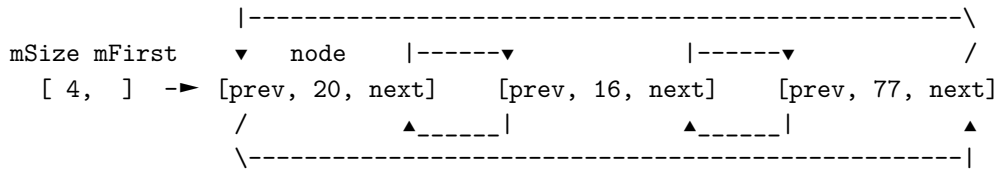
```
CP::node<int> *first = mLast->next;
```



vo.4 Circular Doubly Linked List

- Special linking to last element
 - Fast access to both first and last element
 - Can now easily insert at the end

```
CP::node<int> *last = mFirst->prev;
```



Linked List with Header

Minor Problem

- Code is **not really clean** because of the first element (or the last element)
- Consider insert at the first node and insert at the second node of Doubly Linked List
 - ▶ Assume we have a pointer to that node

```
// assume that s points to the second node
CP::node<int> *tmp =
    new CP::node<int>(99, NULL, NULL);
tmp->next = s;
tmp->prev = s->prev;
s->prev->next = tmp;
s->prev = tmp;

// assume that f points to the first node
CP::node<int> *tmp =
    new CP::node<int>(99, NULL, NULL);
tmp->next = f;
tmp->prev = f->prev;
mFirst = tmp; // Because first node is different
f->prev = tmp;
```

Special First Node Problem

- First node (and last node) is **different from other nodes** in both singly or doubly linked list
- For circular singly and circular doubly, each node look the same **but we also have to adjust mFirst** (or mLast)
- This affects both insert and erase
- Also need to deal **when mFirst is NULL** (when mSize == 0)

```
// circular doubly linked list
void push_front(const T& e) {
    if (mSize == 0) {
        mFirst = new node(e, NULL, NULL);
        mFirst->next = mFirst;
        mFirst->prev = mFirst;
    } else {
        node* tmp =
            new node(e, mFirst->prev, mFirst);
        mFirst->prev->next = tmp;
        mFirst->prev = tmp;
        mFirst = tmp;
    }
}
```

Another Example

- Remove for circular doubly linked list
- Remove is to find and then erase

```
// circular doubly linked list
void remove(const &T e) {
```



```

node *p = mFirst;
for (size_t i = 0; i < mSize; i++, p = p->next) {
    if (p->data == e) {
        p->next->prev = p->prev;
        p->prev->next = p->next;
        if (p == mFirst) {
            mFirst = p->next;
        }
        delete p;
        mSize--;
        break;
    }
}
if (mSize == 0) mFirst = NULL;
}

```

Linked List with Header

- Add a special node that will not be used to stored data

Simpler Code with Header

```

void push_front(const T& e) {
    node* f = mFirst->next;
    node* tmp = new node(e, f, mFirst);
    mFirst->next = tmp;
    f->prev = tmp;
}

void remove(const T& e) {
    node *p = mFirst->next;
    while (p != mHeader && p->data != e)
        p = p->next;
    if (p != mHeader) {
        p->next->prev = p->prev;
        p->prev->next = p->next;
        mSize--;
    }
}

```

- Header simplifies code
 - Because mFirst always points to the header (mFirst never is NULL)

Variant Summary

- **Circular** makes accessing first and last element fast
- **Doubly** makes accessing previous element fast
 - Also making erase at node p easy if we have pointer to p
 - Need more space for prev pointer
- **Header** makes code simpler
 - Need more space for header node

Final Version

- CP::list is “circular doubly linked list with header”
 - Simple code for insert / erase
 - Use most space (two pointers per node, need header node)
 - Can push_back, pop_back, push_front, pop_front
- Also need custom iterator class
 - Iterator just store a pointer to a node

- We cannot directly use a pointer to a node (node*) because we need to override some operator (--, ++, and something else)

CP::list

Layout

- **Inner class** is a class inside another class
 - **Inner class** can access any members of **outer class**
 - **Outer class** cannot access protected or private of the **inner class**
- **Friend class** allows other class to access

```
template <typename T>
class list {
protected:
    class node {
        friend class list;
    public:
        T data;
        node *prev, *next;
        // some functions
    };
    class list_iterator {
        friend class list;
    protected:
        node* ptr;
    public:
        // some functions & operators
    };
public:
    typedef list_iterator iterator;
protected:
    node *mHeader; // pointer to a header node
    size_t mSize;
public:
    // functions
};
```

Doubly Linked List Node

```
class node {
    friend class list;
public:
    T data;
    node *prev;
    node *next;

    node() :
        data( T() ), prev( this ), next( this ) { }
    node(const T& data, node* prev, node* next) :
        data( T(data) ), prev( prev ), next( next ) { }
};
```

Constructor

```
// default constructor
list() : mHeader( new node() ), mSize( 0 ) { }

// copy constructor
```

```

list(list<T>& a) : mHeader( new node() ), mSize( 0 ) {
    for (iterator it = a.begin(); it != a.end(); it++) {
        push_back(*it);
    }
}

list<T>& operator=(list<T> other) {
    using std::swap;
    swap(this->mHeader, other.mHeader);
    swap(this->mSize, other.mSize);
    return *this;
}

~this() {
    clear();
    delete mHeader;
}

```

Small functions

```

// capacity function
bool empty() const { return mSize == 0; }
size_t size() const { return mSize; }

// access
T& front() { return mHeader->next->data; }
T& back() { return mHeader->prev->data; }

// modifier
void push_back(const T& element) {
    insert(end(), element);
}
void push_front(const T& element) {
    insert(begin(), element);
}
void pop_back() {
    erase(iterator(mHeader->prev));
}
void pop_front() {
    erase(begin());
}

```

- Task is delegated to insert and erase
- Need iterator

Iterator

```

class list_iterator {
    friend class list;
protected:
    node* ptr;
public:

    list_iterator() : ptr( NULL ) { }
    list_iterator(node *a) : ptr(a) { }

    list_iterator& operator++() {
        ptr = ptr->next;
    }
}

```

```
    return (*this);
}
```

```
list_iterator& operator--() {
    ptr = ptr->prev;
    return (*this);
}
```

- Has custom constructor that takes node pointer

```
list_iterator& operator++(int) {
    list_iterator tmp(*this);
    operator++;
    return tmp;
}
```

```
list_iterator& operator--(int) {
    list_iterator tmp(*this);
    operator--;
    return tmp;
}
```

- operator++() is an operator for ++it
- operator++(int) is a syntax for it++
- operator++(int) delegates to operator++()
- Same for operator--()

```
T& operator*() { return ptr->data; }
T* operator->() { return &(ptr->data); }
```

```
bool operator==(const list_iterator& other) {
    return other.ptr == ptr;
}
```

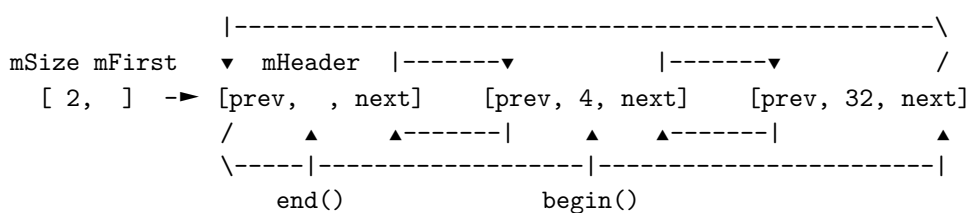
```
bool operator!=(const list_iterator& other) {
    return other.ptr != ptr;
}
};
```

Other small functions

```
iterator begin() {
    return iterator(mHeader->next);
}
```

```
iterator end() {
    return iterator(mHeader);
}
```

```
void clear() {
    while (mSize > 0) erase(begin());
}
```



Insert & Erase

```
iterator insert(iterator it, const T& element) {
    node *n = new node(element, it.ptr->prev, it.ptr);
    it.ptr->prev->next = n;
    it.ptr->prev = n;
    mSize++;
    return iterator(n);
}
```

```
iterator erase(iterator it) {
    iterator tmp(it.ptr->next);
    it.ptr->prev->next = it.ptr->next;
    it.ptr->next->prev = it.ptr->prev;
    delete it.ptr;
    mSize--;
    return tmp;
}
```

- Header make insert / erase very simple

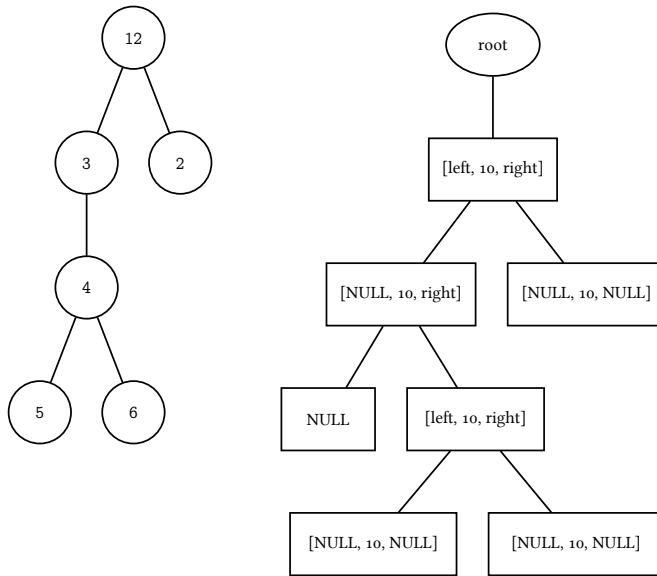
Lecture 15**Binary Tree**

Practicing Pointer & Recursive

Overview

- This is a basic for the next data structure, Binary Search and AVL Tree
- Focus on using Node and Pointer
- Focus on using recursive programming
- Some applications using just Binary Tree
- There is no data in std that is Binary Tree

Binary Tree & Node



- A rooted tree where each node have at most two children
- Tree Node is very similar to a linked list node

```
node
    left data right
    NULL 10 NULL

class node {
public:
    ValueT data;
    node *left, *right;
    node() :
        data( ValueT() ), left( NULL ), right( NULL ) { }
    node(const ValueT& data, node* left, node* right) :
        data( data ), left( left ), right( right ) { }
};
```

Node with parent link

```
class node {
public:
    ValueT data;
    node *left, *right, *parent;
    node() :
        data( ValueT() ), left( NULL ), right( NULL ), parent( NULL ) { }
    node(const ValueT& data, node* left, node* right, node* parent) :
        data( data ), left( left ), right( right ), parent( parent ) { }
};
```

- Sometimes, we need a link to parent
- Root is the only node that parent is NULL

String Encoding

Huffman Coding: Example Application of Tree

- David Huffman proposed this as his term project in Robert Fano's class (co-worker of Claude Shannon) which beats Shannon-Fano encoding
- Encoding = associate meaning to a representation

- ASCII Code
 - Fix length encoding
 - Each char = 8 bits

Variable Length Encoding

Never gonna give you up

16 different characters

Never gonna let you down

Fix-length needs $4 \times 86 = 344$ bits

Never gonna run around and desert you

Variable Length need 327 bits

n	e	o	u	r	a	v	g	d	y	t	w	s	p	l	i
14	11	9	7	7	6	5	5	5	4	3	2	2	2	2	2
0000	0001	0010	0011	0100	0101	0110	0111	1000	1001	1010	1011	1100	1101	1110	1111
11	010	011	0001	0011	0000	1011	1010	1000	0010 1	1001 1	1001 01	0010 001	0010 01	1001 00	0010 000

Encoding "never"

Fix-length 00000001011000010100

Variable-length 1101010110100011

Huffman Coding

Problem Statement

- Input: a string
- Output: encoding of each character in the string such that
 - The total length of encoding is minimum
 - The encoding of each character is *not ambiguous*
 - Any character encoding is not a prefix of any other character

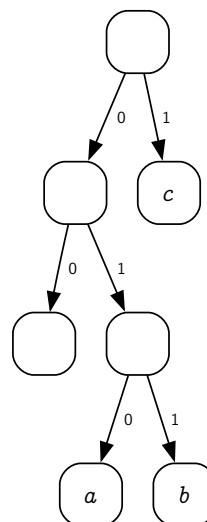
Tree Encoding

- Using a **tree** to represent encoding
- Each character is represented at **leaf nodes**
 - **Leaf node** is a node without children
- Encode by start at the root and **walk toward leaf nodes**
 - The path gives the encoding
 - Going to left child equal to 0
 - Going to right child equal to 1
- Guaranteed to be non-ambiguous

a = 010

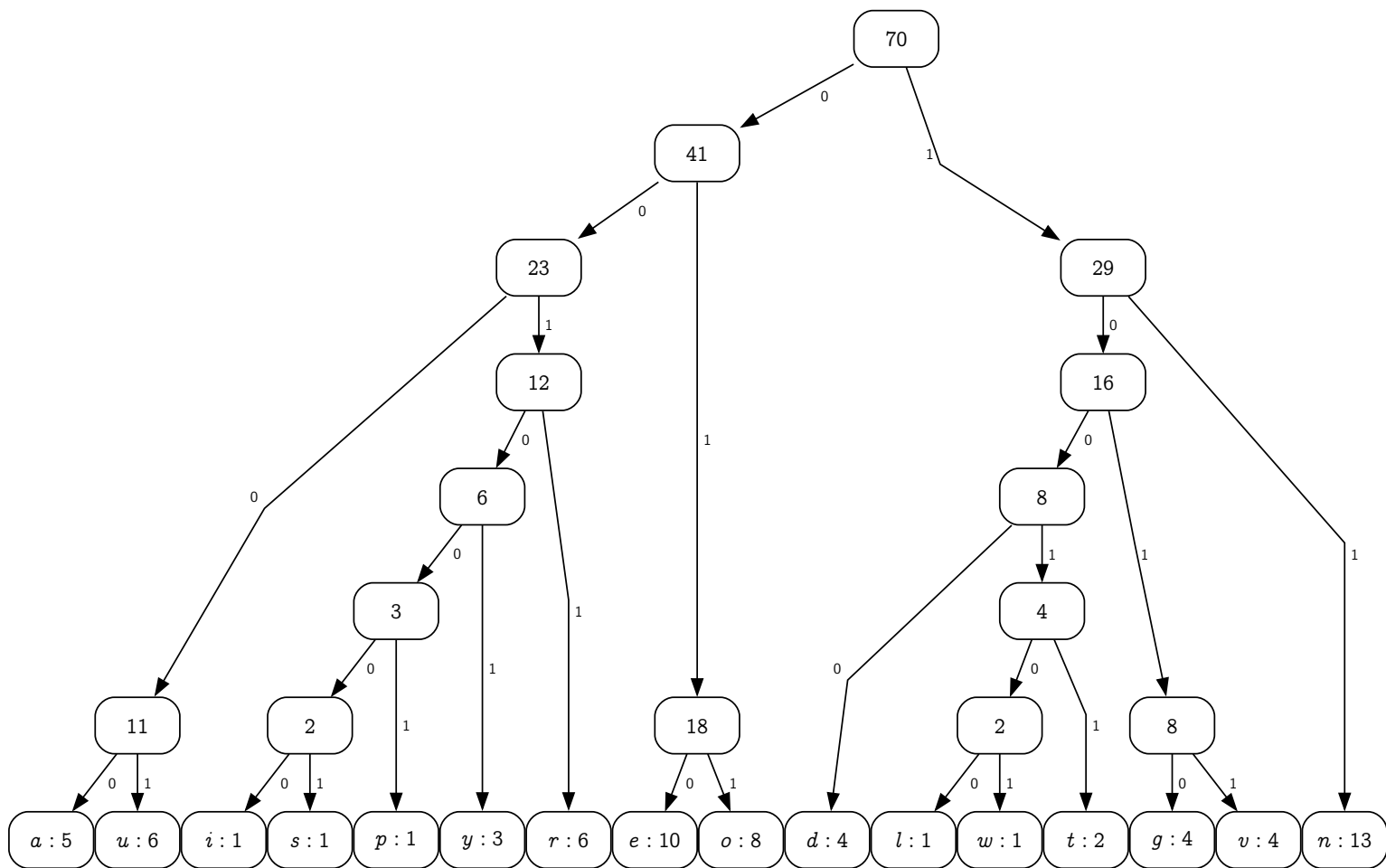
b = 011

c = 1



Huffman Tree

- Find 2 min nodes
- Create a new node with those two nodes as children, set freq equal to summation
- Repeat until only one node left



Huffman Tree

Huffman Tree Node

- Instead of data, we have both character and frequency
- Since we have to pick two nodes with minimum freq, we overload operator< to do so and use priority_queue

Huffman Code: Node

```
class huffman_tree {
protected:
    class huffman_node {
    public:
        char letter;
        int freq;
        huffman_node *left, *right;
        huffman_node() : letter('*'), freq(0), left(NULL), right(NULL) {}
        huffman_node(char letter, int freq, huffman_node *left, huffman_node *right) :
            letter(letter), freq(freq), left(left), right(right) {}

        bool is_leaf() { return left == NULL && right == NULL; }
    };
};
```



```

class node_comparator {
public:
    bool operator()(const huffman_node *a, huffman_node *b) {
        return a->freq > b->freq;
    }
};

```

Huffman Code: Build Tree

```

class huffman_tree {
protected:
    huffman_node *root;
    void build_tree(vector<huffman_node*> data) {
        priority_queue<huffman_node*, vector<huffman_node*>, node_comparator> pq;
        for (auto &x : data) pq.push(x);
        while (pq.size > 1) {
            huffman_node *right = pq.top(); pq.pop();
            huffman_node *left = pq.top(); pq.pop();
            pq.push(new huffman_node('*', left->freq+right->freq, left, right));
        }
        root = pq.top();
    }
public:
    huffman_tree(string s) {
        map<char, int> count;
        for (auto &c : s) {
            count[c]++; // word count
        }
        vector<huffman_node*> nodes;
        for (auto &x : count)
            nodes.push_back(new huffman_node(x.first, x.second, NULL, NULL));
        build_tree(nodes);
    }
};

```

Recursive Programming

Calling itself

Recursive

- A function that call itself
- Must have some input, usually via function argument
- The function must check a condition for execution
 - Result in either **terminating case** where the function won't call itself
 - or **recursion case** where the function will call itself **with different parameters**

```

// calculate sum 0..n
int recur1(int n) {
    if (n <= 0) {
        // terminating case
        return 0;
    } else {
        // recursion case
        return recur1(n-1) + n;
    }
}

```

Why recursion?

- Much simpler code
 - When the task is right
 - Recursion is natural for several mathematical model that is recursi
- Comparing to a normal loop, recursion has the same growth rate but recursion might take more time because function call is costlier than a loop

More Example

```
void print_range1(int step, int goal) {
    if (step < goal) {
        std::cout << step << " ";
        print_range1(step+1, goal);
    }
}

void print_range2(int step, int goal) {
    if (step < goal) {
        print_range2(step+1, goal);
        std::cout << step << " ";
    }
}
```

- Terminating Case do nothing
- Which is the output of print_range1(0,5) and print_range2(0,5)

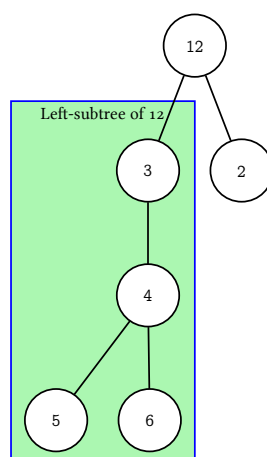
```
1. 0 1 2 3 4 5
2. 0 1 2 3 4
3. 5 4 3 2 1 0
4. 4 3 2 1 0
```

Binary Tree Recursive Definition

- A Binary Tree is
 - A tree with no nodes (root is NULL)
 - A tree with a root
 - Both children of the root must be a binary tree
 - Each child is called left-subtree and right-subtree
- Since binary tree can be defined recursively, operation on a binary tree can be naturally written as a recursion

Subtree

- For any node
 - its left (or right) child and all of the child's descendants is called left-subtree (or right-subtree)



Recursion on Binary Tree

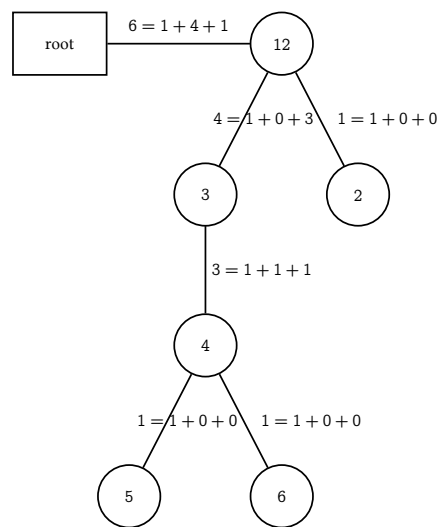
Tree Size by Recursion

- An empty tree has 0 nodes

- A tree with a root has 1 node (the root)
 - Plus the size of its two subtrees
- Easily written as recursive

```
class node {
public:
    int data;
    node *left, *right;
};

int get_size(node* n) {
    if (n == NULL) return 0;
    return 1 + get_size(n->left) + get_size(n->right);
}
```



Tree Height

- Height of a tree is the number of link we have to go to reach it deepest children
- Empty tree has height -1
- Height of a tree is 1 + max of height of its children

```
class node {
public:
    int data;
    node *left, *right;
};

int get_height(node* n) {
    if (n == NULL) return 0;
    return 1 + std::max(get_height(n->left),
                        get_height(n->right));
}
```

Tree Copy

```
class node {
public:
    int data;
    node *left, *right;
}
```

```

    node() : data(0), left(NULL), right(NULL);
    node(int data, node *left, node *right)
        : data(data), left(left), right(right);
};

node* copy(node *n) {
    if (n == NULL) return NULL;
    node *lc = copy(n->left);
    node *rc = copy(n->right);
    node *result = new node(n->data, lc, rc);
}

```

Walk over a tree

- Visiting all nodes (and maybe do something)

```

// Preorder traversal
void preorder(node *n) {
    if (n == NULL) return;
    std::cout << n->data << " ";
    preorder(n->left);
    preorder(n->right);
}

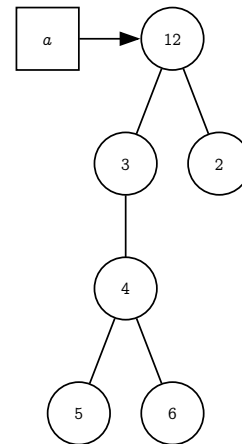
// Inorder traversal
void inorder(node *n) {
    if (n == NULL) return;
    inorder(n->left);
    std::cout << n->data << " ";
    inorder(n->right);
}

```

```

// Postorder traversal
void postorder(node *n) {
    if (n == NULL) return;
    postorder(n->left);
    postorder(n->right);
    std::cout << n->data << " ";
}

```



What is the result of

- preorder(a)
 - 12 3 4 5 6 2
- inorder(a)
 - 5 4 6 3 12 2
- postorder(a)
 - 5 6 4 3 2 12

Huffman Tree: Encoding

```

class huffman_tree {
protected:
    class huffman_node { };
    class node_comparator { };
    huffman_node *root;
public:
    void print(huffman_node *n, string s) {
        if (n->is_leaf()) {
            cout << n->letter << ": " << s << endl;
        } else {
            print(n->left, s+"0");
            print(n->right, s+"1");
        }
    }

    void print() {

```

```

    print(root, "");
}
};

```

- Recursive printing
- Use `s` to store path

```

class huffman_tree {
protected:
    class huffman_node { };
    class node_comparator { };
    huffman_node *root;

    void delete_node(huffman_node *n) {
        if (n == NULL) return;
        delete_node(n->left);
        delete_node(n->right);
        delete n;
    }

public:
    ~huffman_tree() {
        delete_node(root);
    }
};

```

- Recursive delete node
- Use postorder traversal
- Can we use inorder or preorder?

Lecture 16

Binary Search Tree

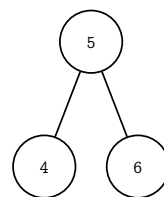
Binary Tree with value condition

Overview

- We add additional **value constraint** to a Binary Tree
- The constraint make finding data in the tree much faster
 - $\mathcal{O}(h)$ where h is the height of the tree
 - The tree is expected to have h be in $\mathcal{O}(\lg n)$, but this is not always true
 - The next tree (AVL tree) will add more constraint so that we can guarantee that $h = \mathcal{O}(\log n)$
- Using the same approach as a binary heap, maintain the constraint during modification

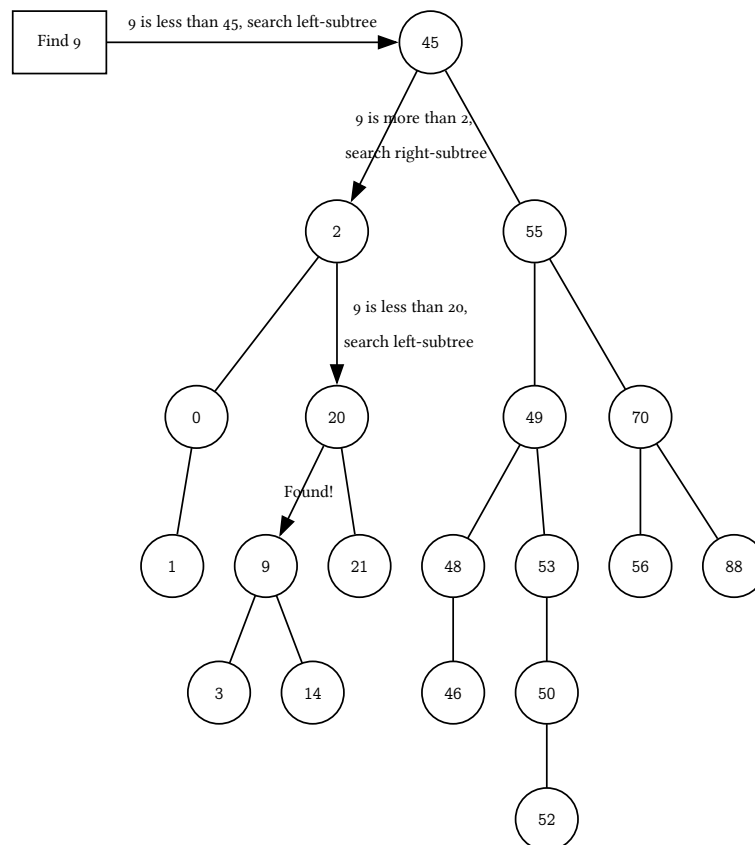
Binary Search Tree

- Structure rule: must be a Binary Tree
- Value rule: for any node `x`
 - data in **left-subtree** must be **less than** the data in `x`
 - data in **right-subtree** must be **more than** the data in `x`
- Recursive Definition
 - An **empty tree** is a Binary Search Tree (BST)
 - A node `X` is a BST when
 - Its subtrees (if any) must be BST and
 - If **left-subtree** exists, `X->data` must be more than `X->left->data`
 - If **right-subtree** exists, `X->data` must be less than `X->right->data`



Finding Value in BST

- Value rules make finding fast
- To find **e**, start from root
 - If the current node is not **e**,
 - search in left-subtree if **e** is **less** than the **current node**
 - search in right-subtree if **e** is **more** than the **current node**
 - Keep going until we find **e** or reach **NULL**
- Other operation is also depends on **find**



Insert & Erase with 0, 1 child

Find Node

```

node* find_node(const ValueT& k, node* r, node* &parent) {
    node *ptr = r;
    while (ptr != NULL) {
        int cmp == compare(k, ptr->data); // return -1 if a < b, 1 if a > b, 0 if equal
        if (cmp == 0) return ptr;
        parent = ptr;
        ptr = cmp < 0 ? ptr->left : ptr->right;
    }
    return NULL;
}

```

Insert

- Assumption: Data in BST is unique
- **insert(e)** by find **e**
 - If **e** is found, don't add any node

- If **e** is not in BST, find must reach **NULL** somewhere, that **NULL** is where to put **e**
- Both structure and value constraints are satisfied

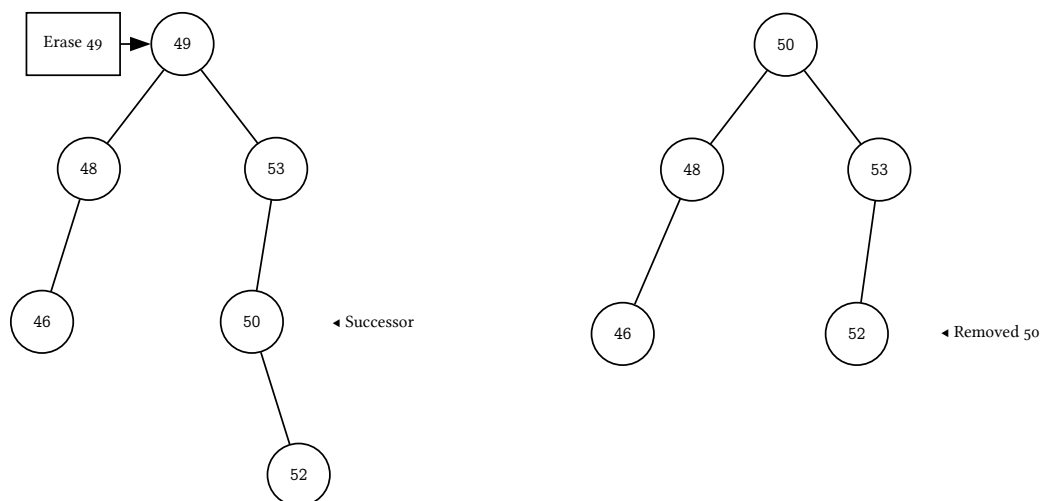
Erase

- **erase(e)** first have to find **e** as well
- If not found, do nothing
- If found at node **X**, there are 3 cases depend on the number of children of **e**
 - If has **no child**, just simply delete **X**
 - If has **one child**, have parent of **X** points (using the same link) to the child of **X** instead
 - If has **two children**, pick either **successor** or **predecessor** of **e**
 - Assume we choose **successor p** (must be in right-subtree), replace **X** with **p** and **erase(p)** from right subtree

Erase with 2 children

Erase node with 2 children

- Replace by **successor** (or **predecessor**) preserves value rules
 - **Successor** is the minimum in the **right-subtree**
 - **Predecessor** is the maximum in the **left-subtree**
- Both exists (because the node has both subtrees)



Find min max

Finding Successor and Predecessor

- If a tree has **left-subtree**, min is the min of **left-subtree**
 - If not, min is the root

```
node* find_min_node(node* r) {
    // r must not be NULL
    node *min = r;
    while (min->left != NULL) {
        min = min->left;
    }
    return min;
}
```

- If a tree has **right-subtree**, max is the max of **right-subtree**
 - If not, max is the root

```
node* find_max_node(node* r) {
    // r must not be NULL
    node *max = r;
    while (max->right != NULL) {
        max = max->right;
    }
    return max;
}
```

Finding Successor and Predecessor (recursive)

```

node* find_min_node(node* r) {
    // r must not be NULL
    if (r->left == NULL) return r;
    return find_min_node(r->left);
}

node* find_max_node(node* r) {
    // r must not be NULL
    if (r->right == NULL) return r;
    return find_max_node(r->right);
}

```

Complexity Analysis

- insert, erase depends on `find`, `find_min` (or `find_max`)
- All finds start from root and in the worst case reach the leaf
 - Hence, $\mathcal{O}(h)$
- Height of the tree can be in the range from $\lg n$ to n
- For 1,000,000 nodes, it's in the range of [20, 999999]
 - $\mathcal{O}(h)$ is, right now, $\mathcal{O}(n)$
 - Will be fixed by AVL tree

CP::map_bst (insert)

Using Binary Search Tree to create associated data structure

Layout

- Need node class
- Also need iterator class
- Template has two types
 - Key Type and Mapped Type
 - ValueType is `pair<KeyType, MappedType>`
- Also need custom comparator

```

template <typename KeyT,
          typename MappedT,
          typename CompareT = std::less<KeyT> >
class map_bst {
protected:
    typedef std::pair<KeyT, MappedT> ValueT;
    class node {
        friend class map_bst;
    protected:
        ValueT data;
        node *left;
        node *right;
        node *parent;
    };
    class tree_iterator {
    protected:
        node* ptr;
    public:
    };
    node *mRoot;
    CompareT mLess;
    size_t mSize;
public:
    typedef tree_iterator iterator;
};

```

Node class

- Data stores both the key type and mapped type (as a pair)

- Map finds by key

```
class node {
    friend class map_bst;
protected:
    ValueT data;
    node *left;
    node *right;
    node *parent;

    node() :
        data( ValueT() ), left( NULL ), right( NULL ), parent( NULL ) { }

    node(const ValueT& data, node* left, node* right, node* parent) :
        data( data ), left( left ), right( right ), parent( parent ) { }
};
```

Ctors, Dtor

```
map_bst(const map_bst<KeyT, MappedT, CompareT> & other) :
    mLess(other.mLess), mSize(other.mSize)
{ mRoot = copy(other.mRoot, NULL); } // recursive copy

map_bst(const CompareT& c = CompareT() ) :
    mRoot(NULL), mLess(c), mSize(0)
{ }

map_bst<KeyT, MappedT, CompareT>& operator=(map_bst<KeyT, MappedT, CompareT> other) {
    using std::swap;
    swap(this->mRoot, other.mRoot);
    swap(this->mLess, other.mLess);
    swap(this->mSize, other.mSize);
    return *this;
}

~map_bst() {
    clear(); // recursive delete
}
```

Actual Find

```
iterator find(const KeyT &key) {
    node *parent;
    node *ptr = find_node(key, mRoot, mParent);
    return ptr == NULL ? end() : iterator(ptr);
}
```

- Find by Key

```
int compare(const KeyT& k1, const KeyT& k2) {
    if (mLess(k1, k2)) return -1;
    if (mLess(k2, k1)) return 1;
    return 0;
}

node* find_node(const KeyT& k, node* r, node* &parent) { // modify parent pointer
    node *ptr = r;
    while (ptr != NULL) {
        int cmp = compare(k, ptr->data.first);
```

```

    if (cmp == 0) return ptr;
    parent = ptr;
    ptr = cmp < 0 ? ptr->left : ptr->right;
}
return NULL;
}

```

Insert

- insert returns pair of iterator and insert result

```

std::pair<iterator, bool> insert(const ValueT& val) {
    node *parent = NULL;
    node *ptr = find_node(val.first, mRoot, parent);
    bool not_found = (ptr == NULL);
    if (not_found) {
        ptr = new node(val, NULL, NULL, parent);
        child_link(parent, val.first) = ptr;
        mSize++;
    }
    return std::make_pair(iterator(ptr), not_found);
}

```

child_link returns a reference (the variable) to the pointer of the appropriate child of the parent with respect to `k`

```

node* &child_link(node* parent, const KeyT& k) {
    if (parent == NULL) return mRoot;
    return mLess(k, parent->data.first) ?
        parent->left : parent->right;
}

```

CP::map_bst (erase and iterator)

Erase

- Handle multiple cases

```

size_t erase(const KeyT &key) {
    if (mRoot == NULL) return 0; // empty tree
    node *parent = NULL;
    node *ptr = find_node(key, mRoot, parent);
    if (ptr == NULL) return 0; // not found
    if (ptr->left != NULL && ptr->right != NULL) {
        // have two children => erase min
        node *min = find_min_node(ptr->right); // successor
        // remove min like case 0 or 1 child
        node * &link = child_link(min->parent, min->data.first);
        link = (min->left == NULL) ? min->right : min->left;
        if (link != NULL) link->parent = min->parent;
        // swap min with ptr
        std::swap(ptr->data.first, min->data.first);
        std::swap(ptr->data.second, min->data.second);
        ptr = min; // we are going to delete this node instead
    } else {
        // case 0 or 1 child => erase ptr
        node * &link = child_link(ptr->parent, key); // find the pointer that holds ptr
        link = (ptr->left == NULL) ? ptr->right : ptr->left; // replace it with ptr child
        if (link != NULL) link->parent = ptr->parent; // if child exists, update parent
    }
}

```

```

    delete ptr;
    mSize--;
    return 1;
}

```

Operator[]

```

MappedT& operator[](const KeyT& key) {
    node *parent = NULL;
    node *ptr = find_node(key, mRoot, parent);
    if (ptr == NULL) {
        ptr = new node(std::make_pair(key, MappedT()), NULL, NULL, parent);
        child_link(parent, key) = ptr;
        mSize++;
    }
    return ptr->data.second;
}

```

- Find node
- If not exists, create one with default MappedTypeReturn MappedType of the node

Iterator

- Just like linked list, we need a class for iterator
 - Because we need custom operator++, -- (and some more)
- Iterator class just store a **pointer to a node**

```

class tree_iterator {
protected:
    node* ptr;

public:
    tree_iterator() : ptr( NULL ) { }
    tree_iterator(node *a) : ptr(a) { }
    // more functions below
};

```

Other Functions

```

tree_iterator operator++(int) { // ++it
    tree_iterator tmp(*this);
    operator++;
    return tmp;
}

```

```

tree_iterator operator--(int) { // --it
    tree_iterator tmp(*this);
    operator--;
    return tmp;
}

```

```

ValueT& operator*() { return ptr->data; }
ValueT* operator->() { return &(ptr->data); }
bool operator==(const tree_iterator& other)
{ return other.ptr == ptr; }
bool operator!=(const tree_iterator& other)
{ return other.ptr != other }

```

Operator++

- Find **successor** of **x**, easy if **x** have **right-subtree**

- ▶ Just find **min** of **right-subtree**
- If not, we have to go up (go toward root) until we find one that is **more** than **x**
 - ▶ This is always the closest **ancestor** of **x** that has **x** in its **left-subtree**

```
tree_iterator& operator++() {
    if (ptr->right == NULL) {
        node *parent = ptr->parent;
        while (parent != NULL &&
            parent->right == ptr) {
            ptr = parent;
            parent = ptr->parent;
        }
        ptr = parent;
    } else {
        ptr = ptr->right;
        while (ptr->left != NULL)
            ptr = ptr->left;
    }
    return (*this);
}
```

Summary

- Binary Search Tree relies on **Value Constraint** to make find fast
 - ▶ Possible to be slow (will be fixed later)
- Erase requires `find_min`, `find_max`
- `CP::map_bst` uses pair to store **KeyT** and **MappedT**
 - ▶ `find` uses key
- **Iterator** is just a pointer
 - ▶ Have a problem of **operator--** at `end()` (will be fixed later)

Lecture 17

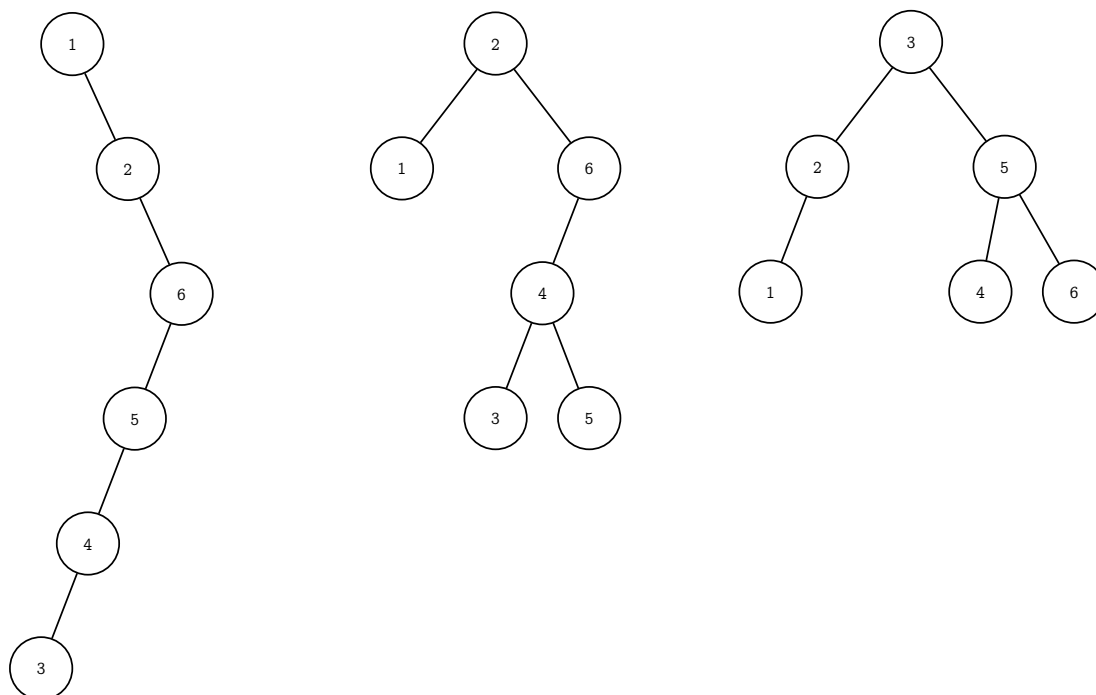
AVL Tree

หัวข้อ

- นิยามต้นไม้เอวีแอล
- การวิเคราะห์ความสูงของต้นไม้เอวีแอล
- การปรับต้นไม้เอวีแอลให้สูงสมดุล
- กระบวนการหมุนปม

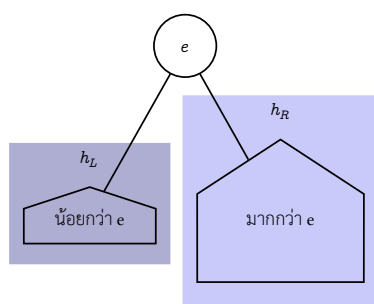
ต้นไม้ค้นหาแบบทวิภาค

- เวลาการทำงานเป็น $\mathcal{O}(h)$
- $\lfloor \log_2 n \rfloor \leq h \leq n - 1$
- โชคดีทำงานเร็ว $\mathcal{O}(\log n)$, โชคร้ายทำงานเป็น $\mathcal{O}(n)$



ต้นไม้เอวีแอล

- AVL = Binary Search Tree + กฎความสูงสมดุล

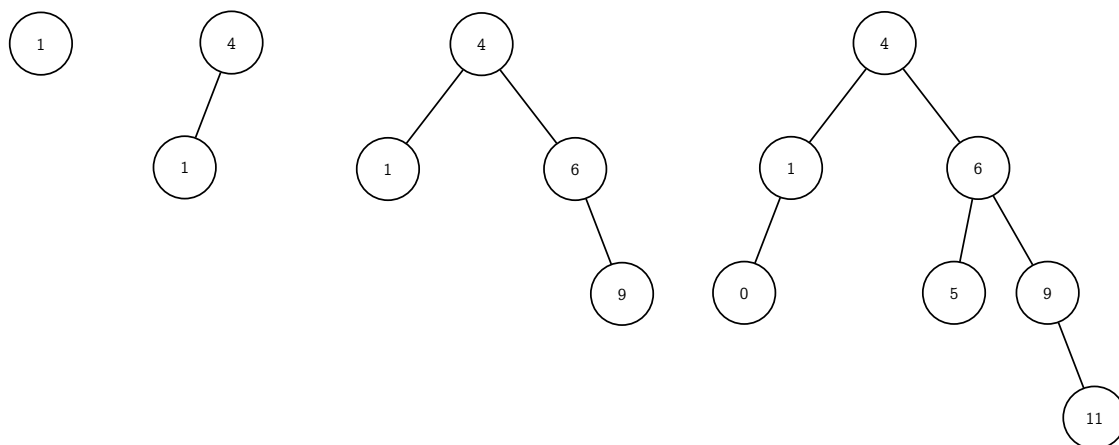


$$|h_L - h_R| \leq 1$$

ต้นไม้ย่อยทุกต้นต้องเป็นไปตามกฎ

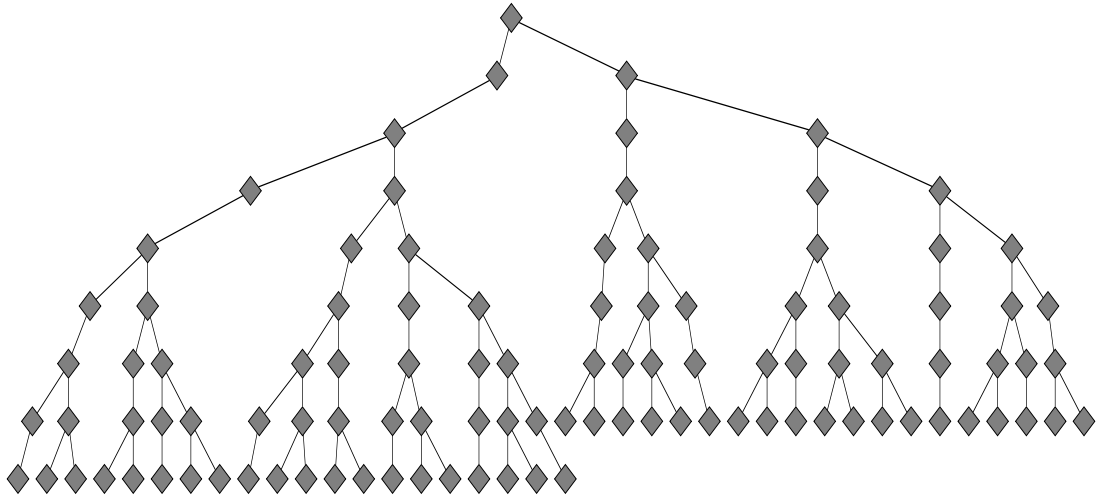
AVL: Adelson-Velskii and Landis

ตัวอย่างต้นไม้ไอบีแอล

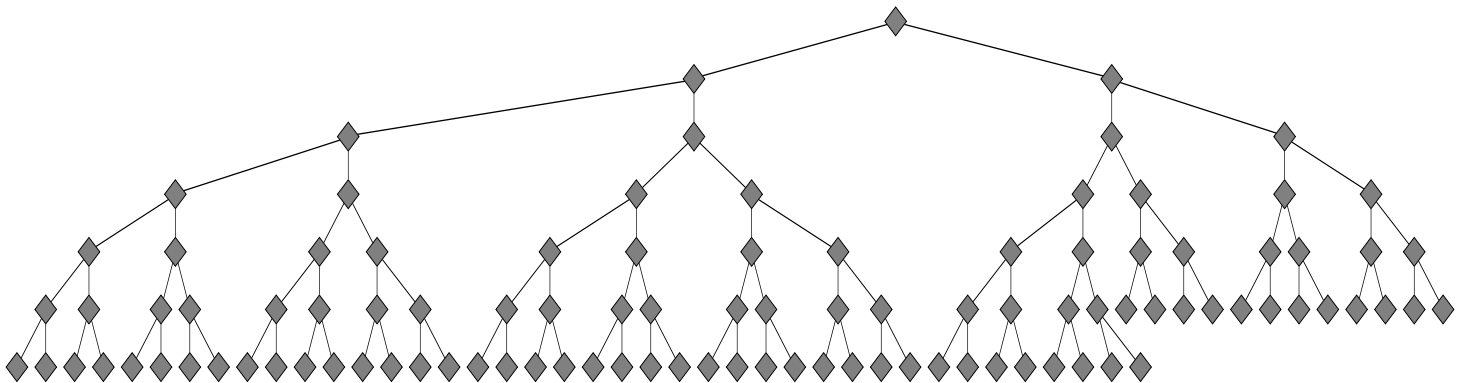


ต้นไม้ว่าง (null) สูง -1

ต้นไม้ค้นหาแบบทวิภาคเอวีแอล



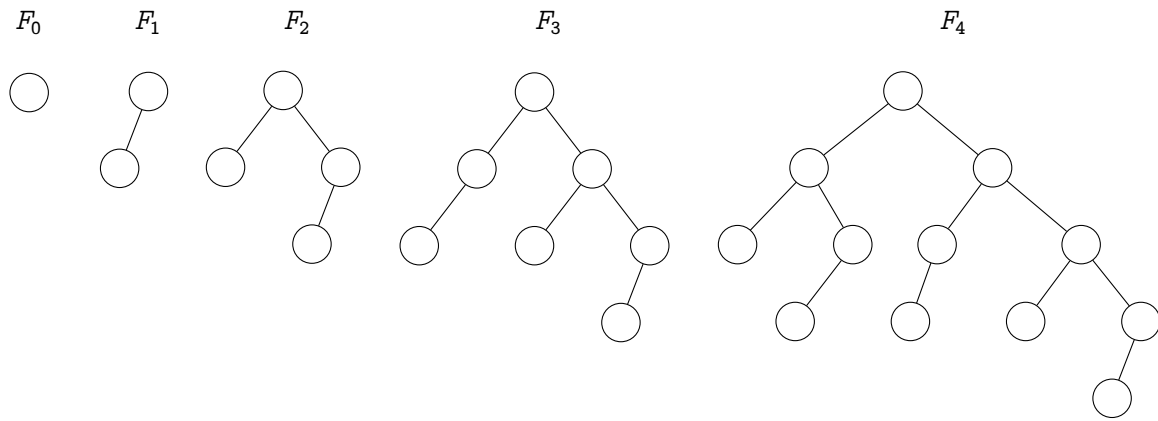
ต้นไม้ค้นหาแบบทวิภาคที่สร้างจากข้อมูลสุ่ม 100 ตัว



ต้นไม้เอวีแอลที่สร้างจากข้อมูลกลุ่ม 100 ตัว

ต้นไม้เอวีแอลสูงเท่าใด ?

- ให้ F_n คือต้นไม้เอวี่แอลซึ่งสูง n ที่มีจำนวนปมน้อยสุด



$$|F_h| = 1 + |F_{h-1}| + |F_{h-2}|$$

Fibonacci Tree

ความสูงของต้นไม้ฟีโบนัชชี

$$|F_h| = 1 + |F_{h-1}| + |F_{h-2}|$$

$$n_h = 1 + n_{h-1} + n_{h-2} \quad h \geq 2, \quad n_0 = 1, \quad n_1 = 2$$

$$n_h = \alpha_1 \phi^h + \alpha_2 \hat{\phi}^h - 1, \quad \phi = 1.618..., \quad \hat{\phi} = -0.618$$

$$n_h \approx \alpha_1 \phi^h$$

$$h \approx \frac{1}{\log_2 \phi} (\log_2 n_h)$$

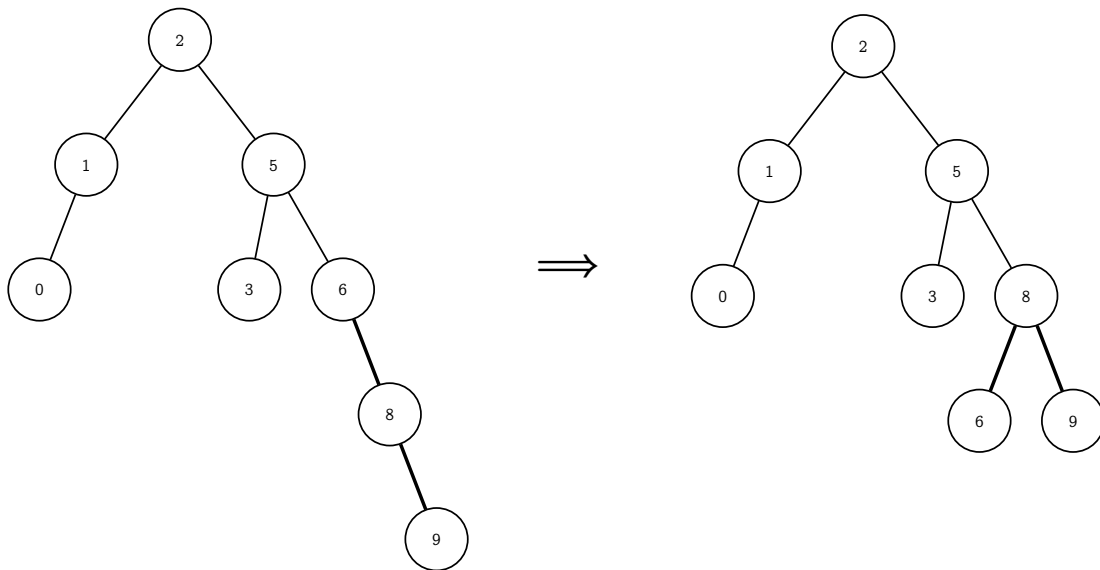
$$h \approx 1.44 (\log_2 n_h)$$

สรุป: ต้นไม้เอวี่แอลที่มี n ปม สูงไม่เกิน $1.44 \log_2 n$

AVL Rotation

ทำอย่างไรให้เป็นไปตามกฎของ AVL

- การเพิ่ม/ลบข้อมูลทำเหมือน BSTree
- แต่หลังการเพิ่ม/ลบ อาจทำให้ผิดกฎสูงสมดุล
- ถ้าผิดกฎต้องปรับต้นไม้



```

map_avl
template <typename KeyT,
          typename MappedT,
          typename CompareT = std::less<KeyT> >
class map_avl {
protected:
    class node {
        friend class map_bst;
        ...
    };

    class tree_iterator { // เหมือน map_bst
        ...
    };
public:
    ...
};

node
class node {
    friend class map_avl;
protected:
    ValueT data;
    node *left;
    node *right;
    node *parent;
    int height; // แต่ละบมมีความสูงกำกับ

    node() :
        data( ValueT() ), left( NULL ), right( NULL ),
        parent( NULL ), height(0) { }

    node(const ValueT& data, node* left,
         node* right, node* parent) : data(data),
        left(left), right(right), parent(parent) {
        set_height();
    }
};

```

```

}

int get_height(node *n) { // ไม้ ๐๐ ?
    return (n == NULL ? -1 : n->height);
}

void set_height() {
    int hL = get_height(this->left);
    int hR = get_height(this->right);
    height = 1 + (hL > hR ? hL : hR);
}

int balance_value() {
    return get_height(this->right) -
           get_height(this->left);
}

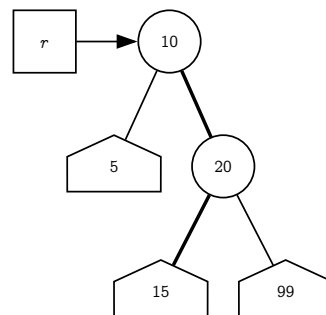
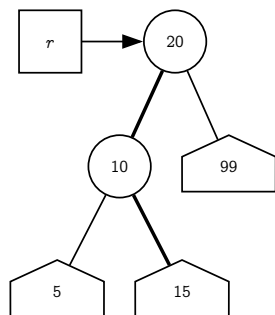
void set_left(node *n) {
    this->left = n;
    if (n != NULL) this->left->parent = this;
}

void set_right(node *n) {
    this->right = n;
    if (n != NULL) this->right->parent = this;
}
};

```

การหมุนปม

- การปรับต้นไม้อาศัยการหมุน (rotation)



`rotate_left_child(r)`
 $h_r = \max(2 + 5, 2 + 15, 1 + 99)$

`rotate_right_child(r)`
 $h_r = \max(1 + 5, 2 + 15, 2 + 99)$

CP::map_avl

rotate_left_child(r)

```

node *rotate_left_child(node *r) {
    node *new_root = r->left;
    r->set_left(new_root->right);
    new_root->set_right(r);

    new_root->right->set_height();
    new_root->set_height();
}

```

```

    return new_root;
}

```

rotate_right_child(r)

```

node *rotate_right_child(node *r) {
    node *new_root = r->right;
    r->set_right(new_root->left);
    new_root->set_left(r);

    new_root->left->set_height();
    new_root->set_height();
    return new_root;
}

```

insert และ erase ใช้ rebalance

```

node* insert(const ValueT& val, node *r, node * &ptr) {
    ... // same as insert in map_bst

    r = rebalance(r); // เพิ่มตามปกติ แล้วค่อยปรับ
    return r;
}

```

```

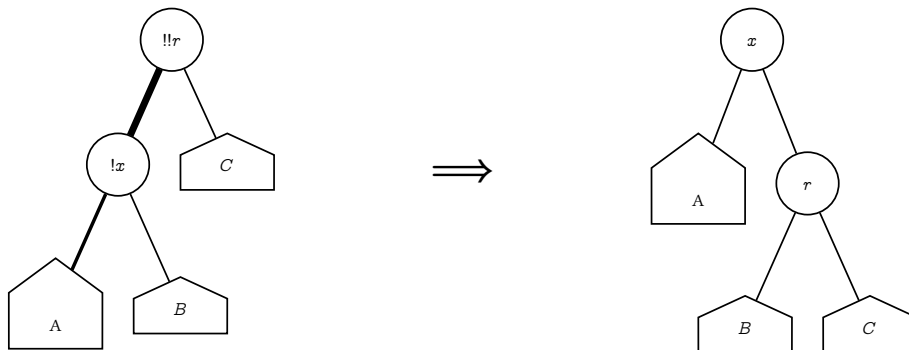
node *erase(const KeyT &key, node *r) {
    ... // same as erase in map_bst

    r = rebalance(r); // ลบตามปกติ แล้วค่อยปรับ
    return r;
}

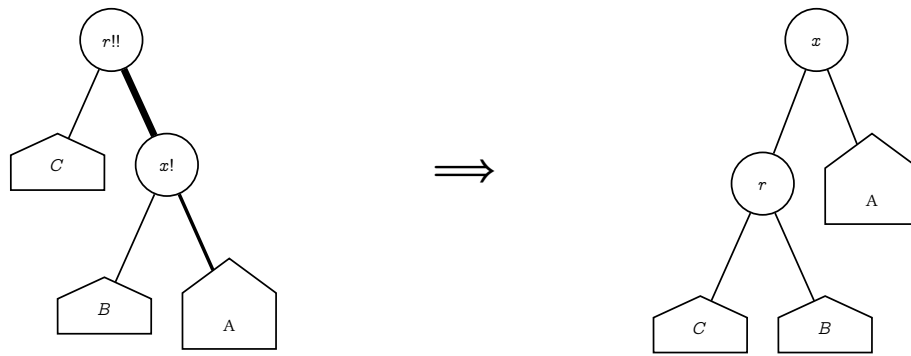
```

rebalance

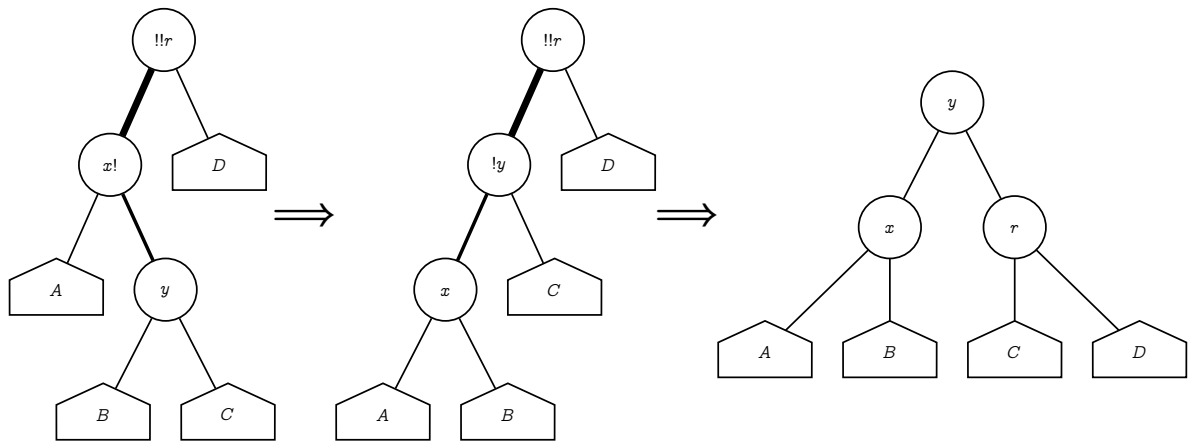
1. r เอียงซ้ายมากเกินไป และ x เอียงซ้าย $\Rightarrow$ rotate_left_child(r);



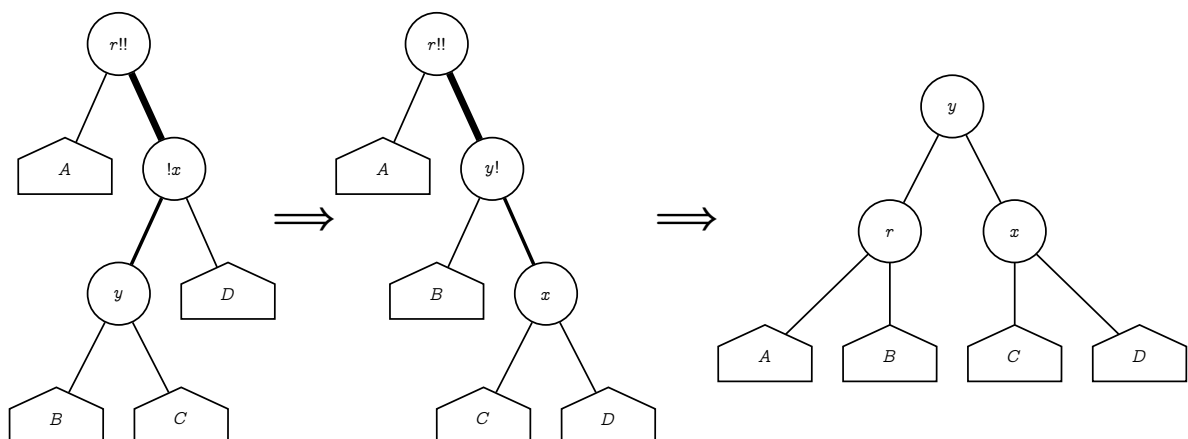
2. r เอียงขวาเกินไป และ x เอียงขวา $\Rightarrow$ `rotate_right_child(r);`



3. r เอียงซ้ายมากเกินไป และ x เอียงขวา $\Rightarrow$ `rotate_right_child(r->left); rotate_left_child(r);`



4. r เอียงขวาเกินไป และ x เอียงซ้าย $\Rightarrow$ `rotate_left_child(r->right); rotate_right_child(r);`



```
node *rebalance(node *r) {
    if (r == NULL) return r;
    int balance = r->balance_value();
```

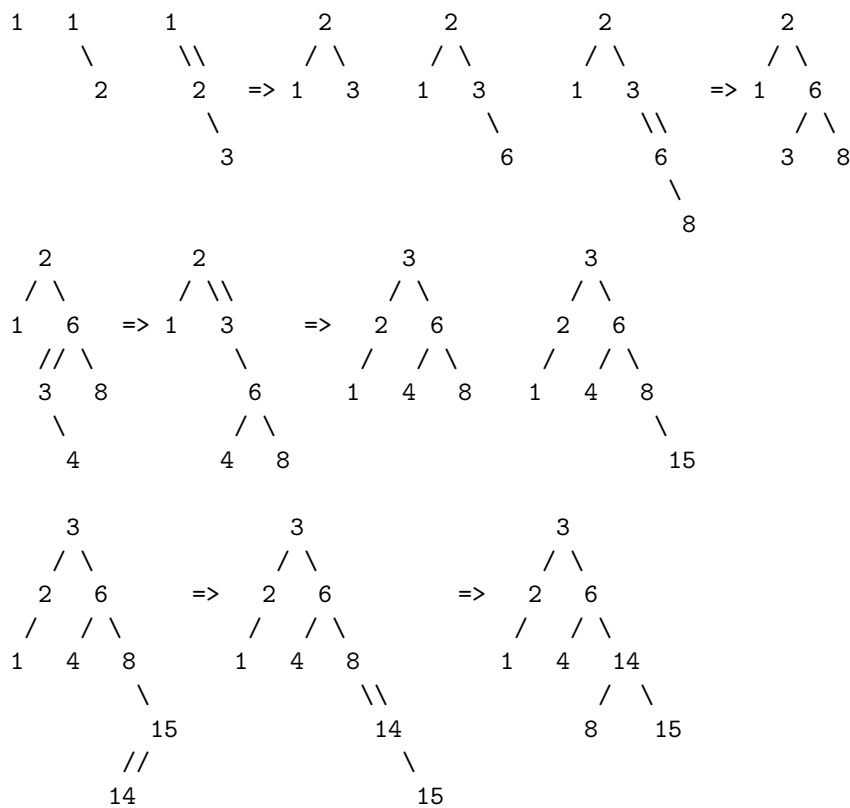
```

if (balance == -2) {
    if (r->left->balance_value() == 1) // case 3
        r->set_left(rotate_right_child(r->left));
    r = rotate_left_child(r);
} else if (balance == 2) {
    if (r->right->balance_value() == -1) // case 4
        r->set_right(rotate_left_child(r->right));
    r = rotate_right_child(r);
}
r->set_height();
return r;
}

```

ตัวอย่าง

1, 2, 3, 6, 8, 4, 15, 14



สรุป

- ต้นไม้เอวีแอลคือต้นไม้ค้นหาที่ถูกรักษาความสูง
- ผลต่างความสูงของลูกสองข้างห้ามเกินหนึ่ง
- พิสูจน์ได้ว่า $\lfloor \log_2 n \rfloor \leq h < 1.44 \log_2 n$
- แต่ละปมเก็บความสูงไว้ตรวจสอบ
- ถ้าหลังเพิ่ม/ลดข้อมูลแล้วผิดกฎให้ปรับต้นไม้
- การปรับต้นไม้อาศัยการหมุนปม
- เวลาการทำงานของเพิ่ม ลง และค้นหาเป็น $\mathcal{O}(\log n)$

ความสูงของต้นไม้ AVL

```

class node {
    friend class map_avl;
protected:
    ValueT data; // int, int 8

```

```

class node {
    friend class map_avl;
protected:
    ValueT data;

```

```

node *left;    // 4
node *right;   // 4
node *parent;  // 4
int height;    // 4
}

node *left;
node *right;
node *parent;
unsigned char height; // 1
}

```

นอกเรื่อง

- `std::map` ใช้ Red Black Tree
- self-balanced tree มี AVL, Red Black, Splay

Lecture 18

Hash Table

หัวข้อ

- การใช้ตารางเก็บข้อมูลด้วยฟังก์ชันดัชนี
- การเก็บข้อมูลแบบแยกกันโยง
- ฟังก์ชันแฮช
- กลวิธีการเขียนฟังก์ชันแฮช
- การแฮชใน C++
- การกำหนดเลขที่อยู่เปิด
- การเกาะกลุ่มของข้อมูล

ใช้ฟังก์ชันดัชนีคำนวณตำแหน่ง

- key ของข้อมูลคือส่วนของข้อมูลที่ใช้ในการค้น
- มีตารางซึ่งแต่ละช่องเป็นที่เก็บข้อมูล
- หา $f(\text{key})$ เพื่อแปลง key ไปเป็น index ของตาราง
- ฟังก์ชันดัชนีหาไม่ยาก ถ้าองตารางขนาดใหญ่

ฟังก์ชันดัชนีนั้นหายาก

- เมื่อต้องเก็บอย่างประหยัด
- เมื่อต้องประกันว่าไม่เกิดการ “ชน”

เปลี่ยนกลยุทธ์: อนุญาตให้ชนกันได้

- จะได้เก็บข้อมูลในตารางที่ไม่ใหญ่มาก
- แต่ต้องหาวิธีแก้ไขปัญหการชน ที่ทำงานได้เร็ว ๆ

Separate Chaining

- จัดเก็บข้อมูลที่ชนกันไว้ในรายการเดียวกัน

```

0 [ ] --> < 14, 7 >
1 [ ] --> < 22, 1 >
2 [ ] --||
3 [ ] --> < 3, 38 >      h(x) = x % 7
4 [ ] --> < 46 >
5 [ ] --> < 40 >
6 [ ] --> < 6, 13, 27, 20 >

```

การกระจายข้อมูล

- ถ้าข้อมูลกระจายทั่วตาราง
 - แต่ละช่องเก็บรายการยาว $\approx \lambda$
 - ถ้า λ น้อย ค้นหาได้เร็ว
- ถ้าไม่กระจาย

load factor

$$\lambda = n/m$$

- มีบางรายการยาวเกิน λ มาก
- การค้นหาซ้ำเหมือนเก็บด้วย list

```

0 [ ] --||
1 [ ] --> < 401, 321, 105, 185, 129 >
2 [ ] --||
3 [ ] --||
4 [ ] --||
5 [ ] --> < 180, 260, 60, 252, 92, 764,
6 [ ] --||           4, 76, 28, 108 >
7 [ ] --||
          h(x) = x % 8

```

n = ปริมาณข้อมูล
 m = ขนาดของตาราง

```

          h(x) = x % 10
0 [ ] --> < 180, 260, 80 >
1 [ ] --> < 401, 321 >
2 [ ] --> < 252, 92 >
3 [ ] --||
4 [ ] --> < 764, 4 >
5 [ ] --> < 105, 185 >
6 [ ] --> < 76 >
7 [ ] --||
8 [ ] --> < 28, 108 >
9 [ ] --> < 129 >

```

Hash Function

ฟังก์ชันแฮช (Hash Function)

- www.webster.com
 - **hash**: to chop (as meat and potatoes) into small pieces
- สอ เสถบุตร
 - สับ แหวก นำมาโขลกเข้าด้วยกัน

ตัวอย่างฟังก์ชันแฮช

```

size_t h1(size_t x) {
    return (2654435769U * x) >> 22;
}

```

```

size_t h2(size_t x) {
    x = ~x + (x << 15);
    x ^= (x >> 11);
    x += (x << 3);
    x ^= (x >> 5);
    x += (x << 10);
    x ^= (x >> 16);
    return x & 0x3FF;
}

```

x	1	2	3	4	5	6	7	8
h1(x)	632	241	874	483	92	725	334	966
h2(x)	500	1001	507	978	486	1014	403	933

การวิเคราะห์เลขโดด (Digit Analysis)

- คัดเลือกเลขโดดบางหลักของคีย์มาพิจารณา
- มั่นใจว่าที่ตัดไปไม่ทำให้เกิดความเอนเอียงในการกระจายของคีย์
- เช่น
 - รหัสนิติศึกษา ป.ตรี มีรูปแบบ: xx3xxxxx21
 - ก็ตัดเลข 3 และ 21 ออกจากการพิจารณา
 - $k = 4830109521$
 - $k_1 = \lfloor k / 100 \rfloor$ // $k_1 = 48301095$
 - $k_2 = \lfloor k_1 / 100 \rfloor$ // $k_2 = 48$
 - $k_3 = k_2 * 10^5 + k_1 \% 10^5$ // $k_3 = 4801095$

การคูณ (Multiplicative Hashing)

- คูณคีย์ด้วยจำนวนจริง A ที่มีค่าระหว่าง $(0, 1)$
- นำเศษมาคูณกับขนาดของตาราง ($m = 2^p$)

$$h(x) = \lfloor m((xA - \lfloor xA \rfloor)) \rfloor$$

การคูณ: Fibonacci Hashing

- ถ้า A = golden ratio 0.6180339887 จะแยกคีย์ที่มีค่าใกล้เคียงกันออกจากกันได้ดี

```
size_t multHash(size_t x, size_t p) {
    size_t s = 2654435769U; // 0.6180339887*2^32
    size_t hash = (s * x);
    return (hash >> (32-p));
}
```

```
for (size_t i = 0; i < 10; i++) {
    cout << multHash(i, 10) << ", ";
}
```

0, 632, 241, 874, 483, 92, 725, 334, 966

การพับ (Folding)

- แบ่งคีย์ออกเป็นส่วน ๆ แล้วนำมา “รวม” กัน
- “รวม” $\equiv$ บวก, xor, ...

การหาร (Modulus Hashing)

- $h(x) = x \% p$
- ไม่ควรเลือก
 - ▶ $p = 10^q$ เพราะเลือกเฉพาะ q หลักขวา ถ้าคีย์เป็นหลักสิบ
 - ▶ $p = 2^q$ เพราะเลือกเฉพาะ q บิตขวา
 - ▶ p ที่มีค่าน้อย ๆ เป็นตัวประกอบ
 - ถ้า c คือตัวประกอบร่วมของ p และ x
 - ค่า $x\%p$ จะเป็นจำนวนเท่าของ c
 - ถ้า c มีค่าน้อย ๆ จะมีคีย์จำนวนมากที่ได้ $x\%p$ มีค่าเป็นจำนวนเท่าของตัวประกอบนั้น ซึ่งไม่กระจาย
 - ▶ โดยทั่วไปเลือก p ที่เป็นจำนวนเฉพาะ

c++11 std::unordered_map

```
#include <iostream>
#include <unordered_map>

using namespace std;

int main() {
    unordered_map<string, int> facultyCode;
    facultyCode["engineering"] = 21;
    facultyCode["accounting"] = 26;
    facultyCode["science"] = 23;

    cout << facultyCode["engineering"] << endl;
    cout << facultyCode["science"] << endl;
    cout << facultyCode["communication"] << endl;

    return 0;
}
```


c++11 std::hash

```
#include <functional>
using namespace std;
int main() {
    hash<string> hStr;
    hash<float> hFloat;
    hash<int> hInt;

    cout << hStr("C++") << endl; // 2262514926
    cout << hFloat(1.2f) << endl; // 2462087341
    cout << hInt(123) << endl; // 123

    return 0;
}

cout << hash<string>()("C++") << endl;
cout << hash<float>()(1.2f) << endl;
cout << hash<int>()(123) << endl;
```

อยากใช้ Book เป็นคีย์

```
class Book {
public:
    string title;
    int edition;
    double price;

    Book(string title, int ed = 1, double price = 199.0) :
        title(title), edition(ed), price(price)
    {}
    ...
};
```

เขียน Hasher class, Equal class

```
class BookHasher {
public:
    size_t operator()(const Book& b) const {
        return hash<string>()(b.title) ^ hash<int>()(b.edition);
    }
};

class BookEqual {
public:
    bool operator()(const Book& b1, const Book b2) const {
        return b1.title==b2.title && b1.edition==b2.edition;
    }
};

unordered_map<Book,string,BookHasher,BookEqual> m;
m[Book("Data Structures", 1, 200)] = "reserved";
m[Book("Algorithm", 5, 200)] = "available";

Book b1("Data Structures, 1");
Book b2("algorithm", 5);
```

```
cout << m[b1] << endl; // "reserved"
cout << m[b2] << endl; // ""
```

ปฏิกิริยาวันเกิด (Birthday Paradox)

- ต้องมีคนในห้องกี่คนขึ้นไป จึงจะเกิดโอกาสเกินครึ่งที่จะมีคนเกิดวันเดียวกันสองคนขึ้นไป

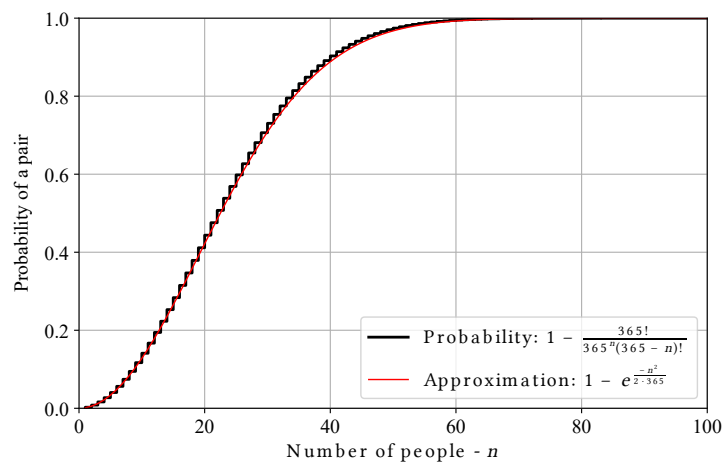
$$1 \text{ คน โอกาสที่มีวันเกิดไม่ซ้ำกัน} = \left(\frac{366}{366}\right)$$

$$2 \text{ คน โอกาสที่มีวันเกิดไม่ซ้ำกัน} = \left(\frac{366}{366}\right) \left(\frac{365}{366}\right)$$

$$3 \text{ คน โอกาสที่มีวันเกิดไม่ซ้ำกัน} = \left(\frac{366}{366}\right) \left(\frac{365}{366}\right) \left(\frac{364}{366}\right)$$

$$k \text{ คน โอกาสที่มีวันเกิดไม่ซ้ำกัน} = \left(\frac{366}{366}\right) \left(\frac{365}{366}\right) \left(\frac{364}{366}\right) \dots \left(\frac{366-k+1}{366}\right)$$

$$1 - \left(\frac{366}{366}\right) \left(\frac{365}{366}\right) \left(\frac{364}{366}\right) \dots \left(\frac{366-k+1}{366}\right) > 0.5$$



CP::unordered_map

```
CP::unordered_map<KeyT, MappedT>
bucket count = 10
[ ] -> [ * , * ] <-- bucket size = 2
[ ] -> [ * , * ]
[ ] -> [ * , * ]
[ ] -> [ * , (*) ] -> pair<key, mappedValue>
[ ] -> [ * , * ]
[ ] -> [ * , * ]
[ ] -> [ ] <-- bucket size = 0
[ ] -> [ ]
[ ] -> [ * , * , * ]
[ ] -> [ * , * , * ]
  ▲          ▲ vector< ValueT > = BucketT
  |
vector< vector< ValueT > > = vector< BucketT >

typedef pair<KeyT, MappedT> ValueT;
typedef vector< ValueT > BucketT;
```

Constructors

```
template <typename KeyT,
          typename MappedT,
```

```

        typename HasherT = std::hash<KeyT>,
        typename EqualT = std::equal_to<KeyT> >
class unordered_map {
protected:
    typedef std::pair<KeyT, MappedT> ValueT;
    typedef std::vector<ValueT> BucketT;
    ...

    std::vector<BucketT> mBuckets;
    size_t mSize;
    HasherT mHasher; // ใช้ในการเปรียบเทียบ ระหว่างการค้น Key
    EqualT mEqual; // hash function เพื่อคำนวณตำแหน่ง bucket
    float mMaxLoadFactor;
    ...

    unordered_map() :
        mBuckets( std::vector<BucketT>(11) ), mSize(0),
        mHasher( HasherT() ), mEqual( EqualT() ),
        mMaxLoadFactor(1.0)
    { }

    unordered_map(const
        unordered_map<KeyT,MappedT,HasherT,EqualT> &other) :
        mBuckets(other.mBuckets), mSize(other.mSize),
        mHasher(other.mHasher), mEqual(other.mEqual),
        mMaxLoadFactor(other.mMaxLoadFactor)
    { }

    unordered_map<KeyT,MappedT,HasherT,EqualT>&
    operator=(unordered_map<KeyT,MappedT,HasherT,EqualT> other) {
        using std::swap;
        swap(this->mBuckets, other.mBuckets);
        swap(this->mSize, other.mSize);
        swap(this->mHasher, other.mHasher);
        swap(this->mEqual, other.mEqual);
        swap(this->mMaxLoadFactor, other.mMaxLoadFactor);
        return *this;
    }
}

```

Functions

```

template < ... >
class unordered_map {
public:
    bool empty() { return mSize == 0; }
    size_t size() { return mSize; }
    size_t bucket_count() { return mBuckets.size(); }
    size_t bucket_size(size_t n) { return mBuckets[n].size(); }
    float load_factor() { return (float)mSize/mBuckets.size(); }
    float max_load_factor() { return mMaxLoadFactor; }
    void max_load_factor(float z) {...}

    iterator begin() {...}
    iterator end() {...}

```

```

MappedT& operator[](const KeyT& key) {...}
void      clear()          {...}
void      rehash(size_t n)  {...}
size_t    erase(const KeyT &key) {...}
pair<iterator, bool> insert(const ValueT& val) {...}
}

operator[]
size_t hash_to_bucket(const KeyT& key) {
    return mHasher(key) % mBuckets.size();
}

ValueIterator find_in_bucket(BucketT& b, const KeyT& key) {
    for (ValueIterator it = begin(); it != b.end(); it++) {
        if (mEqual(it->first, key)) return it;
    }
    return b.end();
}

ValueIterator insert_to_bucket(const ValueT& val, size_t& bi) {
    if ( ตารางแน่นเกินไป ) { ปรับตาราง }
    ++mSize;
    return mBuckets[bi].insert(mBuckets[bi].end(), val);
} // ผลการ insert ใน vector คือ iterator ไปยังข้อมูลที่ถูกเพิ่มใหม่

MappedT& operator[](const KeyT& key) {
    size_t      bi = hash_to_bucket(key);
    ValueIterator it = find_in_bucket(mBuckets[bi], key);

    // ถ้าหาไม่พบ ต้องเพิ่ม pair(key, ค่า default ของ mapped value)
    if (it == mBuckets[bi].end()) {
        it = insert_to_bucket(make_pair(key, MappedT()), bi);
    }

    return it->second;
}

erase
size_t erase(const KeyT & key) {
    size_t      bi = hash_to_bucket(key);
    ValueIterator it = find_in_bucket(mBuckets[bi], key);
    if (it == mBuckets[bi].end()) {
        return 0; // erase 0 element
    } else {
        mBuckets[bi].erase(it);
        mSize--;
        return 1; // erase 1 element
    }
}

```

ผลของการ erase คือ จำนวนข้อมูลที่ถูกลบ

- 0 คือ ไม่พบ ไม่มีการลบเกิดขึ้น
- 1 คือ พบ key มีการลบ pair ของ key และ mapped value ของ key นั้น

insert

```

pair<iterator, bool> insert(const ValueT& val) {
    pair<iterator, bool> result;
    const KeyT& key = val.first;
    size_t bi = hash_to_bucket(key);
    ValueIterator it = find_in_bucket(mBuckets[bi], key);
    result.second = false;
    if (it == mBuckets[bi].end()) {
        it = insert_to_bucket(val, bi); // bi อาจเปลี่ยน ถ้ามีการปรับตาราง
        result.second = true;
    }
    result.first = iterator(it, // iterator ของ unordered_map
                           mBuckets.begin() + bi,
                           mBuckets.end());
    return result;
}

```

val คือ pair(key, mappedValue)

clear

```

void clear() {
    for (vector<BucketT>::iterator it = mBuckets.begin();
         it != mBuckets.end();
         ++it) {
        (*it).clear();
    }
    mSize = 0;
}

void clear() {
    for (BucketT & bucket : mBuckets) { // for each bucket in mBuckets
        bucket.clear();
    }
    mSize = 0;
}

```

destructor

```

class unordered_map {
    ...
    ~unordered_map() {
        clear();
    }
    ...
};

```

Rehashing

ถ้าฟังก์ชันแฮชกระจายดี การลบและค้นใช้เวลา $\mathcal{O}(1)$

ถ้าควบคุม λ ไม่ให้เกินค่าคงตัว k การลบและค้นใช้เวลาคงตัว

ถ้า “แน่น” เกิน ต้อง rehash

```

void rehash(size_t m) {
    if ( m <= mBuckets.size() &&
         load_factor() <= max_load_factor() ) return;
    m = std::max(m, (size_t)(0.5*mSize/mMaxLoadFactor));
    m = *std::lower_bound(PRIMES, PRIMES+N_PRIMES, m); 1
}

```

```

vector<ValueT> tmp;
for (auto& val : *this) tmp.push_back(val);
this->clear();
mBuckets.resize(m);
for (auto& val : tmp) this->insert(val);
}

ValueIterator insert_to_bucket(const ValueT& val, size_t& bi) {
    if (load_factor() > max_load_factor()) {
        rehash(2*bucket_count());
        bi = hash_to_bucket(val.first);
    }
    ++mSize;
    return mBuckets[bi].insert(mBuckets[bi].end(), val);
}

```

ขอปรับขนาดตารางเป็นจำนวนเฉพาะ

```

const size_t      N_PRIMES      = 256;
const unsigned long PRIMES[256] = {
    2ul, 3ul, 5ul, 7ul, 11ul, 13ul, 17ul, 19ul, 23ul, 29ul,
    ...
};

```

Separate Chaining Iterator

iterator

```

class unordered_map {
protected:
    ...
    class hashtable_iterator {
        ...
    public:
        hashtable_iterator() { ... }
        hashtable_iterator& operator++() { ... } // ++it
        hashtable_iterator operator++(int) { ... } // it++
        ValueT & operator*() { ... } // *it
        ValueT * operator->() { ... } // it->first
        bool operator!=(const hashtable_iterator &other) { ... }
        bool operator==(const hashtable_iterator &other) { ... }
    };
public:
    typedef hashtable_iterator iterator;
    ...
};

```

++iterator

```

vector<ValueT>::iterator
                        ValueIterator
vector<BucketT>::iterator ▼
BucketIterator ▶ [ ] -> [ * , * , * ]
                  [ ] -> [ ]
                  [ ] -> [ * , * ]
                  [ ] -> [ ]
                  [ ] -> [ * ]

```

```

class hashtable_iterator{
protected:
    ValueIterator  mCurValueItr;  // ▼
    BucketIterator mCurBucketItr;  // ►
    BucketIterator mEndBucketsItr;  // เก็บ mBuckets.end() ตอนสร้าง iterator

    void to_next_data() {
        while(mCurBucketItr != mEndBucketsItr &&
              mCurValueItr == mCurBucketItr->end())
            mCurBucketItr++;
        if (mCurBucketItr == mEndBucketsItr) break;
        mCurValueItr = mCurBucketItr->begin();
    }

public:
    hashtable_iterator& operator++() {  // it++
        mCurValueItr++;
        to_next_data();
        return (*this);  // ++(++it) it เลื่อนสองครั้ง
    }
    hashtable_iterator operator++(int) {  // ++it
        hashtable_iterator tmp(*this);
        operator++;
        return tmp;  // (it++)++ it เลื่อนครั้งเดียว!
    }

    *it กับ it->
    typedef ValueT & reference;
    typedef ValueT * pointer;

    reference operator*() {
        return *mCurValueItr;  // (*it).first
    }

    pointer operator->() {
        return &(*mCurValueItr);  // it->first
    }

    it1 == it2 กับ it1 != it2
    bool operator==(const hashtable_iterator &other) {
        return mCurValueItr == other.mCurValueItr;
    }
    bool operator!=(const hashtable_iterator &other) {
        return mCurValueItr != other.mCurValueItr;
    }

    constructor
    hashtable_iterator(ValueIterator valueItr,
                      BucketIterator bucketItr,
                      BucketIterator endBucketItr) :
        mCurValueItr(valueItr);
        mCurBucketItr(bucketItr);
        mEndBucketItr(endBucketItr);
    {
        to_next_data();
    }

```

```

    }
};

begin() && end()
iterator begin() {
    return iterator( mBuckets.begin()->begin(),
                    mBuckets.begin(),
                    mBuckets.end() );
}

iterator end() {
    return iterator( mBuckets[mBuckets.size()-1].end(),
                    mBuckets.end(),
                    mBuckets.end() );
}

```

อย่าลืม default constructor

```

class unordered_map {
    ...
    class hashtable_iterator {
        ...
        hashtable_iterator() { }
        ...
    };
    ...
    pair<iterator, bool> insert(const ValueT& val) {
        pair<iterator, bool> result;
        ...
        return result;
    }
    ...
};

```

Open Addressing

การแก้ปัญหาการชนแบบอื่น

- แบบแยกกันโยง (separate addressing)
 - ▶ แต่ละช่องในตารางเก็บรายการโยงของข้อมูล
 - ▶ ข้อมูลที่ชนกันเก็บอยู่ด้วยกัน ไม่กระทบข้อมูลอื่น
- แบบเลขที่อยู่เปิด (open addressing)
 - ▶ แต่ละช่องในตารางเก็บข้อมูล
 - ▶ ถ้าชน ก็หาช่องว่างใหม่ในตารางเพื่อเก็บข้อมูล
 - ▶ $\lambda = n/m \leq 1$ เสมอ ต้องคุมไม่ให้เกินเกณฑ์ ($\lambda \leq 0.5$)
 - ▶ มีหลายวิธีในการหาช่องว่างใหม่ในตาราง เมื่อเกิดการชน
 - การตรวจเชิงเส้น (linear probing)
 - การตรวจกำลังสอง (quadratic probing)
 - การตรวจสองชั้น (double hashing)

การตรวจเชิงเส้น (Linear Probing)

- เมื่อชน หาช่องว่างถัดไปด้วยวิธีดูตัวถัดไปเรื่อย ๆ

ใช้ $h(x) = x \% 13$ แล้วเพิ่มข้อมูลที่มีคีย์ตามลำดับดังนี้

17	32	26	7	4	43	12	11	24				
----	----	----	---	---	----	----	----	----	--	--	--	--

0	1	2	3	4	5	6	7	8	9	10	11	12
---	---	---	---	---	---	---	---	---	---	----	----	----


```

[26, , , , 17, , 32, 7, , , , ]
[26, , , , 17, 4, 32, 7, , , , ] // 4 ไม่ว่าง ลง 5 แทน
[26, , , , 17, 4, 32, 7, 43, , , , ] // 4 ไม่ว่าง ลง 8 แทน
[26, , , , 17, 4, 32, 7, 43, , , 11, 12]
[26, 24, , , , 17, 4, 32, 7, 43, , , 11, 12] // 11 ไม่ว่าง วนไปลง 1 แทน

```

- ให้ $h_j(x)$ คือช่องที่ probe หลังจากชนครั้งที่ j
- $h_0(x) = h(x)$ คือช่องที่ hash เริ่มต้น (home address)

$$h_j(x) = (h(x) + j) \% m$$

$$h_j(x) = (h_{j-1}(x) + 1) \% m$$

- ค้น - หาไม่พบเมื่อเจอช่องว่าง
- ลบ - ถ้าเจอช่องว่างที่เคยมีข้อมูล ให้ค้นต่อไป

สถานะของช่องเก็บข้อมูล (bucket)

- แต่ละช่องมี 3 สถานะ
 - 0: empty : ช่องว่าง ๆ ไม่เคยมีข้อมูลมาก่อนเลย
 - 1: deleted: ช่องที่เก็บข้อมูลที่ถูกลบไปแล้ว
 - 2: data : ช่องที่มีข้อมูลเก็บอยู่

Linear Probing Iterator

ข้อมูลที่เก็บในตาราง

```

template <...>
class unordered_map {
protected:
    typedef pair<KeyT, MappedT> ValueT;
    class BucketT {
    public:
        ValueT      value;
        unsigned char status;

        bool available() { return status < 2; }
        bool empty()     { return status == 0; }
        bool has_data()  { return status == 2; }
        void mark_deleted() { status = 1; }
        void mark_empty()  { status = 0; }
        void mark_data()   { status = 2; }
    };
    vector<BucketT> mBuckets;

```

การเปลี่ยนสถานะของ bucket

- constructor -> empty
- m.insert(val) -> mark_data
- m["X"] = 2 -> mark_data
- m.erase("X") -> mark_deleted
- m.clear() -> mark_empty
- m.rehash(...)
 - clear -> mark_empty
 - insert -> mark_data

มาดู iterator กันก่อน

```

class hashtable_iterator {
protected:
    BucketIterator mCurBucketItr;
    BucketIterator mEndBucketItr;

```

```

    void to_next_data() {
        while ( mCurBucketItr != mEndBucketItr &&
                !mCurBucketItr->has_data() ) {
            mCurBucketItr++;
        }
    }
public:
    hashtable_iterator(BucketIterator bucket,
                      BucketIterator endBucket) :
        mCurBucketItr(bucket), mEndBucketItr(endBucket) {
        to_next_data();
    }
};

public:
    iterator begin() {
        return iterator( mBuckets.begin(), mBuckets.end() );
    }
};

++it ãŸŸ it++
class hashtable_iterator {
protected:
    BucketIterator mCurBucketItr;
    BucketIterator mEndBucketItr;
    void to_next_data() {...}

public:
    ...
    hashtable_iterator& operator++() { // ++it
        mCurBucketItr++;
        to_next_data();
        return (*this);
    }
    hashtable_iterator operator++(int) { // it++
        hashtable_iterator tmp(*this);
        operator++();
        return tmp;
    }
}

*it ãŸŸ it->
    ValueT & operator*() {
        return mCurBucketItr->value;
    }

    ValueT * operator->() {
        return &(mCurBucketItr->value);
    }
};

```

Linear Probing Hash Table

unordered_map (linear probing)

```

template <...>
class unordered_map {
protected:

```

```

typedef pair<KeyT, MappedT> ValueT;
class BucketT {...};
class hashtable_iterator {...};

vector<BucketT> mBuckets;
size_t          mSize;
HasherT         mHasher;           // Will hash_to_bucket
EqualT          mEqual;            // Will find_position
float           mMaxLoadFactor;    // Will insert_to_position < 1
size_t          mUsed;             // # data + # deleted

size_t hash_to_bucket(const KeyT& key) {
    return mHasher(key) % mBuckets.size();
}

size_t find_position(const KeyT& key) {...}
BucketIterator insert_to_position(const ValueT& val, size_t& pos) {...}

```

find_position

```

vector<BucketT> mBuckets;
size_t find_position(const KeyT& key) {
    size_t homePos = hash_to_bucket(key);
    size_t pos      = homePos;
    while ( !mBuckets[pos].empty() &&
            !mEqual(mBuckets[pos].value.first, key) ) {
        pos = (pos + 1) % mBuckets.size();
    }
    return pos; // position of the target or the next empty bucket
}

```

insert

```

BucketIterator insert_to_position(const ValueT& val, size_t& pos) {
    if (load_factor() > max_load_factor()) {
        rehash(2*bucket_count());
        pos = find_position(val.first);
    }
    mSize++;
    mBuckets[pos].value = val;
    if (mBuckets[pos].empty()) mUsed++;
    mBuckets[pos].mark_data();
    return mBuckets.begin()+pos;
}

pair<iterator, bool> insert(const ValueT& val) {
    size_t pos = find_position(val.first);
    if (mBuckets[pos].available()) {
        BucketIterator it = insert_to_position(val, pos);
        result.first = iterator(it, mBuckets.end());
        result.second = true;
    } else {
        result.first = iterator(mBuckets.begin()+pos, mBuckets.end());
        result.second = false;
    }
    return result;
}

```

คำนวณ load factor

```

float load_factor() {
    return (float)mUsed/mBuckets.size();
}

float max_load_factor() {
    return mMaxLoadFactor;
}

void max_load_factor(float z) {
    mMaxLoadFactor = z;
    rehash(bucket_count());
}

```

operator[]

```

MappedT& operator[](const KeyT& key) {
    size_t pos = find_position(key);
    if (mBuckets[pos].available()) { // ไม่มีข้อมูล
        insert_to_position(make_pair(key, MappedT()), pos);
    }
    return mBuckets[pos].value.second;
}

```

erase

```

size_t erase(const KeyT & key) {
    size_t pos = find_position(key);
    if (mBuckets[pos].has_data()) {
        mBuckets[pos].mark_deleted();
        mSize--;
        return 1;
    } else {
        return 0;
    }
}

```

clear

```

void clear() {
    for (auto& bucket : mBuckets) {
        bucket.mark_empty();
    }
    mSize = 0;
    mUsed = 0;
}

```

rehash

```

void rehash(size_t m) {
    if (load_factor() <= max_load_factor() &&
        m <= mBuckets.size()) return;
    m = max(m, (size_t)(0.5+mSize/mMaxLoadFactor));
    m = *lower_bound(PRIMES, PRIMES+N_PRIMES, m);
    vector<ValueT> tmp;
    for (auto& val : this) {
        tmp.push_back(val);
    }
    this->clear();
    mBuckets.resize(m);
    for (auto& val : tmp) {

```

```

        this->insert(val);
    }
}

```

บริการอื่น ๆ

```
bool empty() { return mSize == 0; }
size_t size() { return mSize; }
size_t bucket_count() { return mBuckets.size(); }
size_t bucket_size(size_t n) {
    return mBuckets[n].has_data() ? 1 : 0;
}

};
```

การเกาะกลุ่มปฐมภูมิ (Primary Clustering)

- ถ้าใช้ linear probing แล้วเพิ่มข้อมูลอีกตัวลงตารางข้างล่างนี้ อยากทราบว่า ข้อมูลนี้จะถูกนำไปเก็บไว้ที่ช่องใดด้วยความน่าจะเป็นสูงสุด
- Cookie Monster Effect

Quadratic Probing

การตรวจกำลังสอง (Quadratic Probing)

- ขจัดการเกาะกลุ่มปฐภูมิ
- หลีกเลี่ยงการตรวจช่องติด ๆ กัน
- ให้ตรวจแบบก้าวกระโดดห่าง ๆ

$$h_j(x) = (h(x) + j^2) \% m \qquad h_j(x) = (h_{j-1}(x) - 2j - 1) \% m$$

ใช้ $h(x) = x \% 13$ แล้วเพิ่มข้อมูลที่มีคีย์ตามลำดับดังนี้

4 5 8 0 7 1 17

0	1	2	3	4	5	6	7	8	9	10	11	12
[0,	1,	,	,	4,	5,	,	7,	8,	,	,	,	]

เพิ่ม 17 ลง 4 แต่ไม่ว่าง

```

      ▲ +1
    -   ▲ +3
          - - - - - ▲+5
                - - - - - >

```

> ▲ +7
- - - - - ▲ +9
- - - - - >

```
> -----▼
[ 0, 1, , 17, 4, 5, , 7, 8, , , , ]
```

Linear vs Quadratic

```
size_t find_position(const KeyT& key) {
    size_t homePos = hash_to_bucket(key);
    size_t pos = homePos, m = mBuckets.size(), col_count = 0;
    while ( !mBuckets[pos].empty() &&
            !mEqual(mBuckets[pos].value.first, key) ) {
        col_count++;
        pos = (pos + 2*col_count-1) % m;
        // pos = (homePos + col_count*col_count) % m;
    }
    return pos;
}
```

คลาสเพื่อการคำนวณตำแหน่งถัดไป

```
class LinearProbing {
public:
    size_t operator()(size_t homePos,
                      size_t col_count,
                      size_t bucket_count) {
        return (homePos + col_count) % bucket_count;
    }
};

class QuadraticProbing {
public:
    size_t operator()(size_t homePos,
                      size_t col_count,
                      size_t bucket_count) {
        return (homePos + col_count*col_count) % bucket_count;
    }
};
```

NextAddressT

```
template <typename KeyT,
          typename MappedT,
          typename HasherT = std::hash<KeyT>,
          typename EqualT = std::equal_to<KeyT>,
          typename NextAddressT = LinearProbing >
class unordered_map {
    ...
    NextAddressT    mNextAddress;
```

default & copy constructor

```
unordered_map() :
    mBuckets(vector<BucketT>(11), mSize(0)),
    mHasher(HasherT()), mEqual(EqualT()),
    mMaxLoadFactor(0.5), mUsed(0),
    mNextAddress( NextAddressT() )
{ }

unordered_map(const
    unordered_map<KeyT,MappedT,HasherT,EqualT> &other) :
    mBuckets(other.mBuckets), mSize(other.mSize),
    mHasher(other.mHasher), mEqual(other.mEqual),
    mMaxLoadFactor(other.mMaxLoadFactor), mUsed(other.mUsed),
    mNextAddress( other.mNextAddress )
{ }
};
```

การตรวจกำลังสองไม่ตรวจทุกช่อง

- ลองเพิ่ม 30 อีกตัว ($h(x) = x \% 13$)

$$\begin{array}{ll}
 h(x) = 4 & (4 + 7^2) \% 13 = 1 \\
 (4 + 1^2) \% 13 = 5 & (4 + 8^2) \% 13 = 3 \\
 (4 + 2^2) \% 13 = 8 & (4 + 9^2) \% 13 = 7 \\
 (4 + 3^2) \% 13 = 0 & (4 + 10^2) \% 13 = 0 \\
 (4 + 4^2) \% 13 = 7 & (4 + 11^2) \% 13 = 8 \\
 (4 + 5^2) \% 13 = 3 & (4 + 12^2) \% 13 = 5 \\
 (4 + 6^2) \% 13 = 1 & (4 + 13^2) \% 13 = 4 \\
 & \dots
 \end{array}$$

- มีช่องว่างอาจหาไม่พบ

เมื่อตารางมีขนาดเป็นจำนวนเฉพาะ

- การตรวจกำลังสองจะดูอย่างน้อยครึ่งหนึ่งของตาราง
- ดังนั้น ถ้า **load factor** $\leq 1/2$ ก็สบายใจว่า จะหาช่องว่างพบ เมื่อเพิ่มข้อมูล
- พิสูจน์ : ให้ $0 \leq i \leq j \leq \lfloor m/2 \rfloor$ ถ้าข้างบนไม่จริง ต้องมีการ probe ครั้งที่ i และ j ที่ดูช่องซ้ำกัน

$$\begin{aligned}
 h(x) + j^2 &\equiv h(x) + i^2 \pmod{m} \\
 j^2 &\equiv i^2 \pmod{m} \\
 (j^2 - i^2) &\equiv 0 \pmod{m} \\
 (j - i)(j + i) &\equiv 0 \pmod{m}
 \end{aligned}$$

- เป็นไปไม่ได้: $(j - i)$ ไม่เป็น 0, $(j + i)$ ก็ไม่เป็น m
อีกทั้ง $(j - i)(j + i) \% m \neq 0$ เพราะทั้งสองพจน์ $< m$ และ m เป็นจำนวนเฉพาะ

Double Hashing

การเกาะกลุ่ม

- การเกาะกลุ่มทุติยภูมิ (secondary clustering)
 - ▶ ข้อมูลที่มี $h(x)$ เดียวกัน จะตรวจช่องในตารางเหมือนกัน
 - ▶ ระยะกระโดดของการตรวจแปรตามหมายเลขครั้งที่ชน
 - ▶ $h_j(x) = (h(x) + j) \% m$, $h_j(x) = (h(x) + j^2) \% m$
 - ▶ แก้ปัญหาได้ โดยให้ข้อมูลที่มี $h(x)$ เดียวกัน ไม่จำเป็นต้องมีระยะโดดของการตรวจเหมือนกัน
 - ▶ ให้ระยะกระโดดคำนวณจากค่าของข้อมูล

การแฮชสองชั้น (Double Hashing)

- ใช้ฟังก์ชันแฮชอีกตัวเพื่อคำนวณระยะกระโดด
- ทำให้ชุดข้อมูลที่แฮชไปที่ช่องเดียวกัน อาจมีระยะกระโดดต่างกัน

$$h_j(x) = (h(x) + j \cdot g(x)) \% m \qquad h_j(x) = (h_{j-1}(x) + g(x)) \% m$$

- โดยที่ $g(x) \% m \neq 0$ (เพื่อไม่ให้ย่ำอยู่กับที่) เช่น
 - ▶ $g(x) = R - (x \% R)$, R เป็นจำนวนเฉพาะ และ $R < m$
- และ ตัวหารร่วมมากของ $g(x)$ และ m ต้องเป็น 1 จะได้ตรวจทุกช่องในตาราง
 - ▶ ประกันเงื่อนไขนี้ได้โดยให้ m เป็นจำนวนเฉพาะ
 - ▶ $h(x) = 0, g(x) = 4, m = 8$ จะตรวจช่อง 0 และ 4 เท่านั้น
 - ▶ $h(x) = 0, g(x) = 4, m = 7$ จะตรวจช่อง 0, 4, 1, 5, 2, 6, 3

เปรียบเทียบจำนวนการตรวจเฉลี่ย

	Linear Probing		Quadratic Probing		Double Hashing	
	พบ	ไม่พบ	พบ	ไม่พบ	พบ	ไม่พบ

$\lambda = 0.3$	1.21	1.52	1.21	1.47	1.19	1.43
$\lambda = 0.4$	1.33	1.89	1.31	1.75	1.28	1.67
$\lambda = 0.5$	1.50	2.50	1.43	2.14	1.39	2.02
$\lambda = 0.6$	1.75	3.63	1.59	2.72	1.53	2.54
$\lambda = 0.7$	2.16	6.02	1.82	3.70	1.74	3.44
$\lambda = 0.8$	3.00	12.84	2.16	5.64	2.05	5.32
$\lambda = 0.9$	5.44	49.70	2.79	11.37	2.67	11.63

	จำนวนการตรวจเฉลี่ย	
	หาพบ	หาไม่พบ
แบบแยกกันโยง ($\lambda \geq 0$)	$1 + \frac{\lambda}{2}$	$1 + \lambda$
การตรวจเชิงเส้น ($0 \leq \lambda \leq 1$)	$\frac{1}{2} \left(1 + \frac{1}{1-\lambda} \right)$	$\frac{1}{2} \left(1 + \frac{1}{(1-\lambda)^2} \right)$
การแฮชสองชั้น ($0 \leq \lambda \leq 1$)	$\frac{1}{\lambda} \ln \frac{1}{1-\lambda}$	$\frac{1}{1-\lambda}$

ถาม: เก็บข้อมูลโดยใช้การตรวจเชิงเส้น ถ้าต้องการตรวจโดยเฉลี่ยไม่เกิน 5 ครั้ง ต้องควบคุมให้ตารางแฮชมี λ เป็นเท่าใด

ตอบ:

$$\begin{aligned}
 5 &\geq \frac{1}{2} \left(1 + \frac{1}{(1-\lambda)^2} \right) \\
 9 &\geq \frac{1}{(1-\lambda)^2} \\
 1-\lambda &\geq \sqrt{\frac{1}{9}} \\
 \lambda &\leq \frac{2}{3}
 \end{aligned}$$

ข้อควรระวัง

- ไม่เหมาะกับบริการที่
 - ▶ แจกแจงข้อมูลด้วย iterator
 - ▶ เกี่ยวข้องกับอันดับของข้อมูล `getMin`, `getMax`, ...
 - ▶ ต้องค้นทั้งตาราง $\Theta(m+n)$
- ต้องระวังเรื่องฟังก์ชันแฮช
 - ▶ ฟังก์ชันแฮชไม่ดี ก็ใช้งานได้ แต่ถ้า n มาก อาจช้าเป็น $\mathcal{O}(n)$

```

class BookHasher {
public:
    size_t operator()(const Book& b) const {
        return hash<string>()(b.title) ^
               hash<int>()(b.edition);
    }
};

```

สรุป

- การค้น เพิ่ม ลบข้อมูลในตารางแฮชทำได้รวดเร็ว
- สามารถปรับเวลาการทำงานให้เร็วขึ้นด้วยการใช้เนื้อที่เข้าแลก เพื่อให้ได้ λ ที่เหมาะสม
- ฟังก์ชันแฮชมีผลต่อประสิทธิภาพการทำงาน